

全国第六届研究生数学建模竞赛



题 目 110 警车配置及巡逻方案

摘 要：

本文对某城市中 110 警车的配置和巡逻方案进行抽象、建模和求解，并设计了评价指标，对结果方案进行评估。

第 1 问中，我们针对警车巡逻配置最小化问题建立了带有复杂约束的混合整数非线性规划模型，并将其用简单的矩阵运算表达，然后用分支定界法结合导向局域搜索方法进行全局搜索，得到全局最优解。我们的结果是最少需要 15 辆警车使得地图上点的覆盖率达到 90.88%。

第 2 问，从整体巡逻覆盖率、重点部位照顾以及巡逻公平性三个方面制定了三个指标，并详细论证了巡逻公平性的合理性和正确性。

对于第 3、4、5、6 问，我们构建了离散随机事件模拟系统，根据不同的要求，制定了不同的决策算法，并进行了模拟。通过分析结果数据得到指标值，评估并确定了局部最优化的巡逻方案。

最后，我们对开头提出的假设进行了检验，论证假设的合理性和模型的可扩展性。

关键字：整数非线性规划 分支定界 导向局域搜索 离散随机事件模拟

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110 警车配置及巡逻方案

1 问题重述

某城市拟增加一批配备有 GPS 卫星定位系统及先进通讯设备的 110 警车在街道上巡逻。这样既能震慑犯罪分子，又能增加市民的安全感，同时加快接处警时间，为社会和谐提供有力保障。为此，警车配置及巡逻方案要尽量满足以下要求：

1. 警车在接警后三分钟内赶到现场的比率不低于 90%；而赶到重点部位的时间必须在两分钟之内；
2. 使巡逻效果更显著；
3. 警车巡逻规律应有一定的隐蔽性。

2 问题分析

初步分析题目可知，我们需要解决以下的问题：

1. 为节省费用和开销，新配警车的数量应该尽量少，但又要使接处警时间尽量短，同时还要特别兼顾重点部位。这是有约束的整数非线性规划。约束条件是到达任意地方点或者重要部分的时间限制和覆盖比例限制。在一定的条件限制下找到最少需要配置的警车巡逻数也可以看成是一个带约束的集合覆盖问题。
2. 建立合理的巡逻效果的指标，分析接处警时间、巡逻覆盖率等相关因子之间的联系和制约，并论证指标的正确性和客观性。
3. 安排巡逻方案时，要在警车数量、接处警时间、巡逻覆盖范围以及隐蔽性等多种约束之间找到极值，这是动态情况下的多目标规划问题。根据实际情况，对各种目标和约束条件确定它们之间定性或者定量的关系从而产生不同的策略算法。另外，同时由于车辆的流动性较强，还要建立模拟这种离散随机事件的模型来产生最后的巡逻方案。
4. 通过对题目的理解，综合考虑现实情况，对巡逻和接警中可能出现的其它意外情况给出讨论，基于题目给出的几个要求之上，描述可能的解决方案。

3 模型假设

我们的假设如下：

- 1 以图中顶点为车辆的始发点。
- 2 警车及其上的 GPS、通讯设备不会在巡逻过程中发生故障。

- 3 警车巡逻时匀速运动，不考虑车辆的调头、启动、停止时的加减速过程，不考虑路况、单行道、红绿灯。

4 符号定义与说明

4.1 符号表定义

符号	定义
V	图中所有顶点的集合
E	图中所有路线（边）的集合
$G(V, E)$	抽象的整张无向图
C	图中所有停警车的顶点集合
N	顶点个数 $N=307$
M	边的条数 $M=458$
$\mathbf{M}_r$	图 G 的最短距离矩阵
r_3	警车接到报警后在 3 分钟行驶的路程长度 $r_3 = 40 \times 3/60 = 2000m$
r_2	警车接到报警后在 2 分钟行驶的路程长度为 $r_2 = 40 \times 2/60 = 1333m$
r_3'	速度提高后，警车接到报警后在 3 分钟行驶的路程长度为 $r_3' = 50 \times 3/60 = 2500m$
r_2'	速度提高后，警车接到报警后在 2 分钟行驶的路程长度为 $r_2' = 50 \times 2/60 = 1667m$

4.2 函数表定义

函数定义	说明
$MinD(v, p_i)$	点 v 和重要部位 p_i 之间的最小距离
$BinPSign(A)$	得到一个矩阵，元素是 0 或 1，如果矩阵 A 中元素是正数，则新矩阵中对应位置元素是 1，否则为 0。
$BinNSign(A)$	得到一个矩阵，元素是 0 或 1，如果矩阵 A 中元素是负数，则新矩阵中对应位置元素是 0，否则为 1。

$Sum(\mathbf{x})$	对向量 $\mathbf{x}$ 中的元素求和，返回整数。
$SumR(A)$	按行对矩阵的行向量求和，得到一个行向量。
$Diag(\mathbf{x})$	把向量 $\mathbf{x}$ (长度为 1) 转化成一个 1×1 的方阵， $\mathbf{x}$ 中元素作为方阵的对应的对角线元素。

4.3 名词约定

覆盖：某辆警车的以某距离 r 的覆盖是指，所有与警车距离小于 r 的图中点组成的集合。通常如果不特殊指明，我们所说的覆盖是指以距离 r_3 的覆盖。

有效覆盖：某车辆有且仅有它自己能覆盖到的点集。自己所能覆盖到的所有点集减去与别人重复覆盖的点集。

巡逻公平程度：巡逻公平程度指的是单位长的道路被巡逻次数的平均程度。

5 模型建立和求解

5.1 初步建模

首先，我们将区域地图抽象为图论模型。已知地图上一些点的坐标和它们之间的邻接关系，可以抽象出一个无向图 $G(V, E)$ ，地图上代表交叉路口的点作为无向图中的节点，连接路口的道路作为图中的边，路径长度为边的权值。通过对图的模型的设计，得到一个加权邻接对称矩阵 $\mathbf{M}_l$ ，权值仍然是路径长度。在此基础上，根据弗洛伊德（Floyd）算法，可以用 $\mathbf{M}_l$ 计算出图上任意两点间的最短距离矩阵 $\mathbf{M}_f$ 。在本题中，因为 G 一张无向连通图，所以 $\mathbf{M}_f$ 是对称矩阵，对角线元素全为零，其他元素都是正数。我们用 $m(v_i, v_j)$ 表示 $\mathbf{M}_f$ 中的 i 行 j 列的元素，即 v_i, v_j 的最小距离。

经计算发现，地图中的重点部位都不在路线上，我们假设通过到达离重点部位最近的几个点之一，再继续前行就可以到达重点部位。如图所示，四个点包着一个重点部位，通过它们就可以到达重点部位。

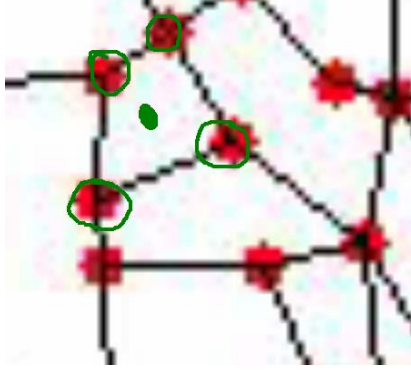


图 1 重点部位与其相邻节点示意图

我们用 S_i 表示到达某重点部位 p_i ($i = 1, 2, 3$) 的最近图上的点的集合， $S_i = \{s_{i,1}, s_{i,2}, s_{i,3}, s_{i,4}\}$ ，我们可以算出 S_i 中点与 p_i 之间的直线距离。用 $d_{i,j}$ 表示 p_i 与 $s_{i,j}$ 的距离，地图上的点与重点部位的最小距离是：

$$\text{MinD}(v, p_i) = \min(\{m(v, s_{i,j}) + d_{i,j} \mid s_{i,j} \in S_{m,i}\})$$

5.2 问题一解答,

5.2.1 分析与建模

我们需要找到一个最小的点集 $C = \{c_1, c_2, \dots, c_k\}$ ($1 \leq k \leq N$)，使得：

约束 (1) $|H_3| \geq |V| \times 90\%$ ，其中 $H_3 = \{v \mid v \in V, \exists c \in C \text{ s.t. } m(c, v) \leq r_3\}$ 。

约束 (2) $\exists c \in C \text{ s.t. } \text{MinD}(c, p_{m,i}) \leq r_2, \quad i = 1, 2, 3$ 。

这是一个最优化问题，我们需要进一步抽象模型，以得到更明确的目标函数和约束。两个不等式左侧的内容都需要重新表述。

为此，我们首先引入覆盖矩阵：

定义 1: 覆盖矩阵 $\mathbf{M}_h$ 是这样一个矩阵，它是 $|V| \times |V|$ 的矩阵，元素取值为 0 或 1，第 i 行第 j 列元素的值表示这两个点之间的最短距离是否小于等于某距离 r ，如果是，则元素值为 1，反之为 0。

覆盖矩阵类似于邻接矩阵，它描述了警车以某距离的覆盖。我们可以通过 $\mathbf{M}_f$ 计算覆盖矩阵，比如对于距离 r_3 ，先取一个全 1 矩阵 $\mathbf{O}_1$ 作数乘运算得到全 r_3 矩阵，再与 $\mathbf{M}_f$ 作差，得到的矩阵中元素值为正值则说明该点与行列序号所表示的点之间的距离大于 r_3 ，再通过一个非线性运算，将正数变为 0，其他数变为 1，得到覆盖矩阵。

$$\mathbf{M}_{h3} = \text{BinNSign}(\mathbf{M}_f - r_3 \cdot \mathbf{O}_1) \text{ 其中 } \mathbf{O}_1 \text{ 表示全 1 矩阵。}$$

同理可以计算得到 $\mathbf{M}_{h2}$ (表示警车在距离 r_2 上的覆盖能力), $\mathbf{M}_{h5}$ (表示警车在距离 r_5 上的覆盖能力)。

按序号给 V 中的点编号为 $1, 2, 3, \dots, |V|$ 。定义 0-1 离散决策向量 $\mathbf{x} = \{x_1, x_2, \dots, x_{|V|}\}$, $x_i = 0/1$, $i=1, 2, \dots, |V|$ 表示 C 中所包含的点在 V 中的序号。令 $\mathbf{x}$ 表示警车所在位置情况, 0 表示该序号的点没有警车, 1 表示该序号的点有警车。我们希望得到一个函数, 输入为 $\mathbf{x}$, 输出是警车对整个图的覆盖情况。

这个函数的具体构造方法如下: 把 $\mathbf{x}$ 变成对角阵, 再左乘以覆盖矩阵 $\mathbf{M}_{h3}$, 将 $\mathbf{M}_{h3}$ 的很多行变成了 0 向量, 从而得到警车所在点之外的其它点的覆盖情况。再对结果矩阵作行累加, 就可以得到每个点的累计覆盖情况。

$$f(\mathbf{x}) = \text{SumR}(\text{Diag}(\mathbf{x}) \times \mathbf{M}_{h3})$$

$f(\mathbf{x})$ 是 $|V|$ 维向量, 它的每一位都表示该序号的点被覆盖的次数。后续我们可以对重复的覆盖做一下处理, 只简单计作覆盖, 得到:

$$g(\mathbf{x}) = \text{BinPSign}(f(\mathbf{x}))$$

$g(\mathbf{x})$ 返回一个长度为 $|V|$ 的 0-1 离散决策向量, 第 i 位为 1 表示 C 中存在警车能够覆盖 v_i 。至此, 约束 (1) 可以表示成:

$$\text{Sum}(g(\mathbf{x})) \geq |V| \times 90\%$$

约束 (2) 的表述也可以通过向量运算来实现。通过遍历 V , 我们可以得到警车能够在 2 分钟内到达重点部位的点。 $h_{m1} = \{h_1, h_2, \dots, h_N\}$, $h_i = 0, 1$ 。 h_i 为 1 表示 V 中 v_i 可以在 2 分钟内到达重点部位, 同理有 h_{m2}, h_{m3} 。

综合以上的推导, 最优化模型可以表述为:

$$\mathbf{x} = \{x_1, x_2, \dots, x_{|V|}\}, x_i = 0, 1 \quad (i = 1, 2, \dots, |V|)$$

$$\text{Min Sum}(\mathbf{x})$$

$$s.t. \begin{cases} Sum(g(\mathbf{x})) \geq |V| \times 90\% \\ \mathbf{h}_{m1} \& \mathbf{x} \neq 0 \\ \mathbf{h}_{m2} \& \mathbf{x} \neq 0 \\ \mathbf{h}_{m3} \& \mathbf{x} \neq 0 \end{cases}$$

显然，所建模型是带有复杂约束的混合整数非线性规划模型，是 NP-Hard 问题[1]。需要考虑分支定界法等最优化方法。

5.2.2 算法与求解

由于问题规模很大，为了能够找到全局最优解，可以使用分支定界法结合导向局域搜索。算法使用了多种技巧来加快搜索。我们使用了如下几个剪枝规则：

规则一，如果 $Sum(\mathbf{x}_{current}) \geq Sum(\mathbf{x}_{optimal})$ ，就不再往下搜索，直接返回到上一次搜索（ $\mathbf{x}_{optimal}$ 是当前的已经搜索到的最优解）。

规则二，在执行搜索算法执行前，计算并保存一个查表函数 $MaxCover(y)$ ， $y=1,2,\dots,k$ 。若在图上放置 y 辆警车，该函数返回该情况下，这 y 辆警车最多能覆盖的点数（重复覆盖的按多次计算）。在搜索算法执行时，当前的 $\mathbf{x}_{current}$ 满足 $Sum(g(\mathbf{x}_{current})) + MaxCover(Sum(\mathbf{x}_{optimal}) - Sum(\mathbf{x}_{current})) < |V| \times 90\%$ ，那就不必再往下搜索，直接返回上一次搜索。

规则三，如果当前的 $\mathbf{x}_{current}$ 还有 a 个重点部分不能在以 r_2 为半径覆盖到，如果 $Sum(\mathbf{x}_{current}) + a \geq Sum(\mathbf{x}_{optimal})$ ，可以直接返回上一次搜索。

规则四，在搜索过程中，如果下一步搜索的可能选择是 $S_{next} = \{\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_q\}$ ，如果对于某个 $\mathbf{x}_s \in S_{next}$ ，存在 $\mathbf{x}_t \in S_{next}$ 使得 $g(\mathbf{x}_t) \& (\sim g(\mathbf{x}_s)) = 1$ ，即 $\mathbf{x}_s$ 所能覆盖的点 $\mathbf{x}_t$ 都能覆盖，则可以舍弃 $\mathbf{x}_s$ 这个选择。按照此规则对 S_{next} 中的元素进行筛选。事实证明这个剪枝非常有效，这个剪枝对搜索效率有重大影响。

规则五，设置一些标志位，防止相同的解空间被重复搜索。

规则六，除了剪枝以外，我们还使用了导向局域搜索（GLS），利用问题的解特征在原目标函数上附加惩罚项，形成增益目标函数来引导算法在解空间的搜索过程。具体说，在搜索过程中，得到剪枝后的下一步搜索可能的选择集 $S_{next}' = \{\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_p\}$ 之后，

对其进行权重的计算和按权重排序，得到一个队列。一般来说，靠前的元素搜索到优化的解得可能性大于靠后元素。

权重的计算步骤如下，以 $\mathbf{x}_1$ 为例：

计算与 $\mathbf{x}_{current}$ 相比， $\mathbf{x}_1$ 所能覆盖的 $\mathbf{x}_{current}$ 没有覆盖的点的个数 u_1 。

计算与 $\mathbf{x}_{current}$ 相比， $\mathbf{x}_1$ 所能覆盖的 $\mathbf{x}_{current}$ 已经覆盖的点的个数 c_1 ，这个数值是一个惩罚项，引导算法去搜索更大范围的有效覆盖。

计算与 $\mathbf{x}_{current}$ 相比， $\mathbf{x}_1$ 是否能多覆盖某个重要点 b_1 (0 或者 1, 这个覆盖以 r_2 为半径)。

权重 $w_1 = u_1 - 2c_1 + 4b_1$ 。

5.2.3 计算结果

经实验，我们的算法可以在可接受的时间内搜索到全局最优解。

解得最少需要 15 辆警车，三个重要部位都能在两分钟内接处警，三分钟内可以到达的点有 279 个，三分钟内无法到达的点为 28 个，覆盖率达到了 90.88%，符合 D1 的要求。每辆车的坐标如下：

1. Node [id=274, Point (6390, 1404)]
2. Node [id=253, Point (11952, 2142)]
3. Node [id=208, Point (4104, 2916)]
4. Node [id=170, Point (10152, 3618)]
5. Node [id=167, Point (1962, 3636)]
6. Node [id=149, Point (7452, 3888)]
7. Node [id=123, Point (9198, 4392)]
8. Node [id=96, Point (4716, 5184)]
9. Node [id=79, Point (13266, 5292)]
10. Node [id=45, Point (11376, 6318)]
11. Node [id=43, Point (1314, 6480)]
12. Node [id=39, Point (14040, 6840)]
13. Node [id=28, Point (6480, 7128)]
14. Node [id=25, Point (9162, 7146)]
15. Node [id=10, Point (3528, 7902)]

5.3 问题二的解答

5.3.1 分析与建模

巡逻效果可以体现在几个方面：（1）整体覆盖率越高，效果越好；（2）警车到达现场越快越好；（3）警车到达重点部位越快越好；（4）巡逻的范围越大越好；（5）一段时间内，各条道路应该有较平均的被巡逻次数。

评价巡逻效果显著程度的有关指标要考虑到所有以上的因素。有些指标只衡量某一个方面的效果，有些可以综合衡量某几个方面的效果。

以下指标都可以用来衡量某个巡逻方案，即 $|C|$ 辆警车在一段时间 T 内巡逻的效果。

指标一，平均覆盖率向量： $\mathbf{h}=\{h_1, h_2 \dots\}$ ， h_i 代表 i 分钟内可以到达的地点的数目的平均值，序号小的元素值越大说明巡逻效果越好。

指标二，三个重要部位的平均可到达时间： t_{m1}, t_{m2}, t_{m3} ，值越小说明巡逻效果越好。

指标三，考虑到各条道路应该有较平均的被巡逻次数，我们定义单条道路单位时间内的巡逻收益：

$$r(e, T) = \frac{\sqrt{c_e \cdot l_e}}{T}$$

其中， l_e 是道路 e 的长度； c_e 是道路 e 被巡逻次数，这个次数不一定是整数，根据实际情况，也可以是0或者小数。

该指标随着道路被巡逻的次数和道路长度的增大而增大，同时随着巡逻次数的增加，该值增加的幅度变小。整个区域所有道路一段时间内的巡逻总收益为：

$$w(E, T) = \sum_{e \in E} r(e, T)$$

定理 1. 单位长的道路被巡逻次数越平均，整个区域所有道路一段时间内的巡逻总收益越高。

证明：一段时间 T 内， $|C|$ 辆警车始终以匀速巡逻，走过的路程是定值，即所有路被走过的总长度为定值，记为 L 。

$$L = \sum_{e \in E} c_e \cdot l_e$$

整个区域所有道路一段时间内的巡逻总收益为：

$$w(E, T) = \sum_{e \in E} r(e, T) = \frac{1}{T} \sum_{e \in E} \sqrt{c_e \cdot l_e} \leq \frac{1}{T} \sqrt{|E| \cdot \sum_{e \in E} \sqrt{c_e \cdot l_e}} = \frac{1}{T} \sqrt{|E| \cdot L}$$

当且仅当 $c_1 l_1 = c_2 l_2 = \dots c_{|E|} l_{|E|} = \frac{L}{|E|}$ 时，不等式取等，此时 $c_e = \frac{L}{l_e \times |E|}$ 。

在加权为各自道路长度时，道路被通过次数达到平均，整个区域在时间 T 内巡逻总收益最高。

假设在某情况下道路巡逻不公平，不妨设：

$$c_1 l_1 < \frac{L}{|E|} < c_2 l_2$$

取一个小量 d (d 是正实数)，使

$$c_1 l_1 < c_1 l_1 + d < \frac{L}{|E|} < c_2 l_2 - d < c_2 l_2$$

此时，道路间被巡逻次数相对平均一些，

$$c_1' = c_1 + \frac{d}{l_1}, \quad c_2' = c_2 - \frac{d}{l_2}$$

总收益的变化为：

$$\begin{aligned} w(E', T) - w(E, T) &= \sqrt{(c_1 + \frac{d}{l_1}) l_1} + \sqrt{(c_2 - \frac{d}{l_2}) l_2} - \sqrt{c_1 l_1} - \sqrt{c_2 l_2} \\ &= \frac{d}{(c_1 + \frac{d}{l_1}) l_1 + c_1 l_1} - \frac{d}{(c_2 - \frac{d}{l_2}) l_2 + c_2 l_2} \\ &= \frac{[(2c_2 l_2 - d) - (2c_2 l_2 + d)]}{(2c_2 l_2 - d)(2c_2 l_2 + d)} > 0 \end{aligned}$$

由此看出，道路被巡逻的次数越平均，能得到越好的道路总收益值，因此命题得证。

通过定理 1 的证明可以看出， $w(E, T)$ 的大小体现了巡逻公平程度，它同时也体现了巡逻的范围。一般来讲，巡逻范围越大，巡逻越公平，值越大。

基于 $w(E, T)$ 我们可以定义另一个重要指标：

指标三，单位警车单位时间的平均巡逻收益：

$$u(E, T, C) = \frac{w(E, T)}{|C|}$$

其值越大表明巡逻效果越好。指标三的 $u(E, T, C)$ 是将 $w(E, T)$ 平均到每一辆警车上。

按照这个指标的衡量，并不是警车越多越好。当地图上的警车稀疏时，各个警车之间的重叠覆盖比较少，仍然会有很多无法被覆盖的地点，这时，添加一辆警车去这些地点巡逻，会增加很大的巡逻收益；但如果警车很稠密，再增加一辆警车也只是在其他警车能覆盖的地方巡逻，增加的收益就远小于前者。所以，警车的数量并不是越多越好，而是让 $u(E, T, C)$ 达到峰值最好。针对不同的巡逻方案， $w(E, T)$ 会不同，所以会有不同的警车最优数量。

5.4 问题三的解答

5.4.1 分析与建模

我们的方案依托于 GPS 定位和先进的通讯设备。车辆巡逻的基本执行的步骤如下：

1. 各警车就位，初始状态满足 D1 条件；
2. 控制中心获得警车的状态，通过仿真模拟得到下一段时间的巡逻路线，通知各车；
3. 各车按控制中心所安排的路线巡逻；
4. 正常巡逻时间到 T 时，回到第 2 步；
5. 当出现报警时，离事故点最近的警车去处理，相当于该位置没有了车来覆盖周围的点，所以指挥中心要做调整，即回到第 2 步。

警车巡逻方案包括巡逻路线的设计和警车配置的确定。首先我们要设计一个安排路线的策略，因为不同策略会有不同的指标值，进而又会有不同的最优警车配置。

为了简化策略，在我们的离散事件模拟环境中，仅当有车辆到达交叉路口事件发生时，系统才会决策该警车的巡逻方向。做决策的依据是当前的全局状态，并不预测下一时刻其他警车的状态，因此这仍然是一个局部最优策略。

参照巡逻效果的指标的定义，我们发现 D1 和 D2 的第一、二条指标有一定的重复。因此，为了简化决策过程，在满足 D1 的情况下，只考虑尽量满足指标三。对于每一个交叉路口事件时，需要决策某辆警车下一次该走的方向，我们给出具体的决策步骤：

1. 判断是否有一个重点部位当前只有自己能在 2 分钟内到达（其它的车都不能），若是，那么任意一条影响自己不能在 2 分钟内到达该重点部位的路径都被舍去。
2. 如果还有选择性，则在剩余的路径中作选择。观察当前的全局覆盖率是否已经超额满足要求（比如设定 95% 覆盖率为阈值），若全局覆盖率不足，则只选择最大化增加自己有效覆盖的路径；否则可以舍弃一些对覆盖率有很大负影响的路径，保留其他路径做下一步决策。这样做是为了满足 D1 的 90% 覆盖率的需

要。

3. 如果还有选择性，则在剩余的路径中作选择。对于每个路径，计算 $\sqrt{c \cdot l}$ 作为比较目标，优先选择值小的。
4. 如果还有选择性，则在剩余的路径中作随机做出选择。

在策略制定之后，我们需要实际的仿真，并对仿真结果计算指标值。一些指标值的增益会变得不明显，一些指标值会出现峰值，通过分析这些我们可以确定警车的数量。

5.4.2 实验结果

我们在满足 D1 且尽量满足 D2 条件下做了一组实验，一共检查了警车数量从 15 到 30 的不同情况，每种情况都有 5 份场景，计算了三个指标值，并统计了其平均值。具体指标如表 1 所示：

表 1 问题三实验结果的指标值

警车数量	平均覆盖率向量	三个重要部位的平均可到达时间	单位警车单位时间的平均巡逻收益
15	[0.19, 0.39, 0.33]	[0.96, 0.92, 1.10]	11.12
16	[0.20, 0.40, 0.32]	[0.91, 0.84, 1.21]	11.09
17	[0.22, 0.41, 0.30]	[1.18, 0.85, 0.95]	11.50
18	[0.22, 0.40, 0.32]	[1.15, 0.86, 0.94]	11.39
19	[0.23, 0.41, 0.29]	[1.15, 0.86, 1.09]	11.49
20	[0.24, 0.42, 0.28]	[1.15, 0.87, 1.12]	11.44
21	[0.25, 0.41, 0.27]	[1.05, 0.95, 0.99]	11.02
22	[0.26, 0.42, 0.26]	[1.15, 0.94, 0.97]	10.92
23	[0.27, 0.43, 0.24]	[1.02, 0.97, 1.00]	10.71
24	[0.29, 0.43, 0.23]	[1.03, 0.93, 1.09]	10.56
25	[0.30, 0.42, 0.22]	[0.96, 0.91, 1.14]	10.40
26	[0.32, 0.41, 0.21]	[0.91, 0.88, 1.09]	10.25
27	[0.33, 0.41, 0.20]	[0.92, 0.93, 1.00]	10.03
28	[0.33, 0.41, 0.20]	[0.92, 0.97, 1.04]	9.84
29	[0.35, 0.40, 0.19]	[0.85, 1.04, 0.99]	9.69
30	[0.36, 0.41, 0.18]	[0.80, 0.93, 0.96]	9.54

通过观察数据我们发现，在 18、19、20 这几辆警车处，覆盖率指标和重点部门平均到达时间指标都已经变化不明显，而指标三也在 19 辆警车处到达了峰值。综合考虑，19 辆警车配置在当前策略下是最优的。

5.5 问题四的解答

5.5.1 分析与建模

巡逻规律的隐蔽性意味尽量不要对道路进行有规律地巡逻。我们可以通过随机化的手段来实现隐蔽性。这个解决方案的框架依然是离散事件模拟，所以整个过程也类似于第三问：从决策算法的设计，到仿真试验，再到警车配置的确定的。

当一个警车运动到一个点时，按如下逻辑作决策：

1. 判断是否有一个重点部位当前只有自己能在 2 分钟内到达（其它的车都不能），若是，那么任意一条可能会导致自己不能在 2 分钟内到达该重点部位的可能方向上的路径都被舍去。
2. 如果还有选择性，则在剩余的路径中作选择。观察当前的全局覆盖率是否已经超额满足要求（比如设定 95%覆盖率为阈值），若全局覆盖率不足，则只选择最大化增加自己有效覆盖的路径；否则可以舍弃一些对覆盖率有很大负影响的路径，保留其他路径做下一步决策。这样做是为了满足 D1 的 90%覆盖率的需要。
3. 如果还有选择性，则在剩余的路径中作选择。对于每个路径，计算权值

$w_i = \frac{1}{\sqrt{c_i l_i}}$ 。得到 $w_1, w_2, \dots, w_m$ 是几个路径的权重，构造 1-m 离散型随机变量

$\mathbf{X}$ ，使得

$$P\{X = i\} = \frac{w_i^2}{\sum_{i=1}^m w_i^2}$$

考虑到随机化程度不能过高，否则基础指标就会比较差，所以采用平方放大权重差异对决策的影响。最后选取第 $\mathbf{X}$ 个路径。

5.5.2 实验结果

我们在满足 D1 且尽量满足 D2、D3 条件下做了一组实验，一共检查了警车数量从 15 到 30 的不同情况，每种情况都有 5 份场景，计算了三个指标值，并统计了其平均值。具体如下表：

表 2 问题四实验结果的指标值

警车数量	平均覆盖率向量	三个重要部位的平均可到达时间	单位警车单位时间的平均巡逻收益
15	[0.21, 0.46, 0.24]	[0.80, 1.14, 1.30]	9.20
16	[0.20, 0.40, 0.34]	[0.74, 0.94, 0.78]	9.75
17	[0.22, 0.43, 0.31]	[1.14, 0.88, 1.12]	9.57
18	[0.21, 0.41, 0.32]	[1.23, 0.81, 1.30]	9.16
19	[0.22, 0.43, 0.30]	[0.93, 0.70, 1.25]	9.68
20	[0.24, 0.44, 0.28]	[0.87, 0.60, 1.14]	9.76
21	[0.24, 0.44, 0.30]	[1.09, 0.78, 1.04]	9.42
22	[0.25, 0.44, 0.29]	[1.09, 0.81, 1.27]	9.14
23	[0.26, 0.44, 0.27]	[1.02, 0.91, 1.12]	9.36
24	[0.26, 0.45, 0.27]	[1.15, 0.78, 1.21]	8.98
25	[0.27, 0.46, 0.25]	[1.09, 0.83, 1.08]	8.95
26	[0.28, 0.46, 0.24]	[1.03, 0.80, 1.01]	9.00
27	[0.29, 0.46, 0.23]	[1.04, 0.80, 0.97]	8.85
28	[0.30, 0.46, 0.23]	[0.95, 0.84, 1.23]	8.93
29	[0.31, 0.46, 0.22]	[0.98, 0.75, 1.04]	8.74
30	[0.31, 0.47, 0.22]	[1.05, 0.77, 1.11]	8.52

可以看到，实验数据不如同等场景的针对第三问的策略，并且数据比较参差，这是随机化的缘故。从以上数据来看，警车配置为 23 辆时最好。

5.6 问题五的解答

5.6.1 分析与建模

根据问题一的解答，我们认为该区域仅配置 10 辆警车是无法完全满足 D1 的覆盖率的要求的。然而，很多未覆盖的地方又给了警车在选择路径的时候更多的自由选择空间。这属于多目标的规划问题，采用化多为少的方法，把各个指标的加上一定的权值最终得出一个值来给车辆的路径选择作决策。因此，决策算法与前面几问相比又会有些变化，主要是各项指标的权重的变化，相对于前两问，我们降低了覆盖率指标、重点部分平均

到达时间指标的权重，提高了巡逻公平性指标的权重。

多目标的规划问题，需要把多目标因子做复合运算，所以它们必须归一化。一般的归一化的方法是：一个因子在各个候选者的权重值分别为 $w_1, w_2, \dots, w_m$ ，则归一化后的权重值为：

$$w_i' = \frac{w_i}{\sum_{j=1}^m w_j} \quad i = 1, 2, \dots, m$$

归一化后的不同因子有相同的上下界，就可以做加权复合运算了。

综合考虑巡逻效果显著性指标，我们从三个方面对当前警车来提取三个因子 x_1', x_2', x_3' 。

1. 覆盖率：可以考虑用有效覆盖来作为因子，即被该警车唯一覆盖的点的个数。
2. 重点部位最短距离：可以考虑使用平均到达某个重点部位的时间来做因子，为了方便，我们使用到达最近重点部位的距离，考虑到这是一个减函数，函数值域还有 0 值，要做一下变换，因子为 $\frac{1}{d+1}$ ， d 表示该警车到达最近重点部位的距离。
3. 巡逻公平：对于每条可选择的路，对应的巡逻公平因子为 $\frac{1}{\sqrt{cl}}$ ， c 表示路被走过的次数， l 为路的长度。

对于每次需要抉择的交叉口，我们都会计算 $\sum_{j=1}^3 x_j' w_j'$

5.6.2 求解和结果

我们在问题三的基础上，重设了车辆在到达交叉路口后的处理方法，该方法通过对上述三个因子进行规范化，降低之间的数据差异性。然后通过计算目标函数 $s = f_1 \cdot w_1 + f_2 \cdot w_2 + f_3 \cdot w_3$ 作为评价下一步行为好坏的标准，其中 f_1 、 f_2 、 f_3 分别代表三个影响因子， w_1 、 w_2 、 w_3 表示这三个因子相对应权重。

在处理决策时，各因子的权重图（见下图）中的 X 坐标表示有效覆盖的权重百分比 w_1 ，最低为 0%，最高为 90%。由于重点部位最小距离 w_2 在 w_1 为 10% 的情况下已经达到我们预定的效果（到达重点部位的平均时间小于 2 分钟），所以其值在测试过程中设

为固定不变。而巡逻公平因子则随着有效覆盖的权重 w_3 以 $(1-w_2-w_1)$ 的方式进行设置。

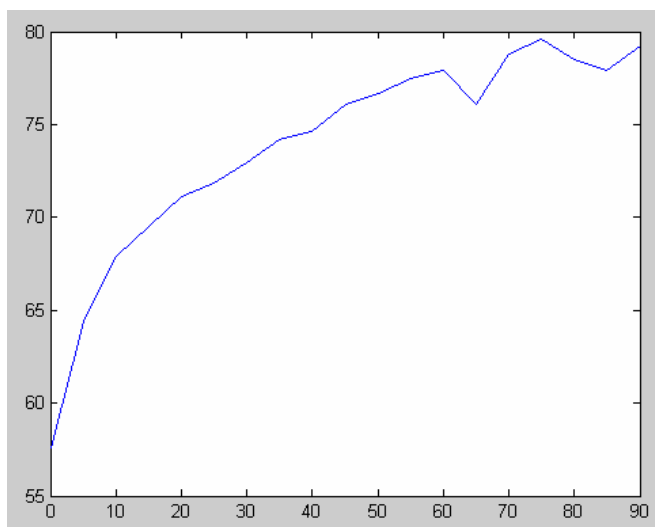


图 2 覆盖率曲线图

上图中的 Y 轴，表示 4 个小时内，警车应对突发状况的平均覆盖率，该值越高表明所覆盖的面积越大，效果越好。由此可见，覆盖率随着其权重的增加而非线性增加，切线斜率在不断的变小。

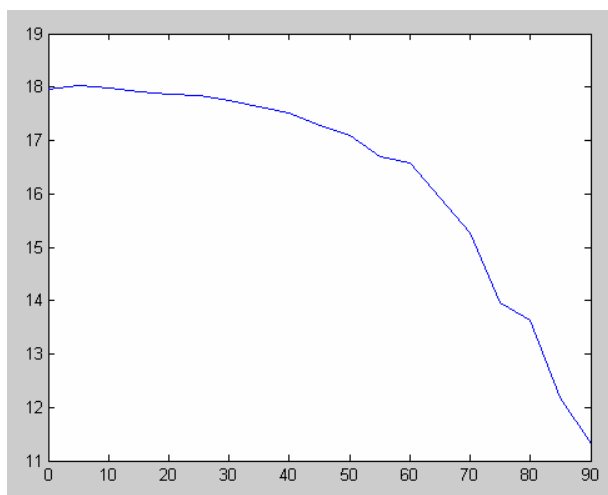


图 3 道路收益值曲线图

上图中 Y 轴，表示整体道路被经过数量的收益 $r(e, T)$ 值，该值越高表明道路总体被经过的数量越多，并且道路之间被经过的次数差异越小。由此可见，随着有效覆盖权重的增加，即巡逻公平的权重的减少，道路总收益值在非线性的减少，其切线的斜率在不断增大。

由此，我们通过分析和比对，从上两幅图的数据中综合得出一个有效覆盖权重的数据平衡点，即将有效覆盖的权重设置在 50%，此时方案的平均覆盖率达到 75%，道路总

收益值达到 17，其综合效果达到峰值。

最后，三个权重因子确定为： $w_1 = 50\%$ 、 $w_2 = 10\%$ 、 $w_3 = 40\%$ 。

5.7 问题六的解答

若想找到满足 D1 且尽量满足 D2 条件的巡逻方案，首先要得到最小配置的满足 D1 的巡逻方案。然后按照第三问的方法做离散事件模拟，再分析指标数据，找出最优警车配置。

1. Node [id=238, Point (7434, 2376)]
2. Node [id=211, Point (1818, 2898)]
3. Node [id=196, Point (5364, 2952)]
4. Node [id=148, Point (9882, 4032)]
5. Node [id=90, Point (5274, 5202)]
6. Node [id=67, Point (1224, 5814)]
7. Node [id=51, Point (7380, 6264)]
8. Node [id=39, Point (14040, 6840)]
9. Node [id=24, Point (10296, 7182)]
10. Node [id=10, Point (3528, 7902)]

按照第三问的方法，我们做了一组实验，一共检查了警车数量从 10 到 20 的不同情况，每种情况都有 5 份场景，计算了三个指标值，并统计了其平均值。观察表中数据可知，平均覆盖率的变化不大，三个部位的平均可到达时间也都很理想，单位警车单位时间的平均巡逻收益在 13 时达到了峰值。所以满足 D1 且尽量满足 D2 的最优警车配置是 13 辆。

5.7.1 求解和结果

表三 问题六实验结果的指标值

警车数量	平均覆盖率向量	三个重要部位的平均可到达时间	单位警车单位时间的平均巡逻收益
10	[0.21, 0.46, 0.24]	[0.62, 0.57, 1.61]	13.06
11	[0.22, 0.47, 0.23]	[0.92, 0.63, 1.17]	13.07
12	[0.22, 0.46, 0.25]	[1.09, 1.15, 1.24]	12.40

13	[0.23, 0.44, 0.27]	[1.11, 0.95, 1.27]	14.25
14	[0.23, 0.44, 0.26]	[1.06, 1.17, 1.14]	13.90
15	[0.26, 0.45, 0.23]	[1.13, 1.15, 1.04]	13.63
16	[1.13, 1.15, 1.04]	[0.94, 1.13, 1.12]	13.28
17	[0.30, 0.44, 0.20]	[0.97, 1.11, 1.12]	12.46
18	[0.29, 0.45, 0.21]	[0.97, 1.18, 1.05]	12.86
19	[0.32, 0.43, 0.19]	[0.96, 1.12, 1.09]	12.22
20	[0.34, 0.42, 0.18]	[0.87, 1.15, 1.05]	11.92

5.8 问题七的解答

我们还需考虑的情况和因素有：

1. 路况状况不理想，在现实中，有些路是单行道，有些路有可能某个方向临时修路等等。

解决方法：将无向图改为有向图，邻接矩阵不再是对称矩阵，根据不同地段路况的不同，增加一个路况系数来调整边的权重。在此基础上，仍然可以利用弗洛伊德（Floyd）算法，本文中所建立的三个模型也都直接适用。

2. 在不同的地方多案件同时并发。

解决方法：如果有若干辆正在巡逻的警车去了案发现场，必然导致巡逻效果变差，因此比较理想的情况是添加警车，以方便更加灵活的调度。

6 假设检验

我们在开始假设了警车始发点只能在图上点的位置，简化了建模的复杂度，同时也降低了精确度。我们近似的找到了汽车停在点上的最优覆盖，但我们仍无法确定是否警车停在边上时能获得更优的解。再比如我们使用点的覆盖率来近似路径的覆盖率。但是根据[2]给出的伪顶点最优性质：对任何网络，存在顶点的一个有限扩充集合至少包含集合覆盖或最大覆盖模型的一个最优解。因此我们可以把原来网络图的边看成是具有有限个（尽管潜在地会很大）顶点的集合。那么最优覆盖集的结果和可以在边上任意地方停靠的结果是一样。因此，尽管我们假设了我们的警车只能停靠在图中顶点上，对于题目中的要求，我们提出的模型仍然适用。

7 模型评价

最小警车配置的最优化模型

优点：

1. 用精确类的分支限界算法的基础上，加入一系列针对实际的问题的剪枝技巧，能在较快的时间内定位解空间并能求得精确的解。
2. 模型的运算用矩阵、向量的运算组成，易于用数学软件求解和验证。

缺点：

1. 从一开始就假设覆盖率以图中的点的为单位计量，而不是以路径长度来计量，虽然简化了模型的建立和求解，但同时也降低了精确性。
2. 解答第一问时，我们假设车辆是静止在某个点上的，这是个比较强的假设。它是在牺牲了巡逻效果的基础上来获得最大的覆盖率。

警车效果显著性指标模型

优点：

1. 用严格的理论来推导和验证指标，使得指标能更加客观地反应其实际价值，减少主观因素。
2. 作为离散事件模拟的决策模型的基础，本模型的提供了决策因素设计的依据。

缺点：

1. 不能表达巡逻规律的隐蔽性。

离散事件模拟的决策模型

优点：

1. 有一套完整而有效的解决方案：决策算法决定了指标，通过分析指标可以得到最优的警车配置方案。
2. 对因子进行归一化处理，使得指标可以量化运算。

缺点：

1. 在动态分析车辆的下一状态的覆盖率来决定导向策略时，仅预测了本次事件触发者的状态，而没有预测其它车辆以后的情况，导致比较时间不一致性，使得计算存在误差。另外，无法验证所有时刻下都是完全符合条件的。

8 参考文献

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- [2] Church, R. L. and Meadows, M. "Location modeling utilizing maximum service distance criteria," Geographical Analysis, 11, 358-373, 1979.
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由于时间原因，论文还有很多需要改进的地方，恳请老师批评指正。

9 附录

1. 第三问中白天连续 4 个小时内的各警车的正常巡逻位置数据

19, 0.22710746482490168, 0.41234270885425817, 0.2925513941638396, 1.153155062141683, 0.8618481859405299, 1.085439849602514, 11.493822818306997

0,(6390.0,1404.0),(432.0,2106.0),(11952.0,2142.0),(4104.0,3348.0),(10152.0,3618.0),(7722.0,3636.0),(9180.0,4086.0),(5292.0,4284.0),(2124.0,4662.0),(8118.0,5220.0),(13266.0,5292.0),(1224.0,5814.0),(5580.0,6246.0),(11376.0,6318.0),(14040.0,6840.0),(2268.0,6858.0),(9594.0,7650.0),(8118.0,7668.0),(3528.0,7902.0)

1,(6061.0,1461.0),(502.0,2431.0),(12285.0,2125.0),(4134.0,3680.0),(9818.0,3618.0),(7482.0,3733.0),(9512.0,4060.0),(5583.0,4356.0),(2167.0,4992.0),(8096.0,4887.0),(13465.0,5559.0),(1557.0,5826.0),(5913.0,6230.0),(11042.0,6318.0),(13855.0,6562.0),(2169.0,7080.0),(9260.0,7650.0),(7786.0,7632.0),(3520.0,7568.0)

2,(6219.0,1433.0),(572.0,2757.0),(12618.0,2109.0),(4409.0,3753.0),(9485.0,3618.0),(7423.0,4060.0),(9844.0,4034.0),(5360.0,4547.0),(2353.0,5180.0),(8075.0,4554.0),(13664.0,5826.0),(1725.0,5666.0),(6246.0,6215.0),(10709.0,6318.0),(13671.0,6284.0),(2467.0,7108.0),(9155.0,7866.0),(7454.0,7596.0),(3512.0,7235.0)

3,(6542.0,1357.0),(690.0,3017.0),(12951.0,2092.0),(4545.0,3758.0),(9152.0,3618.0),(7381.0,3882.0),(9586.0,4054.0),(5574.0,4522.0),(2187.0,4977.0),(8053.0,4221.0),(13863.0,6094.0),(1724.0,5670.0),(6575.0,6250.0),(10376.0,6318.0),(13487.0,6006.0),(2135.0,7076.0),(9369.0,7992.0),(7402.0,7923.0),(3718.0,7110.0)

4,(6871.0,1349.0),(1022.0,2982.0),(12815.0,2099.0),(4212.0,3746.0),(8818.0,3618.0),(7049.0,3854.0),(9254.0,4080.0),(5607.0,4191.0),(2271.0,5172.0),(7830.0,4104.0),(14062.0,6361.0),(1717.0,5703.0),(6906.0,6277.0),(10441.0,6318.0),(13303.0,5728.0),(2247.0,6885.0),(9702.0,7992.0),(7097.0,7956.0),(4051.0,7110.0)

5,(6878.0,1024.0),(1353.0,2947.0),(12483.0,2115.0),(4140.0,4005.0),(8519.0,3474.0),(6881.0,4050.0),(9195.0,4344.0),(5588.0,4054.0),(2184.0,5166.0),(7497.0,4104.0),(14261.0,6629.0),(1594.0,5828.0),(7239.0,6267.0),(10774.0,6318.0),(13119.0,5450.0),(2008.0,7076.0),(10036.0,7995.0),(6764.0,7956.0),(4384.0,7110.0)

6,(6757.0,714.0),(1685.0,2912.0),(12150.0,2132.0),(4193.0,4274.0),(8222.0,3322.0),(6654.0,3922.0),(9163.0,4649.0),(5546.0,3780.0),(1851.0,5175.0),(7166.0,4072.0),(14458.0,6898.0),(1261.0,5815.0),(7572.0,6268.0),(11108.0,6318.0),(13251.0,5295.0),(1675.0,7085.0),(10369.0,7998.0),(6750.0,7637.0),(4718.0,7110.0)

7,(6673.0,391.0),(1856.0,3093.0),(11817.0,2145.0),(4509.0,4321.0),(7925.0,3171.0),(6405.0,3879.0),(9193.0,4318.0),(5213.0,3780.0),(1549.0,5154.0),(7010.0,3902.0),(14438.0,6868.0),(1156.0,5526.0),(7905.0,6276.0),(11390.0,6381.0),(13143.0,5322.0),(1406.0,6988.0),(10702.0,8001.0),(6750.0,7303.0),(5051.0,7110.0)

8,(6595.0,276.0),(1920.0,3421.0),(11483.0,2155.0),(4833.0,4324.0),(8067.0,3243.0),(6515.0,4070.0),(9185.0,4186.0),(5166.0,3493.0),(1564.0,4821.0),(7109.0,3584.0),(14119.0,6840.0),(1079.0,5201.0),(8111.0,6402.0),(11466.0,6706.0),(13251.0,5085.0),(1442.0,7061.0),(11036.0,8004.0),(6574.0,7134.0),(5384.0,7110.0)

9,(6610.0,274.0),(1940.0,3523.0),(11457.0,2155.0),(4993.0,4463.0),(8364.0,3394.0),(6647.0,3882.0),(9190.0,4264.0),(5141.0,3168.0),(1858.0,4721.0),(7257.0,3286.0),(14293.0,6840.0),(1014.0,4874.0),(8229.0,6618.0),(11542.0,7030.0),(13228.0,4753.0),(1757.0,7083.0),(11369.0,8007.0),(6512.0,7225.0),(5611.0,7008.0)

10,(6752.0,270.0),(1876.0,3196.0),(11790.0,2146.0),(4977.0,4794.0),(8661.0,3546.0),(6943.0,3852.0),(9025.0,4088.0),(4934.0,3326.0),(2118.0,4600.0),(7438.0,3007.0),(14537.0,7011.0),(945.0,4548.0),(8465.0,6540.0),(11594.0,7359.0),(13205.0,4420.0),(2090.0,7074.0),(11659.0,7971.0),(6515.0,7233.0),(5597.0,6675.0)

11,(7085.0,270.0),(1813.0,2869.0),(11943.0,1970.0),(5191.0,5002.0),(8977.0,3618.0),(7101.0,3608.0),(8692.0,4093.0),(5157.0,3253.0),(2088.0,4268.0),(7442.0,2678.0),(14476.0,6924.0),(1015.0,4315.0),(8385.0,6295.0),(11630.0,7690.0),(13182.0,4088.0),(1888.0,7305.0),(11624.0,7639.0),(6701.0,7142.0),(5583.0,6341.0)

12,(7419.0,270.0),(1759.0,2540.0),(11926.0,1637.0),(5283.0,5129.0),(9310.0,3618.0),(6810.0,3560.0),(8359.0,4099.0),(5153.0,2931.0),(2023.0,3942.0),(7464.0,2372.0),(14517.0,7234.0),(1273.0,4199.0),(8114.0,6347.0),(11662.0,7997.0),(13159.0,3755.0),(1660.0,7548.0),(11589.0,7308.0),(7034.0,7137.0),(5344.0,6221.0)

13,(7421.0,588.0),(1705.0,2211.0),(11910.0,1304.0),(5492.0,4890.0),(9644.0,3618.0),(6507.0,3694.0),(8047.0,4124.0),(5016.0,2683.0),(1941.0,3639.0),(7737.0,2284.0),(14476.0,7565.0),(1250.0,3867.0),(8095.0,6679.0),(11627.0,7666.0),(13136.0,3423.0),(1590.0,7843.0),(11386.0,7166.0),(7367.0,7128.0),(5060.0,6258.0)

14,(7214.0,731.0),(1660.0,1881.0),(11893.0,971.0),(5566.0,4869.0),(9977.0,3618.0),(6175.0,3725.0),(8068.0,4456.0),(4800.0,2529.0),(1611.0,3688.0),(7724.0,1950.0),(14436.0,7896.0),(1032.0,3752.0),(8072.0,7012.0),(11592.0,7334.0),(13113.0,3090.0),(1572.0,7701.0),(11052.0,7171.0),(7428.0,7445.0),(5238.0,6060.0)

15,(6882.0,749.0),(1671.0,1554.0),(11880.0,638.0),(5559.0,4609.0),(9993.0,3618.0),(5869.0,3841.0),(8209.0,4662.0),(4469.0,2497.0),(1281.0,3738.0),(7451.0,1909.0),(14440.0,7864.0),(827.0,3871.0),(8084.0,7344.0),(11734.0,7147.0),(13090.0,2757.0),(1512.0,7374.0),(10719.0,7176.0),(7425.0,7774.0),(5159.0,5748.0)

16,(6714.0,968.0),(1777.0,1238.0),(11880.0,305.0),(5276.0,4786.0),(9659.0,3618.0),(6019.0,4050.0),(8543.0,4662.0),(4136.0,2485.0),(949.0,3756.0),(7439.0,2242.0),(14480.0,7533.0),(878.0,4200.0),(8118.0,7676.0),(12066.0,7112.0),(13067.0,2425.0),(1437.0,7050.0),(10386.0,7180.0),(7548.0,7959.0),(5069.0,5440.0)

17,(6634.0,1291.0),(1936.0,971.0),(11880.0,424.0),(5062.0,5041.0),(9326.0,3618.0),(5837.0,4269.0),(8876.0,4662.0),(4108.0,2184.0),(764.0,3573.0),(7234.0,2357.0),(14521.0,7202.0),(1090.0,4301.0),(8118.0,7938.0),(12397.0,7078.0),(13050.0,2092.0),(1358.0,6729.0),(10314.0,7424.0),(7881.0,7968.0),(5254.0,5200.0)

18,(6435.0,1390.0),(2114.0,892.0),(11880.0,646.0),(4916.0,5322.0),(8993.0,3618.0),(5628.0,4528.0),(9168.0,4614.0),(4119.0,2282.0),(685.0,3249.0),(6902.0,2327.0),(14458.0,6897.0),(1420.0,4266.0),(8055.0,7661.0),(12729.0,7043.0),(13067.0,2416.0),(1328.0,6558.0),(10224.0,7650.0),(8118.0,7877.0),(4921.0,5184.0)

19,(6656.0,1200.0),(1898.0,988.0),(11880.0,313.0),(4784.0,5184.0),(8675.0,3553.0),(5335.0,4572.0),(9306.0,4399.0),(4104.0,2615.0),(653.0,3122.0),(6854.0,2591.0),(14155.0,6840.0),(1754.0,4266.0),(7724.0,7625.0),(13060.0,7009.0),(13051.0,2110.0),(1386.0,6886.0),(9891.0,7650.0),(8105.0,7544.0),(4693.0,5057.0)

20,(6737.0,876.0),(2153.0,874.0),(11661.0,201.0),(4452.0,5211.0),(8378.0,3402.0),(5002.0,4572.0),(9639.0,4421.0),(4104.0,2883.0),(732.0,3446.0),(6621.0,2739.0),(13826.0,6881.0),(2087.0,4266.0),(7442.0,7537.0),(13391.0,6966.0),(13066.0,2398.0),(1407.0,6990.0),(9597.0,7685.0),(8070.0,7213.0),(4812.0,4886.0)

21,(6847.0,948.0),(1852.0,1013.0),(11328.0,207.0),(4123.0,5220.0),(8081.0,3250.0),(5163.0,4749.0),(9971.0,4443.0),(4128.0,3063.0),(811.0,3770.0),(6525.0,2934.0),(13498.0,6945.0),(2118.0,4597.0),(7392.0,7207.0),(13718.0,6902.0),(12766.0,2439.0),(1727.0,6936.0),(9627.0,7966.0),(7798.0,7139.0),(5128.0,4967.0)

22,(7103.0,1083.0),(1746.0,1329.0),(10995.0,213.0),(4122.0,5053.0),(7806.0,3190.0),(5138.0,4957.0),(9994.0,4742.0),(4149.0,3348.0),(862.0,4099.0),(6737.0,2867.0),(13432.0,6958.0),(2157.0,4928.0),(7380.0,6875.0),(14045.0,6840.0),(12433.0,2450.0),(2056.0,6888.0),(9579.0,7650.0),(7464.0,7130.0),(5275.0,5202.0)

23,(7418.0,1135.0),(1641.0,1645.0),(10917.0,420.0),(4062.0,5205.0),(7615.0,3462.0),(4972.0,4738.0),(9980.0,5075.0),(4430.0,3245.0),(919.0,4427.0),(6492.0,2700.0),(13759.0,6894.0),(2288.0,5174.0),(7380.0,6542.0),(14378.0,6840.0),(12100.0,2461.0),(2347.0,6767.0),(9245.0,7650.0),(7131.0,7135.0),(5608.0,5219.0)

24,(7438.0,1270.0),(1670.0,1977.0),(10989.0,746.0),(4145.0,5111.0),(7649.0,3629.0),(4977.0,4624.0),(9866.0,5286.0),(4517.0,2935.0),(991.0,4752.0),(6261.0,2473.0),(14087.0,6840.0),(2620.0,5196.0),(7382.0,6208.0),(14501.0,6958.0),(11767.0,2459.0),(2566.0,6516.0),(9152.0,7418.0),(6798.0,7144.0),(5632.0,4894.0)

25,(7416.0,938.0),(1644.0,1723.0),(11062.0,1071.0),(4055.0,5250.0),(7753.0,3604.0),(5013.0,4301.0),(9935.0,5266.0),(4606.0,2711.0),(1324.0,4772.0),(6160.0,2156.0),(14419.0,6842.0),(2953.0,5215.0),(7400.0,5875.0),(14525.0,6994.0),(11434.0,2449.0),(2787.0,6266.0),(9163.0,7085.0),(6484.0,7141.0),(5823.0,4629.0)

26,(7416.0,834.0),(1664.0,1920.0),(11134.0,1397.0),(4092.0,5582.0),(7960.0,3565.0),(5330.0,4245.0),(9682.0,5283.0),(4590.0,2378.0),(1561.0,4878.0),(6328.0,2328.0),(14477.0,6924.0),(3042.0,5464.0),(7418.0,5543.0),(14335.0,6840.0),(11309.0,2181.0),(3022.0,6031.0),(9170.0,6752.0),(6357.0,7283.0),(5995.0,4346.0)

27,(7423.0,1167.0),(1437.0,2033.0),(11206.0,1722.0),(4115.0,5913.0),(8266.0,3672.0),(5588.0,4054.0),(9349.0,5265.0),(4385.0,2250.0),(1520.0,5185.0),(6201.0,2142.0),(14187.0,6840.0),(3044.0,5220.0),(7430.0,5210.0),(14003.0,6847.0),(11372.0,2433.0),(3324.0,5977.0),(9177.0,6419.0),(6024.0,7270.0),(5994.0,4050.0)

28,(7452.0,1498.0),(1104.0,2057.0),(11279.0,2047.0),(4000.0,6073.0),(8290.0,3778.0),(5921.0,4050.0),(9162.0,5110.0),(4108.0,2181.0),(1186.0,5197.0),(6125.0,2157.0),(14225.0,6840.0),(3377.0,5233.0),(7401.0,4878.0),(13676.0,6911.0),(11316.0,2209.0),(3657.0,6006.0),(9412.0,6311.0),(5691.0,7257.0),(6181.0,4065.0)

29,(7452.0,1831.0),(772.0,2081.0),(11102.0,2244.0),(4050.0,5917.0),(8308.0,3853.0),(5972.0,3797.0),(9162.0,4776.0),(4080.0,1850.0),(1035.0,4980.0),(6017.0,1878.0),(14498.0,6955.0),(3709.0,5223.0),(7398.0,4778.0),(13348.0,6974.0),(11463.0,2450.0),(3799.0,6036.0),(9745.0,6301.0),(5604.0,7204.0),(6399.0,3962.0)

30,(7708.0,1890.0),(439.0,2105.0),(10795.0,2373.0),(4177.0,5654.0),(8017.0,3888.0),(6292.0,3714.0),(9190.0,4445.0),(4071.0,1516.0),(1009.0,4850.0),(5769.0,1654.0),(14383.0,6840.0),(4042.0,5202.0),(7398.0,4752.0),(13018.0,7013.0),(11796.0,2460.0),(3920.0,5845.0),(9839.0,6033.0),(5598.0,7246.0),(6210.0,3955.0)

31,(7452.0,1858.0),(756.0,2082.0),(10487.0,2502.0),(4480.0,5516.0),(7684.0,3888.0),(6617.0,3657.0),(9162.0,4669.0),(4073.0,1588.0),(1075.0,5177.0),(5628.0,1375.0),(14050.0,6840.0),(4057.0,5173.0),(7419.0,5084.0),(12686.0,7048.0),(12129.0,2460.0),(4034.0,5747.0),(9815.0,6250.0),(5923.0,7266.0),(5900.0,4050.0)

32,(7452.0,1525.0),(1088.0,2058.0),(10179.0,2629.0),(4790.0,5394.0),(7471.0,3788.0),(6899.0,3486.0),(9162.0,5002.0),(4082.0,1921.0),(1388.0,5190.0),(5851.0,1278.0),(13861.0,6570.0),(4111.0,4847.0),(7595.0,5247.0),(12354.0,7082.0),(12463.0,2449.0),(3781.0,5964.0),(10093.0,6311.0),(6257.0,7279.0),(5568.0,4057.0)

33,(7276.0,1362.0),(1421.0,2034.0),(9850.0,2681.0),(4957.0,5155.0),(7492.0,3459.0),(7111.0,3240.0),(9241.0,5260.0),(4120.0,2250.0),(1555.0,5010.0),(6184.0,1278.0),(13677.0,6292.0),(4128.0,4514.0),(7928.0,5229.0),(12023.0,7117.0),(12796.0,2438.0),(3977.0,6071.0),(10242.0,6133.0),(6516.0,7236.0),(5594.0,3888.0)

34,(6942.0,1352.0),(1686.0,2095.0),(9521.0,2733.0),(5198.0,5006.0),(7463.0,3127.0),(7125.0,2907.0),(9574.0,5277.0),(3787.0,2250.0),(1673.0,4763.0),(6448.0,1175.0),(13493.0,6014.0),(4241.0,4265.0),(8261.0,5224.0),(11691.0,7151.0),(13064.0,2368.0),(4303.0,6127.0),(10121.0,5938.0),(6261.0,7123.0),(5580.0,3556.0)

35,(7142.0,1358.0),(1740.0,2424.0),(9192.0,2785.0),(5465.0,4913.0),(7468.0,3180.0),(7374.0,2953.0),(9846.0,5353.0),(3494.0,2250.0),(1998.0,4690.0),(6572.0,866.0),(13309.0,5736.0),(4554.0,4340.0),(8594.0,5236.0),(11359.0,7167.0),(13047.0,2035.0),(4412.0,6160.0),(9839.0,6030.0),(5928.0,7116.0),(5534.0,3235.0)

36,(7475.0,1366.0),(1794.0,2753.0),(9168.0,2742.0),(5592.0,4714.0),(7496.0,3512.0),(7268.0,2822.0),(9846.0,5686.0),(3827.0,2250.0),(1924.0,4707.0),(6578.0,548.0),(13124.0,5459.0),(4630.0,4647.0),(8927.0,5247.0),(11025.0,7171.0),(13030.0,1702.0),(4141.0,6018.0),(9872.0,6302.0),(5613.0,7131.0),(5366.0,2951.0)

37,(7808.0,1344.0),(2004.0,2900.0),(9194.0,2409.0),(5593.0,4387.0),(7711.0,3635.0),(6964.0,2690.0),(9889.0,5970.0),(4114.0,2212.0),(1598.0,4780.0),(6450.0,288.0),(13241.0,5298.0),(4679.0,4977.0),(9162.0,5157.0),(10692.0,7176.0),(13013.0,1369.0),(4261.0,5785.0),(10205.0,6316.0),(5603.0,7463.0),(5394.0,2945.0)

38,(8140.0,1322.0),(2337.0,2905.0),(9183.0,2602.0),(5309.0,4290.0),(7480.0,3743.0),(6709.0,2789.0),(10220.0,5924.0),(4121.0,2245.0),(1266.0,4769.0),(6116.0,288.0),(13450.0,5539.0),(4839.0,5184.0),(9162.0,4824.0),(10359.0,7181.0),(12996.0,1037.0),(4550.0,5635.0),(10242.0,6021.0),(5612.0,7796.0),(5063.0,2916.0)

39,(8274.0,1525.0),(2671.0,2909.0),(9305.0,2767.0),(5002.0,4328.0),(7488.0,3706.0),(6921.0,2905.0),(10242.0,6233.0),(4116.0,2221.0),(1000.0,4806.0),(5832.0,336.0),(13649.0,5806.0),(4922.0,5268.0),(9184.0,4492.0),(10026.0,7173.0),(12979.0,704.0),(4254.0,5789.0),(10009.0,5953.0),(5825.0,7926.0),(5036.0,2809.0)

40,(8294.0,1858.0),(3004.0,2914.0),(9634.0,2715.0),(4971.0,4659.0),(7740.0,3617.0),(7012.0,2819.0),(10257.0,6566.0),(4104.0,2553.0),(1066.0,5133.0),(5832.0,669.0),(13848.0,6074.0),(5179.0,5197.0),(9184.0,4160.0),(9692.0,7162.0),(12967.0,371.0),(4104.0,5704.0),(9827.0,6142.0),(6158.0,7937.0),(5011.0,2632.0)

41,(8014.0,1899.0),(3127.0,2710.0),(9963.0,2663.0),(5262.0,4784.0),(7760.0,3597.0),(6685.0,2751.0),(10278.0,6898.0),(4104.0,2887.0),(1140.0,5458.0),(5807.0,873.0),(14047.0,6341.0),(5280.0,5150.0),(9176.0,3827.0),(9359.0,7152.0),(12964.0,286.0),(4003.0,5960.0),(9634.0,6305.0),(6492.0,7947.0),(4792.0,2736.0)

42,(7723.0,1930.0),(3238.0,2502.0),(10285.0,2586.0),(5546.0,4617.0),(8055.0,3593.0),(6516.0,2878.0),(10299.0,7231.0),(3801.0,2951.0),(1216.0,5783.0),(5796.0,930.0),(14246.0,6609.0),(5090.0,5415.0),(9171.0,3493.0),(9157.0,7281.0),(12975.0,619.0),(3724.0,6005.0),(9301.0,6314.0),(6825.0,7956.0),(4621.0,2884.0)

43,(7736.0,2263.0),(3464.0,2386.0),(10593.0,2458.0),(5394.0,4488.0),(8288.0,3769.0),(6759.0,2928.0),(10325.0,7563.0),(3470.0,2986.0),(1264.0,6112.0),(5999.0,1193.0),(14444.0,6877.0),(5074.0,5536.0),(9167.0,3160.0),(9145.0,7614.0),(12709.0,688.0),(3620.0,5688.0),(8968.0,6312.0),(7158.0,7956.0),(4629.0,3001.0)

44,(7485.0,2369.0),(3670.0,2250.0),(10900.0,2329.0),(5601.0,4302.0),(8104.0,3888.0),(6824.0,2682.0),(10313.0,7403.0),(3444.0,3302.0),(1309.0,6443.0),(6291.0,1278.0),(14452.0,6889.0),(5126.0,5691.0),(9162.0,2827.0),(9159.0,7947.0),(12376.0,693.0),(3517.0,5372.0),(8635.0,6304.0),(7304.0,7956.0),(4648.0,3330.0)

45,(7713.0,2343.0),(4003.0,2250.0),(11208.0,2200.0),(5536.0,4063.0),(7770.0,3888.0),(6603.0,2514.0),(10289.0,7071.0),(3180.0,3315.0),(1366.0,6771.0),(6191.0,1278.0),(14145.0,6840.0),(5238.0,5994.0),(9166.0,3086.0),(9450.0,7992.0),(12042.0,699.0),(3281.0,5229.0),(8654.0,6304.0),(6970.0,7956.0),(4937.0,3299.0)

46,(7435.0,2375.0),(4337.0,2250.0),(11359.0,2382.0),(5292.0,4283.0),(7449.0,3902.0),(6330.0,2335.0),(10268.0,6738.0),(3141.0,2984.0),(1456.0,7089.0),(5936.0,1114.0),(13816.0,6883.0),(5123.0,6241.0),(9170.0,3419.0),(9613.0,7838.0),(11880.0,531.0),(3042.0,5313.0),(8987.0,6313.0),(6750.0,7843.0),(5149.0,3147.0)

47,(7446.0,2043.0),(4590.0,2330.0),(11311.0,2190.0),(5585.0,4356.0),(7411.0,4232.0),(6148.0,2145.0),(10247.0,6406.0),(3251.0,2785.0),(1474.0,7178.0),(5814.0,831.0),(13489.0,6947.0),(4795.0,6230.0),(9175.0,3752.0),(9450.0,7650.0),(11880.0,198.0),(3042.0,5369.0),(9177.0,6177.0),(6750.0,7510.0),(5048.0,2916.0)

48,(7648.0,1894.0),(4602.0,2663.0),(11606.0,2151.0),(5361.0,4545.0),(7401.0,4565.0),(5837.0,2259.0),(9997.0,6307.0),(3456.0,2680.0),(1538.0,7506.0),(5832.0,500.0),(13442.0,6956.0),(4525.0,6244.0),(9180.0,4086.0),(9145.0,7676.0),(12212.0,186.0),(3225.0,5227.0),(9171.0,5844.0),(6750.0,7176.0),(4714.0,2930.0)

49,(7732.0,2149.0),(4564.0,2934.0),(11939.0,2142.0),(5043.0,4572.0),(7405.0,4424.0),(5525.0,2328.0),(9826.0,6154.0),(3456.0,2656.0),(1589.0,7835.0),(5832.0,409.0),(13769.0,6892.0),(4570.0,6574.0),(9194.0,4418.0),(9144.0,7991.0),(12546.0,175.0),(3389.0,5234.0),(9166.0,5510.0),(6490.0,7158.0),(4381.0,2928.0)

50,(7598.0,2356.0),(4429.0,3146.0),(11631.0,2150.0),(4981.0,4846.0),(7403.0,4102.0),(5511.0,1995.0),(9846.0,5822.0),(3275.0,2787.0),(1831.0,7928.0),(5831.0,742.0),(14097.0,6840.0),(4615.0,6904.0),(9074.0,4662.0),(8811.0,7985.0),(12879.0,164.0),(3056.0,5220.0),(9240.0,5260.0),(6503.0,7129.0),(4113.0,2970.0)

51,(7266.0,2360.0),(4132.0,3089.0),(11305.0,2165.0),(5206.0,4872.0),(7073.0,4061.0),(5261.0,1908.0),(9846.0,5489.0),(3176.0,2925.0),(2164.0,7915.0),(5820.0,801.0),(14425.0,6850.0),(4770.0,7110.0),(8740.0,4662.0),(8477.0,7980.0),(12968.0,414.0),(2723.0,5202.0),(9573.0,5277.0),(6748.0,7059.0),(4112.0,3299.0)

52,(6934.0,2330.0),(4104.0,2857.0),(11418.0,2449.0),(5133.0,4963.0),(6977.0,3852.0),(5126.0,1858.0),(9976.0,5255.0),(3460.0,3034.0),(2497.0,7902.0),(5837.0,988.0),(14528.0,7145.0),(5023.0,7028.0),(8407.0,4662.0),(8144.0,7974.0),(12975.0,620.0),(2391.0,5181.0),(9903.0,5275.0),(6741.0,6726.0),(3847.0,3299.0)

53,(7077.0,2343.0),(3831.0,2948.0),(11751.0,2459.0),(5401.0,4966.0),(6644.0,3852.0),(5045.0,1709.0),(10310.0,5243.0),(3405.0,3334.0),(2494.0,7572.0),(6044.0,1250.0),(14487.0,7476.0),(5030.0,6695.0),(8089.0,4662.0),(8424.0,7979.0),(12964.0,287.0),(2159.0,4944.0),(10234.0,5246.0),(6734.0,6392.0),(3800.0,2969.0)

54,(7409.0,2373.0),(3840.0,3248.0),(11855.0,2462.0),(5630.0,4927.0),(6501.0,3824.0),(5001.0,1379.0),(10643.0,5247.0),(3174.0,3254.0),(2486.0,7239.0),(5767.0,1278.0),(14447.0,7807.0),(5037.0,6361.0),(8423.0,4662.0),(8758.0,7985.0),(12752.0,168.0),(2130.0,4710.0),(10567.0,5243.0),(6513.0,6238.0),(3483.0,2985.0)

55,(7443.0,2684.0),(3600.0,3330.0),(11522.0,2452.0),(5575.0,5217.0),(6821.0,3731.0),(4775.0,1288.0),(10975.0,5265.0),(3432.0,3186.0),(2687.0,7110.0),(5640.0,1458.0),(14325.0,8044.0),(5267.0,6213.0),(8756.0,4662.0),(9091.0,7990.0),(12419.0,180.0),(2175.0,5040.0),(10900.0,5261.0),(6372.0,6021.0),(3448.0,3289.0)

56,(7451.0,2958.0),(3268.0,3354.0),(11330.0,2266.0),(5277.0,5171.0),(7102.0,3580.0),(4444.0,1334.0),(11178.0,5436.0),(3296.0,3350.0),(3021.0,7110.0),(5619.0,1307.0),(13992.0,8040.0),(5580.0,6226.0),(9089.0,4662.0),(9148.0,7729.0),(12085.0,191.0),(2388.0,5178.0),(11233.0,5277.0),(6372.0,5688.0),(3179.0,3402.0)

57,(7441.0,2625.0),(3158.0,3117.0),(11358.0,2376.0),(5099.0,4952.0),(6888.0,3497.0),(4114.0,1379.0),(11251.0,5762.0),(3208.0,3186.0),(3354.0,7110.0),(5675.0,1568.0),(13658.0,8036.0),(5592.0,5893.0),(9196.0,4403.0),(9153.0,7396.0),(11752.0,200.0),(2161.0,4959.0),(11566.0,5290.0),(6372.0,5354.0),(3119.0,3730.0)

58,(7437.0,2291.0),(3132.0,2786.0),(11312.0,2195.0),(4779.0,4884.0),(7110.0,3255.0),(4020.0,1103.0),(11324.0,6087.0),(3457.0,3252.0),(3544.0,6935.0),(5922.0,1792.0),(13325.0,8031.0),(5604.0,5560.0),(9519.0,4413.0),(9079.0,7146.0),(11419.0,206.0),(2319.0,5177.0),(11899.0,5302.0),(6391.0,5022.0),(3167.0,3465.0)

59,(7450.0,1958.0),(3111.0,2453.0),(11600.0,2151.0),(4699.0,5089.0),(7124.0,2922.0),(3966.0,774.0),(11398.0,6412.0),(3187.0,3365.0),(3608.0,6608.0),(5890.0,1763.0),(12992.0,8027.0),(5615.0,5227.0),(9851.0,4435.0),(8746.0,7146.0),(11086.0,212.0),(2210.0,5000.0),(12232.0,5315.0),(6392.0,4690.0),(3415.0,3333.0)

60,(7452.0,1625.0),(3098.0,2120.0),(11933.0,2142.0),(4946.0,5167.0),(7360.0,2947.0),(3664.0,758.0),(11474.0,6737.0),(3148.0,3036.0),(3672.0,6281.0),(5650.0,1536.0),(12658.0,8023.0),(5604.0,5546.0),(9831.0,4434.0),(8412.0,7146.0),(10991.0,213.0),(2153.0,5167.0),(12565.0,5327.0),(6694.0,4662.0),(3748.0,3330.0)

61,(7441.0,1293.0),(3294.0,2154.0),(11936.0,1827.0),(5074.0,4940.0),(7678.0,2931.0),(3332.0,782.0),(11550.0,7061.0),(3338.0,2961.0),(3781.0,6030.0),(5688.0,1278.0),(12325.0,8018.0),(5592.0,5879.0),(9498.0,4412.0),(8079.0,7146.0),(11324.0,207.0),(1820.0,5176.0),(12898.0,5340.0),(7027.0,4662.0),(4081.0,3346.0)

62,(7262.0,1100.0),(3416.0,2250.0),(11919.0,1494.0),(4970.0,4662.0),(7761.0,2709.0),(2999.0,805.0),(11521.0,6941.0),(3470.0,3200.0),(4108.0,6080.0),(6022.0,1278.0),(11992.0,8014.0),(5581.0,6212.0),(9196.0,4360.0),(8085.0,6828.0),(11657.0,201.0),(1487.0,5186.0),(13226.0,5301.0),(7361.0,4662.0),(4414.0,3344.0)

63,(6930.0,1065.0),(3262.0,2070.0),(11902.0,1161.0),(5092.0,4723.0),(7989.0,2628.0),(2667.0,829.0),(11446.0,6617.0),(3240.0,3358.0),(4058.0,5909.0),(6355.0,1278.0),(11663.0,8004.0),(5282.0,6214.0),(9238.0,4081.0),(8209.0,6616.0),(11880.0,308.0),(1154.0,5199.0),(13117.0,5329.0),(7694.0,4662.0),(4540.0,3336.0)

64,(6883.0,1350.0),(3262.0,2070.0),(11886.0,828.0),(5393.0,4713.0),(8118.0,2862.0),(2334.0,852.0),(11370.0,6292.0),(3265.0,3186.0),(3791.0,6033.0),(6512.0,1017.0),(11628.0,7673.0),(5238.0,5920.0),(9570.0,4055.0),(8464.0,6560.0),(11880.0,642.0),(1338.0,5192.0),(13373.0,5436.0),(8027.0,4662.0),(4443.0,3227.0)

65,(7217.0,1360.0),(3595.0,2070.0),(11890.0,908.0),(5581.0,4474.0),(8361.0,2967.0),(2016.0,935.0),(11297.0,5967.0),(3462.0,3061.0),(3674.0,6271.0),(6611.0,768.0),(11592.0,7341.0),(5098.0,5641.0),(9889.0,4051.0),(8405.0,6296.0),(11893.0,975.0),(1423.0,5188.0),(13572.0,5704.0),(8099.0,4940.0),(4538.0,2934.0)

66,(7438.0,1270.0),(3929.0,2070.0),(11906.0,1240.0),(5390.0,4323.0),(8686.0,2895.0),(1804.0,1155.0),(11224.0,5642.0),(3203.0,2931.0),(3610.0,6598.0),(6488.0,1077.0),(11538.0,7013.0),(5238.0,5937.0),(9904.0,4094.0),(8115.0,6327.0),(11910.0,1308.0),(1090.0,5201.0),(13771.0,5971.0),(8118.0,5273.0),(4871.0,2923.0)

67,(7416.0,938.0),(4262.0,2078.0),(11923.0,1573.0),(5065.0,4298.0),(9011.0,2823.0),(1699.0,1471.0),(11151.0,5316.0),(3124.0,2655.0),(3546.0,6925.0),(6290.0,1278.0),(11462.0,6689.0),(5179.0,6225.0),(10009.0,4230.0),(8153.0,6611.0),(11926.0,1641.0),(1153.0,5515.0),(13970.0,6238.0),(8118.0,5606.0),(5129.0,2927.0)

68,(7420.0,605.0),(4595.0,2095.0),(11940.0,1906.0),(4734.0,4337.0),(9127.0,2667.0),(1651.0,1793.0),(10852.0,5258.0),(3293.0,2502.0),(3513.0,7255.0),(5998.0,1192.0),(11386.0,6364.0),(4853.0,6238.0),(10343.0,4230.0),(8434.0,6639.0),(11943.0,1973.0),(1217.0,5787.0),(14169.0,6506.0),(8118.0,5940.0),(5014.0,2659.0)

69,(7433.0,272.0),(4928.0,2112.0),(11947.0,2044.0),(4775.0,4331.0),(8795.0,2690.0),(1564.0,2023.0),(10520.0,5241.0),(3468.0,2331.0),(3520.0,7588.0),(5797.0,929.0),(11441.0,6595.0),(4522.0,6192.0),(10167.0,4230.0),(8102.0,6606.0),(11943.0,1977.0),(1141.0,5463.0),(14368.0,6773.0),(8118.0,6273.0),(5031.0,2783.0)

70,(7420.0,600.0),(5150.0,2237.0),(11930.0,1711.0),(4997.0,4405.0),(8462.0,2712.0),(1231.0,2048.0),(10448.0,4966.0),(3294.0,2178.0),(3508.0,7902.0),(5832.0,599.0),(11517.0,6920.0),(4608.0,5876.0),(9913.0,4117.0),(8124.0,6282.0),(11927.0,1644.0),(1067.0,5138.0),(14540.0,7050.0),(8442.0,6298.0),(5255.0,2939.0)

71,(7202.0,731.0),(5157.0,2570.0),(11913.0,1378.0),(5152.0,4572.0),(8130.0,2735.0),(899.0,2072.0),(10436.0,4633.0),(3338.0,2250.0),(3175.0,7902.0),(5853.0,288.0),(11490.0,7165.0),(4581.0,5703.0),(9943.0,4200.0),(8456.0,6298.0),(11910.0,1311.0),(1001.0,4811.0),(14499.0,7381.0),(8462.0,6597.0),(5584.0,2919.0)

72,(6869.0,750.0),(5165.0,2903.0),(11897.0,1045.0),(5483.0,4594.0),(8118.0,2829.0),(566.0,2096.0),(10424.0,4300.0),(3459.0,2447.0),(2841.0,7902.0),(6187.0,288.0),(11157.0,7169.0),(4573.0,5627.0),(9934.0,4177.0),(8461.0,6611.0),(11893.0,978.0),(932.0,4485.0),(14458.0,7712.0),(8475.0,6353.0),(5916.0,2947.0)

73,(6856.0,969.0),(5155.0,3235.0),(11880.0,712.0),(5631.0,4851.0),(7910.0,2915.0),(397.0,1910.0),(10159.0,4230.0),(3734.0,2494.0),(2508.0,7902.0),(6520.0,287.0),(10824.0,7174.0),(4887.0,5515.0),(10231.0,4230.0),(8158.0,6611.0),(11936.0,701.0),(1079.0,4303.0),(14418.0,8043.0),(8198.0,6286.0),(6249.0,2942.0)

74,(6879.0,1295.0),(4932.0,3344.0),(11880.0,379.0),(5704.0,4860.0),(7751.0,2672.0),(340.0,1582.0),(9910.0,4109.0),(4067.0,2485.0),(2494.0,7575.0),(6694.0,473.0),(10490.0,7179.0),(5087.0,5602.0),(10427.0,4372.0),(8115.0,6331.0),(12269.0,695.0),(1268.0,4135.0),(14458.0,7717.0),(7865.0,6275.0),(6516.0,2867.0)

75,(6849.0,1627.0),(5123.0,3170.0),(12031.0,192.0),(6037.0,4860.0),(8027.0,2628.0),(329.0,1517.0),(9940.0,4192.0),(4117.0,2312.0),(2486.0,7241.0),(6783.0,794.0),(10306.0,7320.0),(5081.0,5569.0),(10438.0,4705.0),(8102.0,6565.0),(12603.0,690.0),(1246.0,3802.0),(14498.0,7386.0),(7532.0,6267.0),(6370.0,2605.0)

76,(6840.0,1960.0),(5020.0,3341.0),(12365.0,181.0),(6371.0,4860.0),(8098.0,2676.0),(386.0,1846.0),(9937.0,4184.0),(4104.0,2645.0),(2283.0,7090.0),(6891.0,1106.0),(10329.0,7650.0),(5226.0,5866.0),(10450.0,5038.0),(8080.0,6897.0),(12936.0,684.0),(967.0,3755.0),(14539.0,7056.0),(7199.0,6269.0),(6214.0,2326.0)

77,(6840.0,2293.0),(4687.0,3332.0),(12698.0,170.0),(6390.0,4564.0),(8172.0,2956.0),(446.0,2174.0),(10239.0,4230.0),(4166.0,2918.0),(2199.0,6949.0),(6787.0,1343.0),(9996.0,7650.0),(5238.0,6196.0),(10324.0,5242.0),(7979.0,7143.0),(12967.0,392.0),(985.0,3754.0),(14462.0,6903.0),(6866.0,6278.0),(6277.0,2129.0)

78,(6856.0,2626.0),(4630.0,3041.0),(12962.0,233.0),(6390.0,4231.0),(8199.0,2958.0),(516.0,2499.0),(10271.0,4230.0),(4499.0,2933.0),(2136.0,7032.0),(6793.0,1344.0),(9662.0,7650.0),(5150.0,6233.0),(9991.0,5255.0),(7646.0,7135.0),(12963.0,264.0),(1317.0,3732.0),(14434.0,6863.0),(6536.0,6242.0),(6160.0,1817.0)

79,(7138.0,2752.0),(4605.0,2708.0),(12973.0,566.0),(6404.0,3898.0),(7870.0,2908.0),(587.0,2825.0),(9948.0,4215.0),(4509.0,3032.0),(2386.0,7100.0),(6852.0,1599.0),(9621.0,7913.0),(5076.0,6254.0),(9922.0,5270.0),(7390.0,7194.0),(12975.0,597.0),(1647.0,3683.0),(14113.0,6840.0),(6534.0,6242.0),(6005.0,1523.0)

80,(7407.0,2943.0),(4590.0,2375.0),(12989.0,899.0),(6153.0,3985.0),(7741.0,2632.0),(602.0,2896.0),(9929.0,4162.0),(4285.0,3139.0),(2249.0,7087.0),(6859.0,1524.0),(9376.0,7992.0),(5238.0,6044.0),(9846.0,5546.0),(7440.0,7523.0),(12990.0,930.0),(1964.0,3650.0),(14299.0,6840.0),(6734.0,6413.0),(6120.0,1712.0)

81,(7322.0,3165.0),(4590.0,2457.0),(13006.0,1232.0),(6007.0,4242.0),(8068.0,2628.0),(532.0,2570.0),(9831.0,4035.0),(4109.0,3317.0),(1976.0,7211.0),(6718.0,1338.0),(9155.0,7872.0),(5151.0,5733.0),(9846.0,5879.0),(7413.0,7853.0),(13007.0,1263.0),(2030.0,3977.0),(14541.0,7016.0),(6741.0,6746.0),(6238.0,2024.0)

82,(7371.0,2907.0),(4876.0,2539.0),(13023.0,1565.0),(5889.0,4548.0),(8118.0,2941.0),(461.0,2244.0),(9499.0,4061.0),(4385.0,3307.0),(1748.0,7455.0),(6396.0,1402.0),(9254.0,7650.0),(5080.0,5427.0),(10080.0,5943.0),(7627.0,7961.0),(13024.0,1596.0),(2091.0,4304.0),(14480.0,6929.0),(6748.0,7080.0),(6119.0,2068.0)

83,(7093.0,2728.0),(4998.0,2736.0),(13040.0,1898.0),(5945.0,4444.0),(7799.0,2901.0),(398.0,1917.0),(9166.0,4086.0),(4418.0,3262.0),(1574.0,7715.0),(6705.0,1337.0),(9587.0,7650.0),(5278.0,5166.0),(10242.0,6092.0),(7960.0,7970.0),(13041.0,1929.0),(2121.0,4636.0),(14414.0,6840.0),(6483.0,7128.0),(6089.0,2009.0)

84,(6780.0,2706.0),(4665.0,2736.0),(13057.0,2231.0),(5753.0,4715.0),(7476.0,2981.0),(341.0,1589.0),(8833.0,4091.0),(4583.0,3333.0),(1588.0,7829.0),(6886.0,1187.0),(9628.0,7974.0),(5094.0,4950.0),(10134.0,6313.0),(8118.0,7798.0),(13059.0,2262.0),(2162.0,4967.0),(14081.0,6840.0),(6483.0,7138.0),(6033.0,2203.0)

85,(6516.0,2770.0),(4548.0,2500.0),(13060.0,2296.0),(5493.0,4889.0),(7381.0,2917.0),(294.0,1259.0),(8500.0,4096.0),(4915.0,3346.0),(1537.0,7500.0),(7100.0,1083.0),(9313.0,7992.0),(4774.0,4884.0),(9801.0,6300.0),(8118.0,7871.0),(13059.0,2264.0),(2327.0,5177.0),(13753.0,6895.0),(6158.0,7121.0),(6211.0,2142.0)

86,(6685.0,2946.0),(4215.0,2488.0),(13056.0,2212.0),(5283.0,5128.0),(7314.0,3177.0),(263.0,927.0),(8167.0,4102.0),(5103.0,3172.0),(1473.0,7173.0),(7418.0,1131.0),(9152.0,7810.0),(4631.0,4659.0),(9467.0,6309.0),(8118.0,7743.0),(12893.0,2095.0),(2184.0,5098.0),(13426.0,6959.0),(5824.0,7114.0),(5898.0,2243.0)

87,(7008.0,2879.0),(4104.0,2705.0),(12952.0,2433.0),(5532.0,5215.0),(7163.0,3474.0),(232.0,595.0),(8059.0,4315.0),(5166.0,3455.0),(1453.0,7083.0),(7546.0,1361.0),(9150.0,7477.0),(4565.0,4345.0),(9179.0,6272.0),(8091.0,7411.0),(12560.0,2112.0),(2288.0,5078.0),(13096.0,7005.0),(5600.0,7233.0),(5576.0,2326.0)

88,(7318.0,2928.0),(4104.0,2576.0),(12619.0,2444.0),(5865.0,5220.0),(6921.0,3682.0),(229.0,564.0),(8081.0,4648.0),(5157.0,3779.0),(1623.0,6952.0),(7879.0,1339.0),(9163.0,7146.0),(4253.0,4263.0),(9173.0,5939.0),(7997.0,7144.0),(12227.0,2128.0),(2210.0,5167.0),(12765.0,7039.0),(5600.0,7231.0),(5798.0,2269.0)

89,(7633.0,2942.0),(3863.0,2490.0),(12285.0,2455.0),(6199.0,5220.0),(6609.0,3793.0),(260.0,896.0),(8401.0,4662.0),(4823.0,3768.0),(1953.0,6903.0),(8211.0,1317.0),(9496.0,7156.0),(4321.0,4250.0),(9167.0,5606.0),(7664.0,7135.0),(11894.0,2143.0),(1876.0,5174.0),(12433.0,7074.0),(5405.0,7110.0),(6112.0,2163.0)

90,(7773.0,2753.0),(3530.0,2499.0),(11952.0,2465.0),(6387.0,5060.0),(6531.0,3852.0),(291.0,1228.0),(8735.0,4662.0),(4797.0,3767.0),(2278.0,6846.0),(8279.0,1596.0),(9829.0,7167.0),(4140.0,4135.0),(9162.0,5272.0),(7380.0,7079.0),(11561.0,2152.0),(1548.0,5179.0),(12102.0,7108.0),(5072.0,7110.0),(6320.0,2296.0)

91,(7943.0,2628.0),(3456.0,2761.0),(11619.0,2455.0),(6396.0,4728.0),(6656.0,4052.0),(336.0,1558.0),(9068.0,4662.0),(5130.0,3778.0),(2497.0,6595.0),(8276.0,1907.0),(10162.0,7177.0),(4140.0,3801.0),(8845.0,5245.0),(7380.0,6745.0),(11322.0,2233.0),(1563.0,4846.0),(11770.0,7143.0),(4738.0,7110.0),(6234.0,2142.0)

92,(8092.0,2659.0),(3171.0,2776.0),(11465.0,2450.0),(6475.0,4858.0),(6325.0,4079.0),(393.0,1886.0),(9162.0,4901.0),(5463.0,3780.0),(2717.0,6344.0),(7943.0,1896.0),(10311.0,7381.0),(4140.0,4019.0),(8512.0,5233.0),(7380.0,6412.0),(11348.0,2339.0),(1291.0,4770.0),(11588.0,7298.0),(4664.0,7347.0),(6042.0,1914.0)

93,(8118.0,2922.0),(3131.0,2766.0),(11798.0,2460.0),(6808.0,4852.0),(5993.0,4050.0),(412.0,1997.0),(9162.0,5235.0),(5608.0,3995.0),(2948.0,6105.0),(7610.0,1897.0),(10267.0,7650.0),(4207.0,4271.0),(8179.0,5222.0),(7195.0,6269.0),(11167.0,2217.0),(983.0,4721.0),(11623.0,7630.0),(4692.0,7679.0),(6163.0,2165.0)

94,(8419.0,2954.0),(3111.0,2433.0),(12132.0,2460.0),(7142.0,4846.0),(5835.0,4273.0),(355.0,1668.0),(9473.0,5272.0),(5674.0,4050.0),(3007.0,6046.0),(7452.0,1733.0),(9934.0,7650.0),(4522.0,4327.0),(7846.0,5234.0),(6862.0,6278.0),(10859.0,2346.0),(912.0,4396.0),(11658.0,7961.0),(4659.0,7953.0),(6264.0,2482.0)

95,(8744.0,2882.0),(3097.0,2100.0),(12465.0,2449.0),(7398.0,4764.0),(5899.0,3959.0),(301.0,1339.0),(9806.0,5289.0),(6008.0,4050.0),(2771.0,6282.0),(7452.0,1400.0),(9601.0,7650.0),(4350.0,4253.0),(7513.0,5251.0),(6736.0,6485.0),(10552.0,2475.0),(1168.0,4286.0),(11948.0,8013.0),(4326.0,7938.0),(6198.0,2275.0)

96,(9070.0,2810.0),(3294.0,2174.0),(12798.0,2438.0),(7628.0,4662.0),(5892.0,3631.0),(270.0,1007.0),(9876.0,5283.0),(5872.0,3852.0),(2441.0,6326.0),(7416.0,1069.0),(9628.0,7974.0),(4134.0,4401.0),(7180.0,5247.0),(6743.0,6818.0),(10244.0,2604.0),(1262.0,4044.0),(12282.0,8018.0),(3993.0,7923.0),(6298.0,2207.0)

97,(9164.0,2775.0),(3468.0,2327.0),(13064.0,2366.0),(7962.0,4662.0),(5933.0,3300.0),(240.0,676.0),(9982.0,5021.0),(6164.0,3729.0),(2111.0,6371.0),(7416.0,736.0),(9314.0,7992.0),(4117.0,4734.0),(6847.0,5236.0),(6755.0,7145.0),(9919.0,2670.0),(1244.0,3776.0),(12615.0,8022.0),(3660.0,7908.0),(6489.0,2447.0)

98,(9190.0,2473.0),(3614.0,2497.0),(13052.0,2142.0),(7868.0,4662.0),(5973.0,2969.0),(286.0,412.0),(9997.0,4688.0),(5869.0,3833.0),(1780.0,6416.0),(7428.0,403.0),(9152.0,7811.0),(4126.0,5063.0),(6514.0,5224.0),(7088.0,7136.0),(9590.0,272.0),(1267.0,4108.0),(12948.0,8026.0),(3523.0,7701.0),(6777.0,2616.0)

99,(9198.0,2140.0),(3947.0,2488.0),(13065.0,2384.0),(7535.0,4662.0),(5661.0,2925.0),(619.0,407.0),(10003.0,4536.0),(6029.0,4051.0),(1450.0,6461.0),(7233.0,270.0),(9315.0,7650.0),(4051.0,5202.0),(6391.0,5029.0),(7386.0,7169.0),(9261.0,2774.0),(1453.0,4266.0),(13282.0,8031.0),(3515.0,7368.0),(7088.0,2725.0)

100,(9198.0,1806.0),(4117.0,2307.0),(13013.0,2089.0),(7399.0,4858.0),(5537.0,2700.0),(952.0,402.0),(9989.0,4869.0),(5981.0,4379.0),(1287.0,6286.0),(6900.0,270.0),(9599.0,7704.0),(4149.0,5121.0),(6393.0,4698.0),(7437.0,7499.0),(9165.0,2768.0),(1787.0,4266.0),(13615.0,8035.0),(3435.0,7110.0),(7367.0,2903.0)

101,(9191.0,1474.0),(4397.0,2250.0),(12681.0,2106.0),(7428.0,5190.0),(5526.0,2367.0),(1286.0,397.0),(9974.0,5202.0),(5800.0,4657.0),(1243.0,5956.0),(6567.0,280.0),(9625.0,7947.0),(4056.0,5261.0),(6390.0,4364.0),(7417.0,7828.0),(9190.0,2479.0),(2090.0,4298.0),(13948.0,8039.0),(3101.0,7110.0),(7445.0,2774.0)

102,(9077.0,1161.0),(4590.0,2391.0),(12348.0,2122.0),(7410.0,4989.0),(5513.0,2034.0),(1619.0,392.0),(9955.0,5260.0),(5614.0,4791.0),(1413.0,5821.0),(6581.0,561.0),(9560.0,7650.0),(4093.0,5593.0),(6443.0,4079.0),(7428.0,7753.0),(9187.0,2532.0),(2121.0,4630.0),(14282.0,8044.0),(2768.0,7110.0),(7435.0,2440.0)

103,(8963.0,848.0),(4607.0,2723.0),(12015.0,2138.0),(7398.0,4666.0),(5300.0,1908.0),(1952.0,387.0),(10060.0,5252.0),(5653.0,4498.0),(1702.0,5778.0),(6588.0,590.0),(9226.0,7650.0),(4320.0,5589.0),(6775.0,4050.0),(7624.0,7614.0),(9172.0,2715.0),(2161.0,4961.0),(14442.0,7850.0),(2435.0,7105.0),(7700.0,2344.0)

104,(8850.0,534.0),(4614.0,2812.0),(11682.0,2149.0),(7239.0,4844.0),(5096.0,1832.0),(2285.0,382.0),(10393.0,5240.0),(5851.0,4233.0),(1931.0,5629.0),(6537.0,284.0),(9152.0,7399.0),(4626.0,5458.0),(7006.0,3927.0),(7956.0,7650.0),(9176.0,2690.0),(2321.0,5177.0),(14482.0,7519.0),(2108.0,7074.0),(7728.0,2046.0)

105,(8715.0,254.0),(4568.0,2934.0),(11348.0,2158.0),(6906.0,4850.0),(5401.0,1908.0),(2619.0,377.0),(10726.0,5252.0),(6058.0,4052.0),(2262.0,5593.0),(6615.0,273.0),(9163.0,7066.0),(4916.0,5321.0),(7275.0,3873.0),(8100.0,7498.0),(9165.0,2768.0),(2210.0,5000.0),(14523.0,7188.0),(2440.0,7105.0),(7545.0,1901.0)

106,(8382.0,279.0),(4512.0,3001.0),(11373.0,2439.0),(6573.0,4856.0),(5280.0,1908.0),(2952.0,372.0),(11059.0,5269.0),(6390.0,4085.0),(2594.0,5557.0),(6718.0,566.0),(9170.0,6733.0),(5127.0,5194.0),(7426.0,4042.0),(8066.0,7166.0),(9170.0,2733.0),(2127.0,4689.0),(14450.0,6885.0),(2490.0,7399.0),(7442.0,2147.0)

107,(8049.0,283.0),(4316.0,3140.0),(11297.0,2132.0),(6392.0,4692.0),(5048.0,1730.0),(3285.0,367.0),(11196.0,5518.0),(6506.0,3852.0),(2925.0,5520.0),(6820.0,883.0),(9178.0,6399.0),(5256.0,5038.0),(7400.0,3883.0),(8084.0,6834.0),(9178.0,2671.0),(2096.0,4357.0),(14140.0,6840.0),(2498.0,7732.0),(7537.0,2363.0)

108,(7716.0,276.0),(4114.0,2977.0),(11225.0,1807.0),(6397.0,4963.0),(5004.0,1399.0),(3619.0,362.0),(11269.0,5843.0),(6677.0,4046.0),(3042.0,5291.0),(6840.0,932.0),(8928.0,6311.0),(5248.0,4820.0),(7067.0,3855.0),(8204.0,6616.0),(9167.0,2749.0),(2040.0,4029.0),(13911.0,6646.0),(2338.0,7908.0),(7869.0,2344.0)

109,(7484.0,271.0),(4104.0,2860.0),(11153.0,1481.0),(6372.0,5295.0),(5011.0,1450.0),(3643.0,362.0),(11342.0,6169.0),(6988.0,4026.0),(3042.0,5481.0),(6733.0,894.0),(8595.0,6303.0),(4998.0,4690.0),(7109.0,3581.0),(8384.0,6634.0),(9162.0,2828.0),(1975.0,3702.0),(13727.0,6368.0),(2005.0,7921.0),(8202.0,2355.0)

110,(7817.0,278.0),(4104.0,2872.0),(11080.0,1156.0),(6372.0,5628.0),(5055.0,1781.0),(3310.0,367.0),(11417.0,6493.0),(7175.0,3864.0),(2736.0,5541.0),(6652.0,1217.0),(8693.0,6305.0),(5061.0,4933.0),(6784.0,3573.0),(8096.0,6653.0),(9167.0,3161.0),(1699.0,3675.0),(13543.0,6090.0),(1672.0,7935.0),(8283.0,2084.0)

111,(8151.0,285.0),(4030.0,2924.0),(11008.0,830.0),(6372.0,5962.0),(5071.0,1811.0),(2976.0,372.0),(11493.0,6818.0),(7442.0,3943.0),(2405.0,5577.0),(6417.0,1395.0),(9027.0,6314.0),(4956.0,5156.0),(6479.0,3696.0),(8074.0,6986.0),(9171.0,3494.0),(1369.0,3724.0),(13359.0,5812.0),(1864.0,7927.0),(8285.0,2063.0)

112,(8483.0,271.0),(3699.0,2962.0),(10936.0,505.0),(6286.0,6213.0),(5333.0,1908.0),(2643.0,377.0),(11569.0,7143.0),(7444.0,3935.0),(2074.0,5614.0),(6090.0,1456.0),(9176.0,6498.0),(4635.0,5192.0),(6200.0,3723.0),(8081.0,7318.0),(9176.0,3828.0),(1037.0,3752.0),(13174.0,5534.0),(2197.0,7914.0),(8299.0,2357.0)

113,(8679.0,257.0),(3464.0,3076.0),(10835.0,216.0),(5953.0,6229.0),(5514.0,2066.0),(2310.0,382.0),(11606.0,7474.0),(7511.0,3610.0),(1742.0,5650.0),(6190.0,1438.0),(9168.0,6831.0),(4304.0,5226.0),(6532.0,3691.0),(8116.0,7649.0),(9184.0,4161.0),(826.0,3865.0),(13153.0,5320.0),(2531.0,7902.0),(8633.0,2350.0)

114,(8347.0,281.0),(3364.0,3340.0),(10502.0,222.0),(5620.0,6244.0),(5527.0,2400.0),(1977.0,387.0),(11642.0,7805.0),(7703.0,3337.0),(1826.0,5844.0),(6514.0,1365.0),(9143.0,7146.0),(4069.0,5130.0),(6832.0,3549.0),(8108.0,7973.0),(9184.0,4493.0),(877.0,4195.0),(13054.0,5344.0),(2864.0,7902.0),(8966.0,2344.0)

115,(8262.0,535.0),(3158.0,3516.0),(10168.0,227.0),(5289.0,6215.0),(5538.0,2733.0),(1643.0,392.0),(11536.0,8008.0),(7800.0,3199.0),(2158.0,5873.0),(6842.0,1347.0),(8809.0,7146.0),(4050.0,5199.0),(7063.0,3606.0),(7775.0,7965.0),(9183.0,4500.0),(1085.0,4302.0),(13163.0,5317.0),(3197.0,7902.0),(9198.0,2238.0)

116,(8262.0,869.0),(3198.0,3760.0),(9835.0,233.0),(5518.0,6239.0),(5396.0,2945.0),(1310.0,397.0),(11202.0,8005.0),(7609.0,3471.0),(2490.0,5903.0),(6889.0,1051.0),(8476.0,7146.0),(4128.0,5068.0),(6910.0,3688.0),(7442.0,7957.0),(9168.0,4614.0),(1143.0,4291.0),(13169.0,5526.0),(3524.0,7902.0),(9198.0,1905.0)

117,(8262.0,1202.0),(3531.0,3754.0),(9502.0,239.0),(5591.0,6517.0),(5065.0,2916.0),(977.0,402.0),(10869.0,8002.0),(7659.0,3630.0),(2822.0,5933.0),(6781.0,755.0),(8143.0,7146.0),(4056.0,5260.0),(7076.0,3683.0),(7440.0,7670.0),(9162.0,4947.0),(886.0,4253.0),(13353.0,5804.0),(3191.0,7902.0),(9198.0,1571.0)

118,(8275.0,1535.0),(3864.0,3748.0),(9168.0,244.0),(5605.0,6850.0),(5380.0,2948.0),(643.0,407.0),(10536.0,8000.0),(7746.0,3611.0),(3037.0,5952.0),(7114.0,736.0),(8318.0,7146.0),(4093.0,5592.0),(6864.0,3852.0),(7412.0,7340.0),(9186.0,5257.0),(835.0,3923.0),(13537.0,6082.0),(2858.0,7902.0),(9198.0,1749.0)

119,(8295.0,1867.0),(4134.0,3686.0),(8835.0,250.0),(5606.0,7182.0),(5538.0,2748.0),(310.0,412.0),(10202.0,7997.0),(7489.0,3698.0),(2705.0,5922.0),(7384.0,721.0),(8651.0,7146.0),(4077.0,5452.0),(6661.0,3961.0),(7262.0,7131.0),(9519.0,5274.0),(979.0,3754.0),(13721.0,6360.0),(2524.0,7902.0),(9198.0,2082.0)

120,(8274.0,2200.0),(4104.0,3354.0),(8502.0,270.0),(5605.0,7515.0),(5528.0,2415.0),(238.0,651.0),(9869.0,7994.0),(7428.0,4026.0),(2373.0,5893.0),(7051.0,740.0),(8984.0,7146.0),(4128.0,5221.0),(6436.0,4080.0),(6928.0,7140.0),(9851.0,5290.0),(1246.0,3814.0),(13906.0,6637.0),(2494.0,7591.0),(9121.0,2341.0)

121,(8086.0,2351.0),(3926.0,3335.0),(8262.0,379.0),(5614.0,7848.0),(5515.0,2082.0),(268.0,983.0),(9536.0,7992.0),(7670.0,4104.0),(2041.0,5863.0),(6755.0,708.0),(9156.0,7302.0),(4099.0,5018.0),(6459.0,3852.0),(6595.0,7135.0),(9920.0,5270.0),(1269.0,4146.0),(13950.0,6857.0),(2487.0,7258.0),(8788.0,2347.0)

122,(7753.0,2340.0),(4129.0,3195.0),(8262.0,713.0),(5354.0,7930.0),(5349.0,1908.0),(299.0,1315.0),(9202.0,7992.0),(8003.0,4104.0),(1709.0,5833.0),(6672.0,385.0),(9144.0,7635.0),(4189.0,5236.0),(6669.0,4000.0),(6519.0,7246.0),(10251.0,5245.0),(1491.0,4266.0),(13623.0,6921.0),(2299.0,7092.0),(8455.0,2354.0)

123,(8059.0,2351.0),(4104.0,2865.0),(8262.0,1046.0),(5021.0,7943.0),(5059.0,1801.0),(351.0,1644.0),(9146.0,7700.0),(8064.0,4394.0),(1702.0,5781.0),(6601.0,275.0),(9160.0,7968.0),(4520.0,5204.0),(6961.0,4050.0),(6688.0,7141.0),(10331.0,5242.0),(1825.0,4266.0),(13307.0,6982.0),(2189.0,6963.0),(8273.0,2218.0)

124,(8131.0,2353.0),(3937.0,2935.0),(8265.0,1379.0),(4713.0,7928.0),(5060.0,1802.0),(408.0,1972.0),(9427.0,7650.0),(8077.0,4596.0),(1410.0,5821.0),(6554.0,448.0),(8852.0,7986.0),(4692.0,5049.0),(7000.0,3959.0),(6750.0,7417.0),(9998.0,5255.0),(2158.0,4266.0),(13634.0,6919.0),(2146.0,7019.0),(8276.0,1907.0)

125,(7798.0,2342.0),(3825.0,3143.0),(8286.0,1712.0),(4685.0,7595.0),(5354.0,1908.0),(473.0,2299.0),(9760.0,7650.0),(8056.0,4264.0),(1243.0,5959.0),(6615.0,703.0),(8519.0,7980.0),(4821.0,4886.0),(7223.0,4079.0),(6750.0,7751.0),(9846.0,5468.0),(2491.0,4269.0),(13961.0,6855.0),(1840.0,7081.0),(7942.0,1896.0)

126,(7729.0,2065.0),(3707.0,3330.0),(8287.0,2044.0),(4657.0,7263.0),(5515.0,2087.0),(544.0,2625.0),(10093.0,7650.0),(8057.0,4276.0),(1288.0,6289.0),(6537.0,379.0),(8186.0,7975.0),(4972.0,4728.0),(7411.0,4243.0),(6621.0,7951.0),(9846.0,5801.0),(2825.0,4272.0),(14293.0,6840.0),(1507.0,7090.0),(7609.0,1897.0)

127,(7879.0,1894.0),(3453.0,3268.0),(8281.0,2357.0),(4465.0,7110.0),(5522.0,2259.0),(614.0,2951.0),(10324.0,7555.0),(8078.0,4609.0),(1295.0,6339.0),(6530.0,285.0),(8118.0,7708.0),(4917.0,4892.0),(7400.0,4576.0),(6288.0,7941.0),(9846.0,5817.0),(3158.0,4275.0),(14208.0,6840.0),(1439.0,7054.0),(7626.0,1896.0)

128,(8212.0,1905.0),(3338.0,3344.0),(8614.0,2351.0),(4131.0,7110.0),(5508.0,1926.0),(575.0,2770.0),(10299.0,7222.0),(8362.0,4662.0),(1250.0,6009.0),(6197.0,288.0),(7827.0,7636.0),(4675.0,4953.0),(7150.0,4662.0),(5955.0,7930.0),(9846.0,5483.0),(3491.0,4278.0),(14204.0,6840.0),(1401.0,6971.0),(7484.0,1905.0)

129,(8278.0,2155.0),(3168.0,3187.0),(8947.0,2344.0),(3798.0,7110.0),(5192.0,1908.0),(505.0,2444.0),(10277.0,6890.0),(8695.0,4662.0),(1359.0,5819.0),(5864.0,288.0),(7495.0,7600.0),(4815.0,5184.0),(6817.0,4662.0),(5622.0,7920.0),(9962.0,5258.0),(3825.0,4281.0),(14486.0,6938.0),(1343.0,6642.0),(7440.0,2208.0)

130,(8131.0,2353.0),(3132.0,2856.0),(9193.0,2422.0),(3465.0,7110.0),(5037.0,1642.0),(434.0,2118.0),(10256.0,6557.0),(9028.0,4662.0),(1692.0,5831.0),(5531.0,298.0),(7409.0,7882.0),(4919.0,5292.0),(6483.0,4662.0),(5289.0,7933.0),(9975.0,5174.0),(4158.0,4280.0),(14425.0,6851.0),(1343.0,6645.0),(7269.0,2361.0)

131,(7798.0,2342.0),(3378.0,2799.0),(9088.0,2670.0),(3131.0,7110.0),(4992.0,1312.0),(751.0,2082.0),(10242.0,6224.0),(9188.0,4463.0),(1698.0,5799.0),(5198.0,310.0),(7656.0,7962.0),(5156.0,5195.0),(6629.0,4662.0),(4956.0,7946.0),(9990.0,4841.0),(4476.0,4307.0),(14471.0,6915.0),(1402.0,6973.0),(6937.0,2330.0)

132,(7467.0,2372.0),(3456.0,2552.0),(8755.0,2692.0),(2798.0,7110.0),(4708.0,1298.0),(1083.0,2058.0),(10242.0,5952.0),(9182.0,4131.0),(1844.0,5639.0),(4865.0,322.0),(7989.0,7970.0),(5133.0,5365.0),(6962.0,4662.0),(4623.0,7951.0),(10005.0,4508.0),(4798.0,4328.0),(14531.0,7002.0),(1496.0,7290.0),(6852.0,2557.0)

133,(7445.0,2076.0),(3172.0,2502.0),(8423.0,2715.0),(2465.0,7108.0),(4377.0,1343.0),(1416.0,2034.0),(10242.0,6286.0),(9196.0,4373.0),(2175.0,5602.0),(4531.0,334.0),(8118.0,7768.0),(4853.0,5526.0),(7296.0,4662.0),(4290.0,7936.0),(9737.0,4427.0),(5129.0,4294.0),(14283.0,6840.0),(1559.0,7617.0),(6655.0,2745.0)

134,(7452.0,1743.0),(3102.0,2227.0),(8145.0,2734.0),(2133.0,7076.0),(4064.0,1365.0),(1686.0,2089.0),(10543.0,6318.0),(9171.0,4086.0),(2506.0,5566.0),(4198.0,345.0),(8094.0,7436.0),(4559.0,5647.0),(7398.0,4790.0),(3957.0,7921.0),(9405.0,4405.0),(5412.0,4163.0),(14130.0,6840.0),(1600.0,7928.0),(6364.0,2601.0)

135,(7452.0,1409.0),(3271.0,2070.0),(8477.0,2711.0),(2291.0,6831.0),(4009.0,1036.0),(1739.0,2418.0),(10876.0,6318.0),(8838.0,4091.0),(2838.0,5530.0),(3865.0,357.0),(8023.0,7144.0),(4583.0,5962.0),(7404.0,4457.0),(3624.0,7906.0),(9181.0,4516.0),(5610.0,4138.0),(14444.0,6877.0),(1572.0,7702.0),(6212.0,2319.0)

136,(7416.0,1079.0),(3424.0,2250.0),(8810.0,2689.0),(2511.0,6580.0),(3948.0,710.0),(1793.0,2747.0),(11210.0,6318.0),(8505.0,4096.0),(3042.0,5379.0),(3902.0,604.0),(7689.0,7136.0),(4427.0,6164.0),(7415.0,4123.0),(3522.0,7664.0),(9162.0,4848.0),(5482.0,4356.0),(14524.0,7177.0),(1590.0,7842.0),(6024.0,2207.0)

137,(7416.0,745.0),(3398.0,2250.0),(9143.0,2666.0),(2730.0,6329.0),(3816.0,403.0),(1852.0,3075.0),(11339.0,6154.0),(8171.0,4102.0),(2868.0,5210.0),(3886.0,565.0),(7383.0,7151.0),(4145.0,6033.0),(7105.0,4065.0),(3515.0,7331.0),(9162.0,5181.0),(5411.0,4331.0),(14483.0,7508.0),(1661.0,7548.0),(6249.0,2142.0)

138,(7428.0,412.0),(3731.0,2250.0),(9196.0,2368.0),(2963.0,6090.0),(3512.0,364.0),(1916.0,3402.0),(11266.0,5829.0),(7838.0,4104.0),(2535.0,5191.0),(3688.0,361.0),(7434.0,7480.0),(4104.0,5705.0),(6772.0,4050.0),(3531.0,7000.0),(9166.0,5515.0),(5364.0,4575.0),(14443.0,7839.0),(1889.0,7305.0),(6082.0,2178.0)

139,(7243.0,270.0),(4064.0,2250.0),(9181.0,2644.0),(3241.0,5970.0),(3179.0,369.0),(1980.0,3729.0),(11193.0,5504.0),(7505.0,4104.0),(2203.0,5166.0),(3354.0,366.0),(7419.0,7810.0),(4136.0,5999.0),(6604.0,3852.0),(3595.0,6673.0),(9172.0,5848.0),(5580.0,4476.0),(14433.0,7921.0),(2115.0,7060.0),(6312.0,2263.0)

140,(6910.0,270.0),(4104.0,2525.0),(9197.0,2350.0),(3573.0,5998.0),(2845.0,374.0),(2046.0,4056.0),(11044.0,5268.0),(7660.0,4104.0),(2235.0,5025.0),(3021.0,371.0),(7212.0,7956.0),(3911.0,6067.0),(6286.0,3915.0),(3660.0,6346.0),(9177.0,6181.0),(5428.0,4429.0),(14473.0,7590.0),(2189.0,6869.0),(6268.0,2142.0)

141,(6577.0,279.0),(4104.0,2858.0),(9180.0,2662.0),(3897.0,6066.0),(2512.0,379.0),(2064.0,4148.0),(10711.0,5251.0),(7993.0,4104.0),(2285.0,5174.0),(2688.0,376.0),(6878.0,7956.0),(4104.0,5851.0),(5986.0,4050.0),(3724.0,6019.0),(9175.0,6515.0),(5512.0,4599.0),(14514.0,7259.0),(1859.0,6917.0),(6057.0,1944.0)

142,(6532.0,285.0),(4379.0,2927.0),(9190.0,2340.0),(4140.0,6014.0),(2179.0,384.0),(1999.0,3821.0),(10537.0,5242.0),(7765.0,4104.0),(2156.0,4925.0),(2355.0,381.0),(6954.0,7956.0),(4258.0,5617.0),(5653.0,4050.0),(4002.0,5961.0),(9168.0,6848.0),(5743.0,4390.0),(14491.0,6944.0),(1529.0,6965.0),(6151.0,2133.0)

143,(6697.0,485.0),(4629.0,3020.0),(8857.0,2346.0),(4266.0,5783.0),(1845.0,389.0),(1819.0,3657.0),(10870.0,5259.0),(7431.0,4104.0),(2057.0,4676.0),(2021.0,386.0),(7287.0,7956.0),(4563.0,5483.0),(5580.0,3756.0),(4092.0,5698.0),(9162.0,7110.0),(5901.0,4102.0),(14535.0,7008.0),(1419.0,7015.0),(6212.0,1957.0)

144,(6754.0,808.0),(4620.0,3331.0),(8524.0,2352.0),(4557.0,5642.0),(1512.0,394.0),(1489.0,3706.0),(11202.0,5276.0),(7733.0,4104.0),(1732.0,4750.0),(1688.0,391.0),(7174.0,7956.0),(4873.0,5361.0),(5580.0,3423.0),(3839.0,5914.0),(9170.0,6777.0),(6010.0,4218.0),(14358.0,6760.0),(1635.0,7087.0),(6189.0,1711.0)

145,(6674.0,1131.0),(4287.0,3348.0),(8332.0,2356.0),(4570.0,5676.0),(1179.0,399.0),(1159.0,3747.0),(11536.0,5288.0),(8066.0,4103.0),(1558.0,4950.0),(1355.0,396.0),(6841.0,7956.0),(5017.0,5090.0),(5465.0,3121.0),(3690.0,6192.0),(9177.0,6443.0),(5900.0,4526.0),(14159.0,6493.0),(1968.0,7077.0),(6522.0,1720.0)

146,(6654.0,1208.0),(4128.0,3199.0),(8666.0,2350.0),(4600.0,5904.0),(845.0,404.0),(826.0,3761.0),(11869.0,5301.0),(8400.0,4098.0),(1448.0,5187.0),(1021.0,401.0),(6750.0,7714.0),(5234.0,4837.0),(5497.0,2925.0),(3626.0,6519.0),(9175.0,6525.0),(6002.0,4279.0),(13960.0,6225.0),(2223.0,6917.0),(6840.0,1743.0)

147,(6735.0,885.0),(4058.0,2921.0),(8999.0,2343.0),(4485.0,6182.0),(512.0,409.0),(734.0,3453.0),(12202.0,5314.0),(8733.0,4093.0),(1115.0,5200.0),(688.0,406.0),(6750.0,7380.0),(5503.0,4644.0),(5535.0,2630.0),(3561.0,6846.0),(9168.0,6858.0),(5927.0,4050.0),(13761.0,5958.0),(2112.0,7065.0),(6840.0,2077.0)

148,(6717.0,562.0),(3727.0,2959.0),(9190.0,2474.0),(4166.0,6086.0),(252.0,413.0),(655.0,3130.0),(12535.0,5326.0),(9066.0,4087.0),(1021.0,4909.0),(355.0,411.0),(6651.0,7139.0),(5598.0,4346.0),(5567.0,2329.0),(3575.0,7110.0),(9207.0,7147.0),(5893.0,3935.0),(13562.0,5690.0),(1885.0,7309.0),(6844.0,2410.0)

149,(6673.0,270.0),(3515.0,2981.0),(9178.0,2787.0),(4104.0,5765.0),(585.0,408.0),(852.0,3000.0),(12868.0,5339.0),(9192.0,4305.0),(1023.0,4920.0),(410.0,411.0),(6530.0,7280.0),(5580.0,4055.0),(5889.0,2245.0),(3908.0,7110.0),(9541.0,7158.0),(5844.0,4251.0),(13363.0,5423.0),(1657.0,7552.0),(6935.0,2683.0)

150,(7006.0,270.0),(3805.0,3001.0),(9507.0,2735.0),(4336.0,5582.0),(919.0,403.0),(1184.0,2965.0),(13133.0,5472.0),(9165.0,4636.0),(1090.0,5247.0),(743.0,406.0),(6528.0,7273.0),(5478.0,4097.0),(5811.0,2266.0),(4241.0,7110.0),(9874.0,7168.0),(5641.0,4513.0),(13254.0,5122.0),(1528.0,7453.0),(7241.0,2808.0)

151,(7339.0,270.0),(3853.0,3330.0),(9836.0,2683.0),(4643.0,5451.0),(1252.0,398.0),(1515.0,2930.0),(13317.0,5750.0),(9195.0,4356.0),(1167.0,5571.0),(1076.0,401.0),(6335.0,7124.0),(5356.0,4309.0),(5527.0,2378.0),(4575.0,7110.0),(10207.0,7179.0),(5619.0,4809.0),(13231.0,4790.0),(1464.0,7126.0),(7420.0,3031.0)

152,(7424.0,508.0),(4111.0,3429.0),(10165.0,2631.0),(4918.0,5303.0),(1585.0,393.0),(1847.0,2898.0),(13502.0,6028.0),(9241.0,4081.0),(1308.0,5817.0),(1409.0,396.0),(6001.0,7118.0),(5515.0,4356.0),(5537.0,2711.0),(4908.0,7110.0),(10314.0,7425.0),(5619.0,5140.0),(13208.0,4457.0),(1432.0,7041.0),(7244.0,3313.0)

153,(7294.0,726.0),(4138.0,3726.0),(10474.0,2507.0),(4719.0,5184.0),(1919.0,388.0),(2180.0,2902.0),(13686.0,6306.0),(9574.0,4055.0),(1641.0,5830.0),(1743.0,390.0),(5668.0,7111.0),(5306.0,4572.0),(5539.0,2787.0),(5026.0,6890.0),(10323.0,7541.0),(5362.0,5206.0),(13185.0,4125.0),(1356.0,6719.0),(7124.0,3552.0)

154,(6961.0,745.0),(4108.0,3394.0),(10782.0,2378.0),(5046.0,5190.0),(2252.0,383.0),(2514.0,2907.0),(13870.0,6583.0),(9857.0,4033.0),(1409.0,5821.0),(2076.0,385.0),(5896.0,7115.0),(4983.0,4584.0),(5529.0,2454.0),(5033.0,6556.0),(10298.0,7209.0),(5368.0,4994.0),(13162.0,3792.0),(1329.0,6569.0),(7273.0,3254.0)

155,(6821.0,885.0),(4376.0,3320.0),(11089.0,2249.0),(5287.0,5097.0),(2585.0,378.0),(2847.0,2912.0),(14014.0,6844.0),(9524.0,4059.0),(1243.0,5960.0),(2409.0,380.0),(6229.0,7122.0),(4970.0,4662.0),(5738.0,2284.0),(5000.0,6258.0),(9990.0,7172.0),(5609.0,4863.0),(13139.0,3460.0),(1388.0,6897.0),(7440.0,2983.0)

156,(6885.0,1203.0),(4510.0,3027.0),(11328.0,2257.0),(5033.0,4919.0),(2918.0,373.0),(3083.0,2915.0),(13687.0,6908.0),(9192.0,4085.0),(1288.0,6290.0),(2743.0,375.0),(6562.0,7133.0),(5002.0,4330.0),(5963.0,2226.0),(4669.0,6212.0),(9657.0,7161.0),(5628.0,4963.0),(13116.0,3127.0),(1582.0,7088.0),(7136.0,2847.0)

157,(6689.0,1336.0),(4757.0,2928.0),(11342.0,2314.0),(5204.0,4874.0),(3252.0,368.0),(2749.0,2910.0),(13360.0,6972.0),(9212.0,4392.0),(1294.0,6338.0),(3076.0,370.0),(6604.0,7136.0),(5303.0,4272.0),(5640.0,2310.0),(4542.0,6370.0),(9323.0,7151.0),(5692.0,5220.0),(13093.0,2795.0),(1915.0,7079.0),(7424.0,2975.0)

158,(6688.0,1072.0),(5090.0,2921.0),(11346.0,2328.0),(5064.0,4714.0),(3585.0,363.0),(2416.0,2906.0),(13571.0,6931.0),(9544.0,4415.0),(1250.0,6008.0),(3409.0,365.0),(6496.0,7288.0),(5551.0,4060.0),(5517.0,2125.0),(4587.0,6700.0),(9155.0,7317.0),(6025.0,5220.0),(13070.0,2462.0),(2191.0,6959.0),(7220.0,2796.0)

159,(6770.0,762.0),(5029.0,2771.0),(11586.0,2454.0),(4994.0,4450.0),(3845.0,471.0),(2083.0,2901.0),(13898.0,6867.0),(9877.0,4437.0),(1361.0,5819.0),(3742.0,360.0),(6163.0,7275.0),(5580.0,3784.0),(5391.0,1908.0),(4633.0,7030.0),(9144.0,7650.0),(6358.0,5220.0),(13052.0,2129.0),(2403.0,6702.0),(6911.0,2677.0)

160,(6903.0,1062.0),(5165.0,2937.0),(11919.0,2464.0),(5187.0,4290.0),(4002.0,738.0),(1831.0,2964.0),(14228.0,6840.0),(9999.0,4648.0),(1692.0,5829.0),(3907.0,615.0),(5830.0,7262.0),(5251.0,3780.0),(5088.0,1826.0),(4897.0,7110.0),(9161.0,7983.0),(6403.0,4901.0),(13065.0,2379.0),(2623.0,6451.0),(6727.0,2840.0)

161,(7235.0,1097.0),(5159.0,3269.0),(11687.0,2457.0),(5507.0,4356.0),(4336.0,738.0),(1894.0,3291.0),(14273.0,6840.0),(9984.0,4981.0),(1698.0,5800.0),(3992.0,935.0),(5699.0,7257.0),(5166.0,3532.0),(5087.0,1825.0),(5230.0,7110.0),(9486.0,7992.0),(6116.0,4860.0),(13053.0,2147.0),(2701.0,6362.0),(6553.0,2936.0)

162,(7416.0,964.0),(4936.0,3310.0),(11370.0,2426.0),(5563.0,4596.0),(4669.0,738.0),(1958.0,3618.0),(14139.0,6840.0),(9916.0,5271.0),(1704.0,5767.0),(4047.0,1264.0),(6032.0,7270.0),(5151.0,3199.0),(5019.0,1508.0),(5564.0,7110.0),(9440.0,7992.0),(5783.0,4860.0),(13036.0,1815.0),(2481.0,6613.0),(6594.0,2733.0)

163,(7416.0,808.0),(4886.0,3345.0),(11362.0,2158.0),(5397.0,4483.0),(5002.0,738.0),(1649.0,3682.0),(14449.0,6884.0),
(9847.0,5291.0),(1711.0,5734.0),(4073.0,1596.0),(6365.0,7283.0),(4938.0,3288.0),(4903.0,1271.0),(5614.0,7118.0),(9159.0,
7936.0),(5564.0,4869.0),(13018.0,1482.0),(2259.0,6859.0),(6920.0,2800.0)

164,(7419.0,1141.0),(4733.0,3335.0),(11695.0,2149.0),(5601.0,4296.0),(5336.0,738.0),(1319.0,3732.0),(14447.0,6881.0
,)(9912.0,5272.0),(1718.0,5701.0),(4082.0,1929.0),(6481.0,7133.0),(5161.0,3291.0),(4573.0,1316.0),(5602.0,7433.0),(9190.0
,7650.0),(5561.0,4608.0),(13001.0,1149.0),(1929.0,6907.0),(7011.0,2878.0)

165,(7347.0,1364.0),(4946.0,3214.0),(12028.0,2138.0),(5702.0,4050.0),(5669.0,738.0),(1259.0,3998.0),(14508.0,6968.0
,)(9705.0,5284.0),(1724.0,5668.0),(3893.0,2070.0),(6484.0,7141.0),(5164.0,2959.0),(4243.0,1361.0),(5611.0,7766.0),(9523.0
,7650.0),(5278.0,4785.0),(12984.0,816.0),(1599.0,6955.0),(6740.0,2877.0)

166,(7013.0,1354.0),(5042.0,3177.0),(12361.0,2121.0),(6035.0,4050.0),(5801.0,906.0),(1214.0,4278.0),(14518.0,6984.0
,)(9372.0,5267.0),(1525.0,5825.0),(3560.0,2070.0),(6798.0,7144.0),(5078.0,2919.0),(4042.0,1231.0),(5611.0,7740.0),(9621.0
,7911.0),(5215.0,5015.0),(12777.0,687.0),(1451.0,7078.0),(6764.0,2943.0)

167,(6857.0,1544.0),(5009.0,3342.0),(12694.0,2105.0),(6367.0,4083.0),(5813.0,838.0),(902.0,4350.0),(14348.0,6840.0
,)(9162.0,5133.0),(1216.0,5783.0),(3226.0,2070.0),(7131.0,7135.0),(4893.0,2736.0),(3987.0,903.0),(5602.0,7406.0),(9603.0,7
741.0),(5480.0,4900.0),(12443.0,692.0),(1478.0,7196.0),(6444.0,2936.0)

168,(6840.0,1876.0),(5149.0,3154.0),(13027.0,2089.0),(6480.0,3830.0),(5940.0,1119.0),(973.0,4676.0),(14015.0,6844.0
,)(9162.0,4800.0),(1140.0,5458.0),(3294.0,2074.0),(7380.0,7043.0),(4604.0,2688.0),(3894.0,585.0),(5611.0,7144.0),(9835.0,
7650.0),(5742.0,4728.0),(12110.0,698.0),(1541.0,7524.0),(6111.0,2947.0)

169,(6840.0,2210.0),(5057.0,2913.0),(13066.0,2398.0),(6800.0,3737.0),(5934.0,1278.0),(1245.0,4767.0),(13688.0,6908.
0),(9187.0,4468.0),(1149.0,5199.0),(3294.0,2092.0),(7380.0,6710.0),(4735.0,2520.0),(3832.0,440.0),(5537.0,7110.0),(10168.
0,7650.0),(5937.0,4459.0),(11880.0,599.0),(1591.0,7853.0),(5778.0,2935.0)

170,(6840.0,2100.0),(5006.0,2584.0),(12766.0,2439.0),(7098.0,3588.0),(5617.0,1292.0),(1565.0,4800.0),(13360.0,6972.
0),(9182.0,4136.0),(1482.0,5186.0),(3100.0,2183.0),(7380.0,6376.0),(5010.0,2618.0),(3961.0,747.0),(5204.0,7110.0),(10318.
0,7480.0),(6019.0,4142.0),(11880.0,266.0),(1571.0,7691.0),(5447.0,2935.0)

171,(6840.0,1767.0),(4898.0,2736.0),(12433.0,2450.0),(6873.0,3512.0),(5641.0,1472.0),(1550.0,5133.0),(13030.0,7012.
0),(9196.0,4368.0),(1815.0,5176.0),(3114.0,2516.0),(7391.0,6043.0),(5089.0,2921.0),(4016.0,1076.0),(4871.0,7110.0),(1029
3.0,7147.0),(6005.0,4050.0),(11880.0,463.0),(1601.0,7936.0),(5491.0,3164.0)

172,(6867.0,1435.0),(4611.0,2779.0),(12100.0,2461.0),(7109.0,3276.0),(5634.0,1413.0),(1560.0,4901.0),(12698.0,7046.
0),(9183.0,4085.0),(2148.0,5167.0),(3132.0,2848.0),(7409.0,5710.0),(5160.0,2678.0),(4068.0,1405.0),(4653.0,7215.0),(1027
3.0,6815.0),(5832.0,4283.0),(11974.0,700.0),(1596.0,7617.0),(5580.0,3473.0)

173,(6628.0,1332.0),(4512.0,3003.0),(11767.0,2459.0),(7123.0,2943.0),(5798.0,1680.0),(1780.0,4739.0),(12367.0,7081.
0),(9515.0,4060.0),(2481.0,5188.0),(3391.0,2973.0),(7427.0,5377.0),(5152.0,2345.0),(4077.0,1738.0),(4681.0,7548.0),(1025
2.0,6482.0),(5902.0,3970.0),(12307.0,694.0),(1824.0,7374.0),(5583.0,3806.0)

174,(6703.0,1013.0),(4313.0,3140.0),(11434.0,2449.0),(6899.0,2796.0),(6039.0,1908.0),(2105.0,4666.0),(12035.0,7115.
0),(9847.0,4034.0),(2814.0,5207.0),(3405.0,3186.0),(7422.0,5466.0),(5036.0,2118.0),(4086.0,2072.0),(4709.0,7880.0),(1025
2.0,6486.0),(6049.0,3770.0),(12641.0,689.0),(2052.0,7131.0),(5529.0,4064.0)

175,(6793.0,819.0),(4113.0,3289.0),(11309.0,2180.0),(6573.0,2729.0),(6090.0,2174.0),(2095.0,4349.0),(11703.0,7150.0
,)(9583.0,4054.0),(3146.0,5224.0),(3155.0,3090.0),(7404.0,5799.0),(4703.0,2101.0),(4092.0,2100.0),(4973.0,7945.0),(10273.
0,6818.0),(6374.0,3706.0),(12974.0,684.0),(2258.0,6870.0),(5604.0,4251.0)

176,(6889.0,1132.0),(4377.0,3346.0),(11615.0,2151.0),(6297.0,2550.0),(6240.0,2142.0),(2110.0,4514.0),(11371.0,7166.
0),(9251.0,4080.0),(3479.0,5237.0),(3132.0,2785.0),(7387.0,6132.0),(4370.0,2084.0),(4240.0,2250.0),(5306.0,7932.0),(1029
4.0,7151.0),(6690.0,3620.0),(12966.0,354.0),(2153.0,7078.0),(5376.0,4317.0)

177,(6760.0,1341.0),(4640.0,3263.0),(11948.0,2142.0),(6516.0,2736.0),(6179.0,1869.0),(2147.0,4846.0),(11038.0,7171.
0),(9176.0,3824.0),(3812.0,5216.0),(3158.0,3116.0),(7581.0,6268.0),(4095.0,2117.0),(4573.0,2250.0),(5615.0,7896.0),(1031
9.0,7483.0),(6957.0,3428.0),(12819.0,166.0),(2482.0,7109.0),(5463.0,4112.0)

178,(6671.0,1141.0),(4622.0,2934.0),(11622.0,2151.0),(6516.0,2798.0),(6034.0,1570.0),(2131.0,4723.0),(10704.0,7176.0),(9171.0,3490.0),(3954.0,5207.0),(3267.0,3354.0),(7914.0,6277.0),(4088.0,2084.0),(4525.0,2499.0),(5606.0,7563.0),(10165.0,7650.0),(7026.0,3359.0),(12486.0,177.0),(2150.0,7078.0),(5776.0,4050.0)

179,(6752.0,818.0),(4289.0,2924.0),(11319.0,2159.0),(6756.0,2720.0),(6092.0,1665.0),(2099.0,4391.0),(10371.0,7180.0),(9167.0,3157.0),(3622.0,5228.0),(3461.0,3186.0),(8118.0,6151.0),(4109.0,2413.0),(4192.0,2487.0),(5600.0,7230.0),(9831.0,7650.0),(6908.0,3555.0),(12152.0,188.0),(1817.0,7082.0),(6100.0,4012.0)

180,(6700.0,495.0),(4128.0,3061.0),(11652.0,2150.0),(6706.0,2783.0),(6218.0,1972.0),(1880.0,4266.0),(10553.0,7178.0),(9162.0,2824.0),(3531.0,5413.0),(3171.0,3225.0),(8118.0,5818.0),(4366.0,2493.0),(4132.0,2250.0),(5606.0,7187.0),(9603.0,7744.0),(7068.0,3705.0),(11880.0,258.0),(1484.0,7091.0),(6395.0,3858.0)

181,(6542.0,284.0),(4061.0,3344.0),(11985.0,2140.0),(6915.0,2907.0),(6319.0,2290.0),(1547.0,4266.0),(10887.0,7173.0),(9166.0,3089.0),(3634.0,5730.0),(3376.0,3338.0),(8118.0,5485.0),(4598.0,2611.0),(4465.0,2250.0),(5605.0,6855.0),(9621.0,7907.0),(6841.0,3852.0),(11880.0,591.0),(1428.0,7033.0),(6390.0,4171.0)

182,(6567.0,280.0),(3728.0,3330.0),(12318.0,2123.0),(7123.0,2955.0),(6241.0,2142.0),(1214.0,4278.0),(11220.0,7168.0),(9170.0,3422.0),(3754.0,5987.0),(3462.0,3063.0),(8186.0,5222.0),(4616.0,2934.0),(4590.0,2459.0),(5591.0,6522.0),(9519.0,7650.0),(6507.0,3852.0),(11880.0,479.0),(1487.0,7246.0),(6390.0,4504.0)

183,(6638.0,270.0),(3395.0,3336.0),(12651.0,2107.0),(7110.0,3263.0),(6089.0,2175.0),(902.0,4350.0),(11553.0,7164.0),(9175.0,3755.0),(4007.0,5770.0),(3204.0,2932.0),(8519.0,5233.0),(4489.0,3164.0),(4878.0,2539.0),(5523.0,6240.0),(9186.0,7650.0),(6390.0,4084.0),(11827.0,198.0),(1551.0,7573.0),(6405.0,4837.0)

184,(6724.0,270.0),(3164.0,3485.0),(12984.0,2091.0),(7124.0,2930.0),(6158.0,2148.0),(973.0,4676.0),(11607.0,7475.0),(9179.0,4083.0),(4084.0,5885.0),(3124.0,2657.0),(8852.0,5245.0),(4418.0,3261.0),(5026.0,2760.0),(5284.0,6214.0),(9154.0,7359.0),(6670.0,4008.0),(11494.0,204.0),(1597.0,7903.0),(6400.0,4772.0)

185,(7057.0,270.0),(3123.0,3710.0),(13064.0,2355.0),(7353.0,2944.0),(6259.0,2466.0),(935.0,4502.0),(11642.0,7806.0),(9175.0,3749.0),(3937.0,6020.0),(3291.0,2502.0),(9138.0,5255.0),(4304.0,3140.0),(5231.0,2936.0),(5544.0,6242.0),(9041.0,7146.0),(6673.0,4023.0),(11161.0,210.0),(1900.0,7926.0),(6383.0,5104.0)

186,(7390.0,270.0),(3183.0,3382.0),(13054.0,2171.0),(7671.0,2933.0),(6486.0,2695.0),(874.0,4174.0),(11535.0,8008.0),(9170.0,3416.0),(4129.0,5971.0),(3468.0,2333.0),(8805.0,5243.0),(4112.0,3298.0),(5527.0,2919.0),(5213.0,6216.0),(8708.0,7146.0),(6374.0,4084.0),(10827.0,216.0),(2233.0,7912.0),(6154.0,5220.0)

187,(7422.0,559.0),(3453.0,3269.0),(13063.0,2337.0),(7997.0,2930.0),(6585.0,2731.0),(823.0,3845.0),(11201.0,8005.0),(9166.0,3083.0),(3940.0,6069.0),(3456.0,2499.0),(8472.0,5232.0),(4386.0,3346.0),(5449.0,3094.0),(5036.0,6417.0),(8375.0,7146.0),(6042.0,4051.0),(10494.0,222.0),(2500.0,7837.0),(5821.0,5220.0)

188,(7243.0,729.0),(3460.0,3036.0),(13055.0,2190.0),(8328.0,2968.0),(6307.0,2557.0),(848.0,4008.0),(10868.0,8002.0),(9162.0,2830.0),(3692.0,5907.0),(3786.0,2492.0),(8139.0,5220.0),(4719.0,3334.0),(5580.0,3392.0),(5029.0,6750.0),(8041.0,7145.0),(5870.0,3843.0),(10161.0,228.0),(2492.0,7504.0),(5488.0,5213.0)

189,(6910.0,748.0),(3291.0,3186.0),(13062.0,2318.0),(8118.0,2877.0),(6511.0,2714.0),(899.0,4337.0),(10535.0,8000.0),(9167.0,3163.0),(3588.0,5590.0),(4105.0,2468.0),(8430.0,5230.0),(4945.0,3228.0),(5580.0,3725.0),(5022.0,7083.0),(7708.0,7136.0),(5907.0,3513.0),(9828.0,233.0),(2485.0,7170.0),(5196.0,5292.0)

190,(6720.0,571.0),(3214.0,3361.0),(13056.0,2208.0),(8106.0,2702.0),(6268.0,2496.0),(848.0,4009.0),(10201.0,7997.0),(9171.0,3496.0),(3485.0,5273.0),(4099.0,2138.0),(8763.0,5242.0),(5158.0,3259.0),(5301.0,3780.0),(4714.0,7110.0),(7380.0,7132.0),(5947.0,3182.0),(9494.0,239.0),(2212.0,7084.0),(4944.0,5494.0)

191,(6620.0,273.0),(3464.0,3225.0),(12837.0,2098.0),(8037.0,2937.0),(6167.0,2178.0),(790.0,3681.0),(9868.0,7994.0),(9176.0,3830.0),(3178.0,5225.0),(4115.0,2330.0),(9096.0,5253.0),(4935.0,3320.0),(4967.0,3773.0),(4906.0,7110.0),(7431.0,7461.0),(6077.0,2948.0),(9161.0,244.0),(2241.0,6893.0),(4630.0,5606.0)

192,(6542.0,284.0),(3294.0,3350.0),(12504.0,2114.0),(7785.0,2797.0),(6322.0,2305.0),(711.0,3357.0),(9535.0,7992.0),(9184.0,4163.0),(3042.0,5416.0),(4104.0,2662.0),(9166.0,5523.0),(5146.0,3168.0),(4634.0,3761.0),(5239.0,7110.0),(7422.0,7791.0),(6410.0,2937.0),(8828.0,250.0),(2086.0,7074.0),(4604.0,5760.0)

193,(6567.0,280.0),(3162.0,3144.0),(12171.0,2131.0),(7780.0,2780.0),(6577.0,2499.0),(632.0,3033.0),(9201.0,7992.0),(9184.0,4495.0),(3042.0,5265.0),(4117.0,2995.0),(9172.0,5856.0),(5069.0,2917.0),(4301.0,3749.0),(5573.0,7110.0),(7564.0,7960.0),(6743.0,2950.0),(8495.0,270.0),(1753.0,7083.0),(4624.0,5810.0)

194,(6642.0,273.0),(3028.0,2914.0),(11838.0,2145.0),(7606.0,2949.0),(6865.0,2665.0),(706.0,3338.0),(9146.0,7698.0),(9183.0,4498.0),(3042.0,5507.0),(4108.0,2940.0),(9177.0,6190.0),(4736.0,2929.0),(4124.0,3573.0),(5603.0,6819.0),(7898.0,7968.0),(6781.0,2706.0),(8162.0,285.0),(1441.0,7058.0),(4688.0,5585.0)

195,(6726.0,596.0),(2695.0,2910.0),(11505.0,2154.0),(7466.0,3161.0),(6959.0,2689.0),(785.0,3662.0),(9158.0,7934.0),(9168.0,4616.0),(3086.0,5221.0),(3798.0,2952.0),(9385.0,6312.0),(4848.0,2924.0),(3997.0,3340.0),(5590.0,6486.0),(8118.0,7860.0),(6845.0,2418.0),(7829.0,278.0),(1412.0,6982.0),(5002.0,5473.0)

196,(6725.0,593.0),(2362.0,2905.0),(11435.0,2156.0),(7495.0,3493.0),(6738.0,2762.0),(755.0,3538.0),(8886.0,7987.0),(9212.0,4392.0),(3419.0,5235.0),(3845.0,3282.0),(9718.0,6302.0),(5180.0,2930.0),(4039.0,3343.0),(5583.0,6338.0),(7978.0,7652.0),(6852.0,2558.0),(7496.0,271.0),(1742.0,6934.0),(5076.0,5544.0)

197,(6642.0,270.0),(2028.0,2900.0),(11769.0,2147.0),(7714.0,3635.0),(7065.0,2830.0),(676.0,3214.0),(8553.0,7981.0),(9545.0,4415.0),(3752.0,5220.0),(3566.0,3330.0),(9836.0,6059.0),(5509.0,2922.0),(4189.0,3134.0),(5597.0,6671.0),(7646.0,7617.0),(7078.0,2720.0),(7704.0,275.0),(2072.0,6886.0),(5214.0,5271.0)

198,(6310.0,288.0),(1798.0,2777.0),(12102.0,2134.0),(7995.0,3576.0),(6870.0,2921.0),(766.0,3009.0),(8220.0,7975.0),(9877.0,4437.0),(4014.0,5204.0),(3235.0,3358.0),(10092.0,5942.0),(5841.0,2940.0),(4501.0,3127.0),(5611.0,7004.0),(7431.0,7731.0),(7359.0,2895.0),(8038.0,283.0),(2186.0,6966.0),(5207.0,5010.0)

199,(5977.0,288.0),(1744.0,2448.0),(12435.0,2118.0),(8274.0,3708.0),(6541.0,2935.0),(1098.0,2974.0),(8118.0,7742.0),(9998.0,4648.0),(3682.0,5224.0),(3270.0,3186.0),(10060.0,5946.0),(6173.0,2945.0),(4594.0,2934.0),(5608.0,7172.0),(7413.0,7850.0),(7446.0,2785.0),(8371.0,279.0),(1969.0,7219.0),(4964.0,5148.0)

200,(5832.0,476.0),(1690.0,2119.0),(12767.0,2101.0),(8282.0,3742.0),(6516.0,2809.0),(1429.0,2939.0),(8118.0,7926.0),(9984.0,4981.0),(3512.0,5356.0),(3462.0,3056.0),(9833.0,6091.0),(6507.0,2934.0),(4632.0,3082.0),(5613.0,7129.0),(7440.0,7520.0),(7436.0,2452.0),(8703.0,255.0),(1741.0,7462.0),(4927.0,5224.0)

201,(5819.0,808.0),(1651.0,1788.0),(13047.0,2037.0),(8315.0,3885.0),(6516.0,2725.0),(1761.0,2904.0),(8118.0,7688.0),(9915.0,5272.0),(3615.0,5673.0),(3197.0,2930.0),(9933.0,6305.0),(6743.0,2884.0),(4558.0,3335.0),(5592.0,7110.0),(7389.0,7191.0),(7688.0,2346.0),(8846.0,523.0),(1575.0,7725.0),(4702.0,5108.0)

202,(5922.0,1095.0),(1657.0,1855.0),(13030.0,1704.0),(8170.0,3627.0),(6744.0,2887.0),(1870.0,3169.0),(8085.0,7356.0),(9847.0,5291.0),(3719.0,5990.0),(3253.0,2785.0),(10266.0,6318.0),(6849.0,2668.0),(4432.0,3242.0),(5259.0,7110.0),(7380.0,6858.0),(7728.0,2058.0),(8959.0,836.0),(1587.0,7819.0),(4761.0,4883.0)

203,(6167.0,1278.0),(1656.0,1844.0),(13013.0,1371.0),(7850.0,3533.0),(7021.0,2875.0),(1934.0,3496.0),(8072.0,7024.0),(9775.0,5288.0),(3755.0,6021.0),(3456.0,2678.0),(10600.0,6318.0),(6541.0,2723.0),(4516.0,2933.0),(5024.0,7013.0),(7380.0,6525.0),(7557.0,1900.0),(9073.0,1150.0),(1535.0,7490.0),(4977.0,4788.0)

204,(6442.0,1191.0),(1683.0,1519.0),(12996.0,1038.0),(7825.0,3532.0),(6910.0,2798.0),(1999.0,3823.0),(8094.0,6692.0),(9443.0,5270.0),(3424.0,5986.0),(3299.0,2502.0),(10933.0,6318.0),(6606.0,2940.0),(4183.0,2919.0),(5031.0,6680.0),(7308.0,6265.0),(7678.0,1892.0),(9186.0,1463.0),(1471.0,7163.0),(4968.0,4674.0)

205,(6566.0,882.0),(1788.0,1203.0),(12979.0,705.0),(7535.0,3620.0),(6583.0,2731.0),(2064.0,4150.0),(8113.0,6359.0),(9162.0,5204.0),(3099.0,5958.0),(3123.0,2649.0),(11266.0,6318.0),(6709.0,2790.0),(3851.0,2945.0),(5038.0,6347.0),(6974.0,6275.0),(8012.0,1899.0),(9198.0,1795.0),(1449.0,7074.0),(5215.0,4572.0)

206,(6582.0,565.0),(1813.0,1128.0),(12993.0,995.0),(7447.0,3916.0),(6306.0,2556.0),(1872.0,4266.0),(8103.0,6537.0),(9162.0,4870.0),(3431.0,5986.0),(3341.0,2795.0),(11427.0,6536.0),(6963.0,2690.0),(4075.0,2919.0),(5034.0,6513.0),(6643.0,6264.0),(8295.0,1860.0),(9198.0,2128.0),(1408.0,6992.0),(5546.0,4605.0)

207,(6527.0,334.0),(1708.0,1444.0),(13011.0,1328.0),(7274.0,4086.0),(6523.0,2719.0),(1539.0,4266.0),(8363.0,6632.0),(9162.0,4786.0),(3762.0,6024.0),(3456.0,2589.0),(11503.0,6860.0),(7267.0,2822.0),(4118.0,3003.0),(5027.0,6847.0),(6427.0,6221.0),(8274.0,1528.0),(9191.0,2461.0),(1347.0,6665.0),(5535.0,4356.0)

208,(6605.0,659.0),(1649.0,1766.0),(13028.0,1661.0),(6991.0,4009.0),(6749.0,2901.0),(1207.0,4279.0),(8472.0,6405.0),(9162.0,5119.0),(4088.0,6079.0),(3473.0,2257.0),(11576.0,7185.0),(7402.0,3055.0),(4106.0,3332.0),(4951.0,7110.0),(6709.0,6277.0),(8143.0,1321.0),(9158.0,2790.0),(1457.0,6460.0),(5215.0,4288.0)

209,(6530.0,972.0),(1665.0,1934.0),(13045.0,1994.0),(7074.0,3688.0),(6516.0,2906.0),(904.0,4357.0),(8251.0,6288.0),(9165.0,5453.0),(4410.0,6159.0),(3294.0,2103.0),(11611.0,7517.0),(7409.0,3046.0),(4394.0,3295.0),(4640.0,7084.0),(6382.0,6212.0),(7810.0,1344.0),(8833.0,2863.0),(1787.0,6415.0),(4995.0,4422.0)

210,(6404.0,1278.0),(1643.0,1709.0),(13062.0,2326.0),(6914.0,3686.0),(6657.0,2746.0),(975.0,4683.0),(8317.0,6291.0),(9170.0,5786.0),(4578.0,5979.0),(3197.0,2070.0),(11646.0,7848.0),(7300.0,3200.0),(4512.0,2996.0),(4595.0,6754.0),(6050.0,6224.0),(7478.0,1366.0),(8508.0,2935.0),(2118.0,6370.0),(5038.0,4704.0)

211,(6070.0,1278.0),(1671.0,1990.0),(13055.0,2200.0),(6601.0,3795.0),(6852.0,2555.0),(1252.0,4768.0),(8468.0,6472.0),(9176.0,6119.0),(4581.0,5968.0),(3530.0,2070.0),(11645.0,7839.0),(7151.0,3498.0),(4788.0,2927.0),(4549.0,6423.0),(5717.0,6239.0),(7758.0,1347.0),(8521.0,2932.0),(2448.0,6325.0),(5344.0,4743.0)

212,(5737.0,1278.0),(1638.0,1659.0),(12829.0,2098.0),(6540.0,3852.0),(6845.0,2421.0),(1565.0,4807.0),(8468.0,6478.0),(9177.0,6183.0),(4618.0,6205.0),(3864.0,2070.0),(11610.0,7508.0),(6872.0,3550.0),(5120.0,2926.0),(4423.0,6163.0),(5586.0,6049.0),(8090.0,1325.0),(8846.0,2860.0),(2776.0,6277.0),(5583.0,4681.0)

213,(5644.0,1488.0),(1742.0,1342.0),(12496.0,2115.0),(6873.0,3852.0),(6779.0,2618.0),(1549.0,5140.0),(8323.0,6292.0),(9172.0,5849.0),(4948.0,6251.0),(4107.0,2179.0),(11575.0,7176.0),(6571.0,3680.0),(5160.0,3001.0),(4144.0,6029.0),(5598.0,5716.0),(8271.0,1475.0),(9163.0,2779.0),(3012.0,6041.0),(5626.0,5007.0)

214,(5854.0,1731.0),(1847.0,1026.0),(12163.0,2131.0),(6986.0,4035.0),(6492.0,2449.0),(1837.0,5175.0),(8110.0,6409.0),(9166.0,5516.0),(5238.0,6174.0),(4383.0,2250.0),(11500.0,6851.0),(6831.0,3550.0),(5166.0,3333.0),(4104.0,5700.0),(5610.0,5383.0),(8291.0,1808.0),(9074.0,2809.0),(2943.0,6110.0),(5495.0,5213.0)

215,(5958.0,1825.0),(1754.0,1305.0),(11830.0,2145.0),(6675.0,4037.0),(6299.0,2210.0),(2170.0,5166.0),(8090.0,6742.0),(9162.0,5183.0),(5215.0,5846.0),(4463.0,2250.0),(11425.0,6527.0),(7093.0,3630.0),(5166.0,3667.0),(3860.0,5896.0),(5624.0,5050.0),(8064.0,1900.0),(8749.0,2881.0),(2712.0,6350.0),(5287.0,5091.0)

216,(5711.0,1601.0),(1649.0,1621.0),(11497.0,2154.0),(6496.0,3852.0),(6094.0,2019.0),(2223.0,5013.0),(8068.0,7074.0),(9162.0,4849.0),(5076.0,5547.0),(4129.0,2250.0),(11350.0,6202.0),(7002.0,3950.0),(5386.0,3780.0),(3695.0,6165.0),(5522.0,4875.0),(7731.0,1890.0),(8424.0,2953.0),(2492.0,6600.0),(5526.0,4875.0)

217,(5618.0,1296.0),(1667.0,1951.0),(11337.0,2295.0),(6399.0,4084.0),(6114.0,2058.0),(2302.0,5175.0),(7802.0,7139.0),(9181.0,4517.0),(5120.0,5681.0),(4110.0,2393.0),(11277.0,5877.0),(6752.0,4050.0),(5598.0,3918.0),(3631.0,6492.0),(5573.0,4647.0),(7734.0,2213.0),(8118.0,2927.0),(2273.0,6851.0),(5776.0,4686.0)

218,(5930.0,1278.0),(1405.0,2035.0),(11552.0,2453.0),(6625.0,4056.0),(6082.0,2178.0),(1969.0,5172.0),(7469.0,7130.0),(9403.0,4405.0),(5238.0,5983.0),(3932.0,2250.0),(11204.0,5552.0),(6699.0,3852.0),(5598.0,4052.0),(3567.0,6819.0),(5382.0,4508.0),(7534.0,2364.0),(8090.0,2654.0),(2160.0,7079.0),(5963.0,4411.0)

219,(6263.0,1278.0),(1073.0,2059.0),(11885.0,2463.0),(6294.0,4076.0),(6237.0,2024.0),(1636.0,5181.0),(7624.0,7134.0),(9736.0,4427.0),(5134.0,6238.0),(3599.0,2250.0),(11093.0,5271.0),(7033.0,3853.0),(5362.0,4213.0),(3510.0,7147.0),(5593.0,4386.0),(7442.0,2144.0),(8365.0,2719.0),(2492.0,7110.0),(6025.0,4088.0)

220,(6219.0,1278.0),(740.0,2083.0),(11721.0,2458.0),(5962.0,4050.0),(6120.0,1712.0),(1559.0,4939.0),(7957.0,7143.0),(10005.0,4507.0),(4807.0,6231.0),(3294.0,2222.0),(10760.0,5253.0),(7365.0,3880.0),(5511.0,4356.0),(3518.0,7480.0),(5632.0,4523.0),(7452.0,1810.0),(8697.0,2696.0),(2826.0,7110.0),(6291.0,3913.0)

221,(5954.0,1136.0),(427.0,2082.0),(11387.0,2448.0),(5629.0,4050.0),(6005.0,1523.0),(1743.0,4748.0),(8087.0,7371.0),(9990.0,4840.0),(4529.0,6152.0),(3295.0,2250.0),(10456.0,5207.0),(7420.0,4077.0),(5612.0,4109.0),(3525.0,7813.0),(5839.0,4264.0),(7452.0,1477.0),(9030.0,2674.0),(3159.0,7110.0),(6601.0,3795.0)

222,(5809.0,859.0),(370.0,1754.0),(11328.0,2159.0),(5580.0,3732.0),(6160.0,1817.0),(2068.0,4674.0),(8114.0,7632.0),(9975.0,5173.0),(4620.0,5831.0),(3628.0,2250.0),(10445.0,4874.0),(7479.0,3749.0),(5561.0,4059.0),(3772.0,7913.0),(5896.0,3949.0),(7420.0,1146.0),(9169.0,2734.0),(3492.0,7110.0),(6914.0,3686.0)

223,(5622.0,738.0),(313.0,1426.0),(11661.0,2150.0),(5580.0,3398.0),(6277.0,2129.0),(2098.0,4386.0),(8080.0,7301.0),(9846.0,5411.0),(4539.0,6116.0),(3961.0,2250.0),(10433.0,4541.0),(7704.0,3634.0),(5310.0,4265.0),(4105.0,7928.0),(6070.0,3762.0),(7416.0,813.0),(9178.0,2671.0),(3193.0,7110.0),(7074.0,3688.0)

224,(5288.0,738.0),(279.0,1094.0),(11949.0,2099.0),(5453.0,3100.0),(6419.0,2407.0),(2107.0,4476.0),(8075.0,6969.0),(9846.0,5744.0),(4273.0,6118.0),(4108.0,2422.0),(10400.0,4230.0),(7711.0,3635.0),(4986.0,4304.0),(4438.0,7943.0),(5963.0,3800.0),(7416.0,959.0),(9184.0,2577.0),(2860.0,7110.0),(6991.0,4010.0)

225,(4955.0,738.0),(248.0,762.0),(11933.0,1766.0),(5520.0,2920.0),(6707.0,2575.0),(2142.0,4808.0),(8097.0,6636.0),(9834.0,6077.0),(3945.0,6069.0),(4375.0,2494.0),(10067.0,4230.0),(7579.0,3513.0),(4655.0,4347.0),(4771.0,7953.0),(5931.0,4050.0),(7239.0,1097.0),(9198.0,2244.0),(2527.0,7110.0),(7049.0,3764.0)

226,(4622.0,738.0),(217.0,430.0),(11916.0,1433.0),(5852.0,2941.0),(7011.0,2702.0),(2191.0,5138.0),(8116.0,6303.0),(9919.0,6304.0),(3825.0,6045.0),(4599.0,2620.0),(9885.0,4040.0),(7771.0,3241.0),(4626.0,4620.0),(5104.0,7940.0),(5795.0,4050.0),(6908.0,1063.0),(9198.0,1911.0),(2490.0,7399.0),(6986.0,3648.0)

227,(4288.0,738.0),(245.0,729.0),(11899.0,1100.0),(5767.0,2934.0),(7310.0,2846.0),(2318.0,5108.0),(7806.0,6274.0),(10242.0,6307.0),(4084.0,5885.0),(4624.0,2915.0),(9908.0,4105.0),(7733.0,3295.0),(4651.0,4805.0),(5437.0,7927.0),(5595.0,3897.0),(6873.0,1380.0),(9198.0,1578.0),(2498.0,7733.0),(6678.0,3772.0)

228,(3964.0,738.0),(275.0,1061.0),(11883.0,767.0),(5436.0,2937.0),(7581.0,2955.0),(2177.0,5057.0),(7473.0,6266.0),(10242.0,5974.0),(4376.0,5726.0),(4761.0,2736.0),(9998.0,4230.0),(7541.0,3567.0),(4606.0,4474.0),(5770.0,7924.0),(5365.0,3780.0),(6841.0,1712.0),(9113.0,1259.0),(2666.0,7902.0),(6459.0,3852.0)

229,(4297.0,738.0),(307.0,1392.0),(11893.0,969.0),(5106.0,2921.0),(7906.0,2914.0),(2418.0,5185.0),(7619.0,6269.0),(10242.0,6203.0),(4603.0,5757.0),(5014.0,2663.0),(10331.0,4230.0),(7452.0,3884.0),(4394.0,4272.0),(6104.0,7935.0),(5031.0,3775.0),(6870.0,1412.0),(8999.0,946.0),(2999.0,7902.0),(6668.0,4000.0)

230,(4631.0,738.0),(364.0,1721.0),(11910.0,1302.0),(5011.0,2635.0),(8235.0,2961.0),(2750.0,5203.0),(7953.0,6277.0),(10023.0,6308.0),(4483.0,5670.0),(4780.0,2526.0),(10179.0,4230.0),(7782.0,3888.0),(4209.0,4271.0),(6437.0,7946.0),(4698.0,3763.0),(6606.0,1337.0),(8886.0,633.0),(3333.0,7902.0),(6960.0,4050.0)

231,(4964.0,738.0),(422.0,2049.0),(11926.0,1635.0),(5036.0,2806.0),(8135.0,2953.0),(3042.0,5261.0),(8118.0,6113.0),(9823.0,6181.0),(4188.0,5824.0),(4590.0,2361.0),(9917.0,4128.0),(8115.0,3888.0),(4524.0,4328.0),(6750.0,7935.0),(4365.0,3752.0),(6284.0,1422.0),(8772.0,319.0),(3524.0,7763.0),(7291.0,4088.0)

232,(5297.0,738.0),(490.0,2375.0),(11943.0,1968.0),(5278.0,2941.0),(8086.0,2641.0),(2955.0,5517.0),(8118.0,5780.0),(9832.0,6101.0),(4104.0,5746.0),(4367.0,2250.0),(9947.0,4211.0),(8286.0,3758.0),(4348.0,4252.0),(6750.0,7602.0),(4032.0,3745.0),(5995.0,1472.0),(8487.0,271.0),(3517.0,7430.0),(7450.0,3898.0)

233,(5631.0,738.0),(560.0,2701.0),(11792.0,2146.0),(5481.0,2928.0),(7763.0,2628.0),(2624.0,5553.0),(8118.0,5446.0),(9675.0,6303.0),(4155.0,6073.0),(4104.0,2164.0),(9930.0,4166.0),(8045.0,3590.0),(4133.0,4404.0),(6750.0,7268.0),(3699.0,3751.0),(6324.0,1415.0),(8262.0,395.0),(3522.0,7110.0),(7130.0,3861.0)

234,(5832.0,605.0),(633.0,3023.0),(11459.0,2155.0),(5473.0,3135.0),(7842.0,2903.0),(2292.0,5590.0),(8111.0,5113.0),(9342.0,6313.0),(3926.0,6030.0),(3848.0,2070.0),(10219.0,4230.0),(7752.0,3605.0),(4117.0,4737.0),(6745.0,6935.0),(3365.0,3757.0),(6603.0,1338.0),(8262.0,728.0),(3856.0,7110.0),(6963.0,4050.0)

235,(5832.0,303.0),(965.0,2988.0),(11481.0,2155.0),(5514.0,3203.0),(8118.0,2898.0),(1961.0,5626.0),(8089.0,4781.0),(9176.0,6488.0),(4126.0,5957.0),(3515.0,2070.0),(10291.0,4230.0),(7732.0,3625.0),(4127.0,5065.0),(6738.0,6602.0),(3128.0,3681.0),(6495.0,1371.0),(8262.0,1062.0),(4189.0,7110.0),(6630.0,4055.0)

236,(5832.0,637.0),(1296.0,2953.0),(11814.0,2145.0),(5403.0,2944.0),(8288.0,2724.0),(1708.0,5748.0),(8095.0,4875.0),(9169.0,6821.0),(4108.0,5887.0),(3294.0,2182.0),(9957.0,4230.0),(7486.0,3717.0),(4050.0,5201.0),(6719.0,6279.0),(3198.0,3364.0),(6822.0,1346.0),(8262.0,1232.0),(4522.0,7110.0),(6390.0,4177.0)

237,(5801.0,905.0),(1628.0,2918.0),(12147.0,2132.0),(5357.0,2951.0),(8620.0,2701.0),(1939.0,5854.0),(8117.0,5208.0),(9161.0,7154.0),(4002.0,5774.0),(3467.0,2335.0),(9763.0,4041.0),(7483.0,3732.0),(4150.0,5123.0),(6392.0,6214.0),(3460.0,3241.0),(7155.0,1358.0),(8262.0,899.0),(4662.0,7321.0),(6390.0,4510.0)

238,(5802.0,898.0),(1795.0,2757.0),(12480.0,2115.0),(5089.0,2919.0),(8953.0,2679.0),(2271.0,5883.0),(8439.0,5231.0),
(9149.0,7488.0),(3749.0,5991.0),(3289.0,2502.0),(9431.0,4066.0),(7721.0,3635.0),(4066.0,5141.0),(6678.0,6271.0),(3311.0,
3348.0),(7446.0,1331.0),(8262.0,565.0),(4690.0,7653.0),(6406.0,4843.0)

239,(5832.0,643.0),(1741.0,2428.0),(12813.0,2099.0),(5420.0,2940.0),(9185.0,2557.0),(2603.0,5913.0),(8772.0,5242.0),
(9150.0,7478.0),(3424.0,5986.0),(3107.0,2344.0),(9184.0,4167.0),(7622.0,3627.0),(4172.0,5181.0),(7010.0,6274.0),(3164.0,
3160.0),(7432.0,1230.0),(8317.0,283.0),(4686.0,7954.0),(6091.0,4860.0)

240,(5832.0,310.0),(1687.0,2099.0),(12953.0,2092.0),(5537.0,2709.0),(9191.0,2454.0),(2935.0,5943.0),(9105.0,5254.0),
(9161.0,7146.0),(3093.0,5960.0),(3154.0,2070.0),(9183.0,4499.0),(7504.0,3601.0),(3925.0,5209.0),(7343.0,6265.0),(3045.0,
2914.0),(7424.0,1176.0),(8649.0,259.0),(4353.0,7939.0),(5758.0,4860.0)

End

2. 第五问中白天连续 4 个小时内的各警车的正常巡逻位置数据

10, 0.164, 0.339, 0.272, 1.200, 2.099, 1.896, 17.05988242601705
0,(7434.0,2376.0),(1818.0,2898.0),(5364.0,2952.0),(9882.0,4032.0),(5274.0,5202.0),(1224.0,5814.0),(7380.0,6264.0),(1
4040.0,6840.0),(10296.0,7182.0),(3528.0,7902.0)
1,(7443.0,2709.0),(2151.0,2902.0),(5535.0,3238.0),(9911.0,4114.0),(5230.0,5199.0),(1147.0,5489.0),(7397.0,5930.0),(1
3712.0,6903.0),(10321.0,7514.0),(3520.0,7568.0)
2,(7413.0,2949.0),(2484.0,2907.0),(5452.0,3099.0),(9941.0,4196.0),(4928.0,5217.0),(1072.0,5164.0),(7415.0,5597.0),(1
3384.0,6967.0),(10316.0,7452.0),(3512.0,7235.0)
3,(7438.0,2974.0),(2818.0,2911.0),(5522.0,2920.0),(9935.0,4180.0),(4890.0,5184.0),(1007.0,4837.0),(7433.0,5264.0),(1
3546.0,6936.0),(10234.0,7180.0),(3302.0,7110.0)
4,(7348.0,3130.0),(3132.0,2896.0),(5237.0,2937.0),(9905.0,4097.0),(4688.0,5027.0),(1038.0,4993.0),(7416.0,5579.0),(1
3873.0,6872.0),(9901.0,7169.0),(2968.0,7110.0)
5,(7307.0,2923.0),(3140.0,2980.0),(5123.0,2926.0),(10146.0,4230.0),(4693.0,5056.0),(1200.0,5197.0),(7398.0,5912.0),(
14203.0,6840.0),(9568.0,7158.0),(2635.0,7110.0)
6,(7252.0,2899.0),(3180.0,3310.0),(5013.0,2653.0),(10424.0,4287.0),(4919.0,5184.0),(1533.0,5184.0),(7380.0,6245.0),(
14486.0,6937.0),(9234.0,7148.0),(2302.0,7092.0)
7,(7370.0,2906.0),(3443.0,3307.0),(5032.0,2789.0),(10435.0,4620.0),(5252.0,5200.0),(1562.0,4865.0),(7065.0,6272.0),(
14425.0,6850.0),(9152.0,7406.0),(2187.0,6965.0)
8,(7492.0,2977.0),(3265.0,3186.0),(5158.0,3028.0),(10447.0,4954.0),(5420.0,4951.0),(1815.0,4731.0),(6732.0,6281.0),(
14471.0,6916.0),(9148.0,7739.0),(2148.0,7017.0)
9,(7808.0,2898.0),(3136.0,2952.0),(5135.0,3332.0),(10408.0,5239.0),(5608.0,4771.0),(2126.0,4679.0),(6738.0,6615.0),(
14177.0,6840.0),(9157.0,7911.0),(1926.0,7265.0)
10,(7485.0,2979.0),(3121.0,2619.0),(4946.0,3219.0),(10075.0,5252.0),(5401.0,4581.0),(2169.0,5009.0),(6745.0,6948.0),
(14235.0,6840.0),(9215.0,7650.0),(1698.0,7508.0)
11,(7223.0,2798.0),(3329.0,2502.0),(5159.0,3268.0),(9940.0,5264.0),(5068.0,4572.0),(2370.0,5181.0),(6614.0,7136.0),(
14504.0,6963.0),(9548.0,7650.0),(1583.0,7788.0)
12,(6915.0,2678.0),(3456.0,2708.0),(5040.0,3177.0),(9934.0,5266.0),(5002.0,4321.0),(2177.0,5056.0),(6525.0,7263.0),(
14373.0,6840.0),(9624.0,7936.0),(1579.0,7756.0)
13,(6609.0,2736.0),(3223.0,2782.0),(5011.0,3341.0),(9605.0,5279.0),(5028.0,4572.0),(2319.0,5109.0),(6669.0,7140.0),(
14040.0,6840.0),(9352.0,7992.0),(1523.0,7428.0)
14,(6714.0,2805.0),(3115.0,2531.0),(5166.0,3508.0),(9272.0,5261.0),(5362.0,4574.0),(2191.0,5137.0),(6497.0,7129.0),(
14373.0,6840.0),(9154.0,7849.0),(1459.0,7101.0)
15,(6937.0,2901.0),(3101.0,2198.0),(5166.0,3718.0),(9162.0,5033.0),(5597.0,4733.0),(2142.0,4808.0),(6488.0,7152.0),(
14504.0,6963.0),(9277.0,7650.0),(1421.0,7018.0)

16,(7251.0,2898.0),(3294.0,2077.0),(5166.0,3384.0),(9162.0,4699.0),(5450.0,4925.0),(2107.0,4476.0),(6786.0,7144.0),(14522.0,6989.0),(9595.0,7666.0),(1632.0,7087.0)

17,(7385.0,3079.0),(3454.0,2250.0),(4888.0,3345.0),(9162.0,4957.0),(5276.0,5183.0),(2099.0,4387.0),(7120.0,7135.0),(14341.0,6840.0),(9629.0,7985.0),(1965.0,7077.0)

18,(7426.0,3022.0),(3460.0,2441.0),(4555.0,3335.0),(9162.0,5221.0),(5068.0,5441.0),(2131.0,4718.0),(7306.0,7130.0),(14071.0,6840.0),(9594.0,7654.0),(2297.0,7092.0)

19,(7476.0,3277.0),(3332.0,2250.0),(4434.0,3240.0),(9162.0,4887.0),(5263.0,5213.0),(2230.0,5020.0),(6973.0,7139.0),(14405.0,6840.0),(9922.0,7650.0),(2338.0,7096.0)

20,(7505.0,3609.0),(3409.0,2070.0),(4523.0,2934.0),(9162.0,4769.0),(5591.0,5218.0),(2291.0,5174.0),(6859.0,7142.0),(14486.0,6937.0),(10256.0,7650.0),(2165.0,6994.0)

21,(7478.0,3294.0),(3742.0,2070.0),(4624.0,2919.0),(9162.0,5102.0),(5631.0,4911.0),(2355.0,5180.0),(7192.0,7133.0),(14540.0,7015.0),(10312.0,7393.0),(2170.0,6988.0)

22,(7433.0,2969.0),(4075.0,2070.0),(4596.0,2587.0),(9162.0,5075.0),(5389.0,4976.0),(2023.0,5170.0),(7402.0,7272.0),(14309.0,6840.0),(10174.0,7178.0),(2241.0,6892.0)

23,(7418.0,2954.0),(4077.0,1746.0),(4835.0,2534.0),(9162.0,4742.0),(5281.0,5141.0),(1689.0,5180.0),(7451.0,7590.0),(14103.0,6840.0),(9841.0,7167.0),(2125.0,7075.0)

24,(7331.0,3153.0),(4068.0,1413.0),(5020.0,2718.0),(9162.0,4914.0),(5286.0,5098.0),(1356.0,5191.0),(7400.0,7260.0),(14428.0,6855.0),(9508.0,7156.0),(2457.0,7107.0)

25,(7360.0,2896.0),(4370.0,1344.0),(4706.0,2736.0),(9162.0,5248.0),(5293.0,5056.0),(1068.0,5146.0),(7180.0,7133.0),(14527.0,7151.0),(9174.0,7146.0),(2178.0,7080.0)

26,(7080.0,2721.0),(4700.0,1299.0),(4627.0,2970.0),(9162.0,4930.0),(5562.0,4870.0),(1003.0,4819.0),(6847.0,7143.0),(14487.0,7482.0),(9150.0,7466.0),(1927.0,7264.0)

27,(7350.0,2886.0),(4992.0,1305.0),(4642.0,3303.0),(9162.0,4726.0),(5562.0,4609.0),(1253.0,4768.0),(6750.0,7381.0),(14446.0,7813.0),(9293.0,7650.0),(1699.0,7507.0)

28,(7446.0,2797.0),(5036.0,1635.0),(4337.0,3348.0),(9162.0,5060.0),(5520.0,4356.0),(1585.0,4783.0),(6750.0,7714.0),(14430.0,7947.0),(9597.0,7682.0),(1539.0,7514.0)

29,(7436.0,2464.0),(5185.0,1908.0),(4120.0,3249.0),(9299.0,5263.0),(5380.0,4577.0),(1911.0,4710.0),(6750.0,7863.0),(14470.0,7616.0),(9627.0,7969.0),(1476.0,7187.0)

30,(7441.0,2620.0),(5508.0,1918.0),(4104.0,2921.0),(9632.0,5280.0),(5583.0,4460.0),(2138.0,4776.0),(6750.0,7530.0),(14511.0,7286.0),(9582.0,7650.0),(1383.0,6870.0)

31,(7450.0,2953.0),(5522.0,2251.0),(4104.0,2803.0),(9960.0,5259.0),(5377.0,4318.0),(2185.0,5105.0),(6750.0,7197.0),(14506.0,6966.0),(9249.0,7650.0),(1325.0,6542.0)

32,(7742.0,2915.0),(5763.0,2278.0),(4105.0,2470.0),(9974.0,5196.0),(5051.0,4299.0),(2197.0,4987.0),(6468.0,7127.0),(14239.0,6840.0),(9152.0,7421.0),(1361.0,6745.0)

33,(7558.0,2961.0),(6080.0,2179.0),(4100.0,2140.0),(9989.0,4863.0),(5001.0,4572.0),(2449.0,5187.0),(6135.0,7120.0),(14174.0,6840.0),(9163.0,7088.0),(1444.0,7065.0)

34,(7320.0,3168.0),(6159.0,2142.0),(4115.0,2328.0),(10004.0,4530.0),(5334.0,4572.0),(2782.0,5205.0),(5801.0,7113.0),(14469.0,6913.0),(9170.0,6755.0),(1520.0,7090.0)

35,(7457.0,3054.0),(6164.0,2142.0),(4104.0,2660.0),(9996.0,4694.0),(5543.0,4356.0),(3115.0,5223.0),(5598.0,7251.0),(14427.0,6853.0),(9177.0,6422.0),(1853.0,7081.0)

36,(7486.0,3387.0),(6115.0,2061.0),(4026.0,2925.0),(9982.0,5027.0),(5614.0,4077.0),(3448.0,5236.0),(5603.0,7208.0),(14488.0,6940.0),(8951.0,6312.0),(2025.0,7076.0)

37,(7486.0,3717.0),(6093.0,2016.0),(3695.0,2962.0),(10076.0,5252.0),(5703.0,4050.0),(3781.0,5218.0),(5608.0,7166.0),(14207.0,6840.0),(8617.0,6303.0),(1692.0,7085.0)

38,(7425.0,4045.0),(6298.0,2206.0),(3464.0,3080.0),(10409.0,5239.0),(6023.0,4050.0),(4112.0,5217.0),(5339.0,7110.0),

(14205.0,6840.0),(8284.0,6290.0),(1413.0,7003.0)

39,(7462.0,3834.0),(6292.0,2185.0),(3428.0,2981.0),(10173.0,5248.0),(5838.0,4266.0),(4076.0,5105.0),(5038.0,7110.0),
(14487.0,6939.0),(8108.0,6448.0),(1473.0,6973.0)

40,(7618.0,3627.0),(6180.0,1872.0),(3161.0,2922.0),(9846.0,5292.0),(5630.0,4525.0),(4186.0,5217.0),(5371.0,7110.0),(
14408.0,6827.0),(8088.0,6780.0),(1803.0,6925.0)

41,(7503.0,3630.0),(6036.0,1573.0),(3487.0,2984.0),(9846.0,5626.0),(5593.0,4384.0),(4092.0,5044.0),(5612.0,7021.0),(
14209.0,6560.0),(8066.0,7113.0),(2133.0,6877.0)

42,(7381.0,3882.0),(6191.0,1438.0),(3818.0,2949.0),(9846.0,5959.0),(5594.0,4569.0),(3964.0,5207.0),(5598.0,6688.0),(
14010.0,6292.0),(8094.0,7445.0),(2149.0,7015.0)

43,(7049.0,3854.0),(6515.0,1365.0),(4104.0,2869.0),(9811.0,6290.0),(5807.0,4313.0),(3631.0,5228.0),(5584.0,6355.0),(
13811.0,6025.0),(8118.0,7777.0),(1927.0,7264.0)

44,(7322.0,3877.0),(6843.0,1347.0),(4130.0,3077.0),(9845.0,5977.0),(5911.0,4006.0),(3298.0,5230.0),(5357.0,6222.0),(
13612.0,5758.0),(8118.0,7837.0),(1699.0,7507.0)

45,(7418.0,4088.0),(6889.0,1050.0),(4045.0,3343.0),(10174.0,5931.0),(5905.0,3984.0),(3042.0,5296.0),(5047.0,6261.0)
,(13413.0,5490.0),(7955.0,7650.0),(1539.0,7514.0)

46,(7471.0,3790.0),(6764.0,770.0),(3832.0,3191.0),(10242.0,6187.0),(5971.0,4050.0),(3042.0,5386.0),(5238.0,6073.0),(
13182.0,5312.0),(7623.0,7614.0),(1476.0,7187.0)

47,(7492.0,3461.0),(6683.0,1093.0),(3888.0,2941.0),(10039.0,6309.0),(6023.0,4108.0),(2875.0,5210.0),(5165.0,5759.0)
,(13240.0,5298.0),(7428.0,7754.0),(1455.0,7086.0)

48,(7464.0,3129.0),(6711.0,1338.0),(4104.0,2816.0),(9706.0,6302.0),(5953.0,4430.0),(2542.0,5192.0),(5053.0,5455.0),(
13096.0,5416.0),(7417.0,7827.0),(1544.0,7089.0)

49,(7316.0,2852.0),(6886.0,1181.0),(4337.0,2926.0),(9373.0,6312.0),(5763.0,4702.0),(2210.0,5167.0),(4739.0,5567.0),(
13280.0,5694.0),(7436.0,7497.0),(1877.0,7080.0)

50,(7344.0,2940.0),(6812.0,864.0),(4505.0,3086.0),(9177.0,6178.0),(5598.0,4735.0),(2147.0,4850.0),(4604.0,5760.0),(1
3464.0,5972.0),(7386.0,7168.0),(2210.0,7083.0)

51,(7033.0,2824.0),(6713.0,547.0),(4231.0,3136.0),(9171.0,5844.0),(5364.0,4575.0),(2111.0,4519.0),(4550.0,6079.0),(1
3648.0,6249.0),(7087.0,7136.0),(2425.0,7104.0)

52,(6707.0,2756.0),(6595.0,276.0),(4104.0,2892.0),(9166.0,5511.0),(5623.0,4534.0),(2009.0,4266.0),(4310.0,6129.0),(1
3833.0,6527.0),(6754.0,7145.0),(2097.0,7082.0)

53,(6850.0,2527.0),(6517.0,287.0),(4127.0,3054.0),(9084.0,5253.0),(5834.0,4277.0),(1675.0,4266.0),(4115.0,5915.0),(1
4017.0,6805.0),(6498.0,7183.0),(1869.0,7325.0)

54,(6966.0,2333.0),(6592.0,277.0),(4068.0,3345.0),(8751.0,5241.0),(5832.0,4050.0),(1342.0,4266.0),(4104.0,5792.0),(1
4332.0,6840.0),(6529.0,7131.0),(1641.0,7569.0)

55,(7298.0,2363.0),(6552.0,439.0),(3735.0,3330.0),(8418.0,5230.0),(5502.0,4073.0),(1013.0,4316.0),(3913.0,6042.0),(1
4528.0,6997.0),(6637.0,7138.0),(1593.0,7870.0)

56,(7630.0,2352.0),(6614.0,762.0),(3446.0,3295.0),(8115.0,5187.0),(5325.0,4297.0),(946.0,4550.0),(3846.0,6052.0),(14
276.0,6840.0),(6655.0,7139.0),(1569.0,7674.0)

57,(7518.0,2366.0),(6490.0,1071.0),(3477.0,2985.0),(8094.0,4854.0),(5600.0,4306.0),(1116.0,4759.0),(4153.0,6065.0),(
14136.0,6840.0),(6512.0,7130.0),(1620.0,7937.0)

58,(7443.0,2128.0),(6296.0,1278.0),(3809.0,2950.0),(8222.0,4662.0),(5605.0,3974.0),(1449.0,4780.0),(4008.0,5955.0),(
14447.0,6882.0),(6492.0,7166.0),(1953.0,7923.0)

59,(7447.0,2020.0),(5963.0,1278.0),(4104.0,2879.0),(8555.0,4662.0),(5598.0,3915.0),(1556.0,5004.0),(3726.0,6012.0),(
14449.0,6884.0),(6772.0,7145.0),(2286.0,7910.0)

60,(7434.0,2353.0),(5629.0,1278.0),(4104.0,2546.0),(8889.0,4662.0),(5595.0,4053.0),(1394.0,5189.0),(3978.0,5795.0),(
14509.0,6971.0),(7105.0,7135.0),(2499.0,7784.0)

61,(7445.0,2065.0),(5686.0,1579.0),(4119.0,2286.0),(9162.0,4722.0),(5364.0,4211.0),(1076.0,5183.0),(4104.0,5856.0),(14244.0,6840.0),(7438.0,7129.0),(2491.0,7451.0)

62,(7627.0,1896.0),(5934.0,1803.0),(4104.0,2619.0),(9162.0,5055.0),(5508.0,4356.0),(1010.0,4856.0),(4389.0,5719.0),(14168.0,6840.0),(7771.0,7138.0),(2484.0,7117.0)

63,(7483.0,1905.0),(6121.0,2072.0),(4104.0,2879.0),(9294.0,5262.0),(5612.0,4112.0),(941.0,4530.0),(4608.0,5771.0),(14466.0,6908.0),(8022.0,7144.0),(2160.0,7079.0)

64,(7691.0,1892.0),(6096.0,2171.0),(4129.0,3067.0),(9627.0,5280.0),(5886.0,4050.0),(1033.0,4312.0),(4471.0,5677.0),(14430.0,6858.0),(7689.0,7136.0),(2262.0,6865.0)

65,(7448.0,2001.0),(6236.0,2396.0),(4108.0,3397.0),(9956.0,5260.0),(5886.0,3907.0),(1272.0,4181.0),(4175.0,5830.0),(14491.0,6945.0),(7380.0,7104.0),(2056.0,7126.0)

66,(7435.0,2334.0),(6226.0,2362.0),(4138.0,3728.0),(9919.0,5271.0),(6107.0,4057.0),(1249.0,3849.0),(4187.0,6092.0),(14213.0,6840.0),(7380.0,6771.0),(1828.0,7369.0)

67,(7445.0,2084.0),(6258.0,2142.0),(4111.0,3427.0),(9850.0,5290.0),(6393.0,4036.0),(1014.0,3753.0),(4506.0,6188.0),(13882.0,6870.0),(7380.0,6437.0),(1600.0,7613.0)

68,(7608.0,1897.0),(6032.0,2203.0),(4354.0,3348.0),(9909.0,5273.0),(6276.0,3920.0),(829.0,3889.0),(4836.0,6235.0),(13555.0,6934.0),(7380.0,6423.0),(1512.0,7372.0)

69,(7730.0,2109.0),(6212.0,2142.0),(4020.0,3342.0),(9708.0,5284.0),(6084.0,4055.0),(881.0,4218.0),(4913.0,6246.0),(13227.0,6991.0),(7380.0,6756.0),(1505.0,7090.0)

70,(7842.0,2343.0),(6229.0,2001.0),(4015.0,3341.0),(9375.0,5267.0),(6416.0,4082.0),(1108.0,4298.0),(4583.0,6201.0),(12895.0,7026.0),(7380.0,7090.0),(1838.0,7081.0)

71,(8175.0,2355.0),(6108.0,1690.0),(4165.0,3133.0),(9281.0,5262.0),(6665.0,3981.0),(1439.0,4266.0),(4591.0,5935.0),(12564.0,7060.0),(7675.0,7135.0),(2145.0,7021.0)

72,(8281.0,2111.0),(6018.0,1544.0),(4498.0,3149.0),(9614.0,5279.0),(6440.0,3852.0),(1772.0,4266.0),(4566.0,5629.0),(12232.0,7095.0),(8008.0,7144.0),(2139.0,6876.0)

73,(8169.0,1903.0),(6259.0,1713.0),(4268.0,3348.0),(9846.0,5393.0),(6709.0,3852.0),(2084.0,4248.0),(4880.0,5517.0),(11901.0,7129.0),(8082.0,6868.0),(1810.0,6924.0)

74,(7835.0,1893.0),(6592.0,1721.0),(4131.0,3181.0),(9846.0,5726.0),(7016.0,3873.0),(2019.0,3921.0),(5086.0,5596.0),(11569.0,7164.0),(8095.0,6675.0),(1480.0,6972.0)

75,(7730.0,2109.0),(6848.0,1642.0),(4167.0,2918.0),(9846.0,5892.0),(7137.0,4069.0),(1970.0,3677.0),(5130.0,5369.0),(11236.0,7168.0),(8073.0,7008.0),(1358.0,6731.0)

76,(7842.0,2343.0),(6836.0,1347.0),(4500.0,2933.0),(9846.0,5558.0),(7414.0,4156.0),(2035.0,4003.0),(5161.0,5196.0),(10903.0,7173.0),(8084.0,7340.0),(1303.0,6402.0)

77,(8175.0,2355.0),(6508.0,1367.0),(4833.0,2925.0),(9779.0,5288.0),(7403.0,4489.0),(2075.0,4201.0),(4920.0,5286.0),(10569.0,7178.0),(8117.0,7663.0),(1258.0,6072.0)

78,(8508.0,2353.0),(6589.0,1342.0),(5165.0,2933.0),(9446.0,5270.0),(7403.0,4501.0),(2009.0,3874.0),(4821.0,5184.0),(10300.0,7241.0),(8083.0,7332.0),(1233.0,5885.0)

79,(8841.0,2346.0),(6871.0,1394.0),(5029.0,2775.0),(9162.0,5207.0),(7413.0,4168.0),(1872.0,3649.0),(4930.0,5196.0),(10326.0,7573.0),(7918.0,7142.0),(1278.0,6216.0)

80,(9175.0,2340.0),(6840.0,1726.0),(5087.0,2920.0),(9162.0,4874.0),(7443.0,3937.0),(1543.0,3698.0),(4912.0,5184.0),(10075.0,7650.0),(7584.0,7133.0),(1325.0,6545.0)

81,(9180.0,2650.0),(6871.0,1397.0),(4754.0,2928.0),(9162.0,4782.0),(7252.0,4083.0),(1243.0,3772.0),(4692.0,5049.0),(9741.0,7650.0),(7251.0,7131.0),(1384.0,6874.0)

82,(9170.0,2728.0),(6893.0,1065.0),(4509.0,3030.0),(9162.0,5116.0),(6995.0,3988.0),(1266.0,4105.0),(4689.0,5035.0),(9408.0,7650.0),(6918.0,7141.0),(1412.0,7001.0)

83,(9164.0,2773.0),(6768.0,756.0),(4374.0,3322.0),(9165.0,5449.0),(7081.0,3667.0),(1450.0,4266.0),(4534.0,5202.0),(9

147.0,7718.0),(6746.0,6981.0),(1650.0,7086.0)

84,(9192.0,2441.0),(7099.0,737.0),(4504.0,3093.0),(9170.0,5782.0),(6868.0,3549.0),(1783.0,4266.0),(4203.0,5237.0),(9221.0,7992.0),(6739.0,6647.0),(1983.0,7077.0)

85,(9185.0,2570.0),(7399.0,720.0),(4620.0,2868.0),(9176.0,6116.0),(7124.0,3553.0),(2090.0,4294.0),(4516.0,5204.0),(9554.0,7992.0),(6732.0,6314.0),(2316.0,7094.0)

86,(9177.0,2678.0),(7066.0,739.0),(4592.0,2536.0),(9311.0,6314.0),(7273.0,3255.0),(2120.0,4626.0),(4848.0,5184.0),(9602.0,7735.0),(6738.0,6582.0),(2319.0,7094.0)

87,(9170.0,2727.0),(6759.0,789.0),(4291.0,2490.0),(9644.0,6304.0),(7193.0,3414.0),(1833.0,4727.0),(5181.0,5197.0),(9841.0,7650.0),(6745.0,6915.0),(2177.0,6979.0)

88,(9164.0,2774.0),(6678.0,1112.0),(4104.0,2629.0),(9828.0,6133.0),(7063.0,3720.0),(1563.0,4847.0),(5280.0,5152.0),(10175.0,7650.0),(6853.0,7143.0),(2158.0,7003.0)

89,(9175.0,2696.0),(6731.0,1339.0),(4111.0,2962.0),(9846.0,5800.0),(6985.0,4043.0),(1548.0,5180.0),(5557.0,5216.0),(10155.0,7650.0),(7186.0,7133.0),(2253.0,6877.0)

90,(9173.0,2709.0),(6858.0,1537.0),(4301.0,3140.0),(9846.0,5467.0),(6674.0,4029.0),(1877.0,5174.0),(5341.0,5205.0),(9822.0,7650.0),(7401.0,7265.0),(2471.0,6625.0)

91,(8915.0,2844.0),(6840.0,1869.0),(4511.0,3016.0),(9946.0,5263.0),(6795.0,3852.0),(2210.0,5167.0),(5277.0,5177.0),(9605.0,7754.0),(7451.0,7595.0),(2690.0,6375.0)

92,(8590.0,2916.0),(6840.0,2203.0),(4482.0,2932.0),(9877.0,5282.0),(7128.0,3861.0),(2288.0,5078.0),(5073.0,5436.0),(9620.0,7897.0),(7401.0,7267.0),(2633.0,6439.0)

93,(8438.0,2950.0),(6840.0,2107.0),(4149.0,2917.0),(9882.0,5281.0),(7450.0,3896.0),(2141.0,4799.0),(5259.0,5218.0),(9510.0,7650.0),(7187.0,7133.0),(2414.0,6690.0)

94,(8764.0,2878.0),(6840.0,1774.0),(4115.0,2986.0),(9951.0,5261.0),(7436.0,3982.0),(2148.0,4855.0),(5252.0,5200.0),(9176.0,7650.0),(6854.0,7143.0),(2201.0,6946.0)

95,(9089.0,2806.0),(6867.0,1442.0),(4109.0,3315.0),(10283.0,5244.0),(7397.0,4101.0),(2214.0,5167.0),(4930.0,5196.0),(9159.0,7950.0),(6521.0,7130.0),(2210.0,6934.0)

96,(9162.0,2785.0),(6891.0,1109.0),(4403.0,3345.0),(10299.0,5243.0),(7067.0,4060.0),(2285.0,5075.0),(4744.0,5412.0),(9146.0,7700.0),(6188.0,7121.0),(2125.0,7048.0)

97,(9244.0,2776.0),(6785.0,798.0),(4736.0,3335.0),(9977.0,5255.0),(7011.0,3899.0),(2185.0,5103.0),(4566.0,5482.0),(9154.0,7367.0),(5855.0,7114.0),(1805.0,7082.0)

98,(9573.0,2725.0),(6877.0,1022.0),(4947.0,3211.0),(10310.0,5243.0),(6803.0,4050.0),(2343.0,5133.0),(4876.0,5360.0),(9050.0,7146.0),(5604.0,7203.0),(1471.0,7091.0)

99,(9902.0,2673.0),(6874.0,1349.0),(4787.0,3338.0),(10643.0,5247.0),(6885.0,4050.0),(2150.0,4876.0),(4917.0,5313.0),(8717.0,7146.0),(5598.0,7246.0),(1472.0,7168.0)

100,(10143.0,2634.0),(6706.0,1337.0),(4635.0,3140.0),(10976.0,5265.0),(7030.0,3819.0),(2113.0,4545.0),(4916.0,5320.0),(8383.0,7146.0),(5598.0,7289.0),(1536.0,7495.0)

101,(9814.0,2686.0),(6713.0,1338.0),(4499.0,2933.0),(10974.0,5265.0),(7037.0,3620.0),(2092.0,4318.0),(4786.0,5184.0),(8077.0,7146.0),(5607.0,7622.0),(1686.0,7521.0)

102,(9485.0,2738.0),(6391.0,1403.0),(4166.0,2918.0),(10641.0,5247.0),(6936.0,3674.0),(2122.0,4650.0),(4978.0,5186.0),(8410.0,7146.0),(5615.0,7884.0),(1914.0,7278.0)

103,(9162.0,2784.0),(6062.0,1460.0),(4104.0,2861.0),(10308.0,5243.0),(7066.0,3712.0),(1810.0,4732.0),(5278.0,5165.0),(8744.0,7146.0),(5606.0,7551.0),(2159.0,7079.0)

104,(9191.0,2452.0),(6104.0,1684.0),(4132.0,3085.0),(9975.0,5255.0),(7205.0,3867.0),(1483.0,4782.0),(5283.0,5122.0),(9077.0,7146.0),(5633.0,7255.0),(2477.0,7109.0)

105,(9185.0,2560.0),(6226.0,1994.0),(4115.0,3281.0),(9986.0,4926.0),(7468.0,3803.0),(1150.0,4762.0),(5498.0,4886.0),(8913.0,7146.0),(5967.0,7268.0),(2145.0,7077.0)

106,(9192.0,2434.0),(6205.0,2142.0),(4110.0,2952.0),(10001.0,4593.0),(7649.0,3629.0),(1023.0,4920.0),(5573.0,4868.0),
(8579.0,7146.0),(6300.0,7281.0),(2253.0,6877.0)

107,(9198.0,2101.0),(6327.0,2324.0),(3809.0,2950.0),(9999.0,4631.0),(7753.0,3604.0),(1126.0,5200.0),(5396.0,4970.0),
(8246.0,7146.0),(6434.0,7286.0),(2158.0,7004.0)

108,(9198.0,1768.0),(6138.0,2150.0),(3843.0,3270.0),(9984.0,4964.0),(7491.0,3688.0),(1459.0,5187.0),(5443.0,4931.0),
(8074.0,6995.0),(6101.0,7273.0),(2177.0,6978.0)

109,(9198.0,1553.0),(6186.0,2142.0),(4083.0,3346.0),(9931.0,5267.0),(7477.0,3761.0),(1792.0,5177.0),(5316.0,5037.0),
(8096.0,6663.0),(5768.0,7260.0),(2318.0,7094.0)

110,(9198.0,1886.0),(6105.0,2041.0),(3952.0,3337.0),(9863.0,5287.0),(7489.0,3431.0),(2126.0,5167.0),(5528.0,4875.0),
(8374.0,6633.0),(5613.0,7127.0),(2317.0,7094.0)

111,(9198.0,2219.0),(6103.0,2036.0),(4285.0,3348.0),(9980.0,5072.0),(7461.0,3099.0),(2329.0,5178.0),(5622.0,5085.0),
(8473.0,6394.0),(5608.0,7170.0),(2023.0,7162.0)

112,(9186.0,2552.0),(6060.0,2189.0),(4105.0,3359.0),(9994.0,4738.0),(7470.0,3208.0),(2393.0,5183.0),(5417.0,5209.0),
(8239.0,6288.0),(5603.0,7212.0),(1795.0,7405.0)

113,(9175.0,2696.0),(6246.0,2047.0),(4135.0,3691.0),(10006.0,4486.0),(7499.0,3541.0),(2160.0,4952.0),(5463.0,5211.0),
(8106.0,6493.0),(5598.0,7252.0),(1567.0,7648.0)

114,(9196.0,2363.0),(6129.0,1735.0),(4114.0,3465.0),(9991.0,4819.0),(7455.0,3869.0),(2312.0,5176.0),(5435.0,5210.0),
(8112.0,6386.0),(5603.0,7209.0),(1596.0,7896.0)

115,(8888.0,2345.0),(5992.0,1502.0),(4139.0,3135.0),(9976.0,5152.0),(7139.0,3861.0),(2216.0,5006.0),(5288.0,5084.0),
(8346.0,6293.0),(5608.0,7166.0),(1549.0,7567.0)

116,(8555.0,2352.0),(6209.0,1712.0),(4469.0,3148.0),(9982.0,5026.0),(6996.0,4051.0),(2190.0,5131.0),(5059.0,5190.0),
(8467.0,6501.0),(5614.0,7124.0),(1486.0,7239.0)

117,(8265.0,2317.0),(6542.0,1720.0),(4414.0,3344.0),(9996.0,4693.0),(7327.0,4092.0),(2141.0,4801.0),(4918.0,5303.0),
(8268.0,6622.0),(5612.0,7138.0),(1431.0,7039.0)

118,(8291.0,1985.0),(6843.0,1692.0),(4541.0,3336.0),(10004.0,4531.0),(7456.0,3863.0),(2106.0,4470.0),(5144.0,5195.0),
(8109.0,6441.0),(5543.0,7110.0),(1443.0,7062.0)

119,(8282.0,1653.0),(6875.0,1360.0),(4442.0,3228.0),(9989.0,4864.0),(7489.0,3699.0),(2099.0,4393.0),(5337.0,5020.0),
(8291.0,6290.0),(5210.0,7110.0),(1647.0,6948.0)

120,(8262.0,1320.0),(6641.0,1263.0),(4498.0,2933.0),(9974.0,5197.0),(7479.0,3749.0),(2131.0,4725.0),(5617.0,4862.0),
(8470.0,6445.0),(4877.0,7110.0),(1976.0,6900.0)

121,(8262.0,987.0),(6721.0,940.0),(4165.0,2918.0),(9702.0,5284.0),(7489.0,3419.0),(2177.0,5055.0),(5569.0,4553.0),(8
323.0,6628.0),(4744.0,7110.0),(2244.0,6889.0)

122,(8262.0,653.0),(6732.0,617.0),(4104.0,2644.0),(9370.0,5266.0),(7460.0,3087.0),(2371.0,5181.0),(5579.0,4488.0),(8
092.0,6714.0),(5023.0,7054.0),(2035.0,7148.0)

123,(8262.0,320.0),(6648.0,294.0),(4104.0,2657.0),(9162.0,5131.0),(7281.0,2829.0),(2352.0,5180.0),(5352.0,4573.0),(8
070.0,7046.0),(5030.0,6721.0),(1807.0,7392.0)

124,(8562.0,265.0),(6335.0,288.0),(4104.0,2841.0),(9162.0,4797.0),(6979.0,2694.0),(2685.0,5200.0),(5019.0,4572.0),(7
830.0,7139.0),(5037.0,6388.0),(1579.0,7635.0)

125,(8798.0,390.0),(6002.0,288.0),(4135.0,3105.0),(8964.0,4662.0),(7059.0,2714.0),(3017.0,5218.0),(5286.0,4572.0),(7
496.0,7131.0),(5234.0,6211.0),(1506.0,7342.0)

126,(8911.0,703.0),(5832.0,451.0),(4104.0,2828.0),(8631.0,4662.0),(7345.0,2881.0),(2733.0,5202.0),(5495.0,4356.0),(7
596.0,7133.0),(5167.0,6229.0),(1423.0,7022.0)

127,(9025.0,1016.0),(5823.0,783.0),(4104.0,2495.0),(8297.0,4662.0),(7446.0,2805.0),(2400.0,5184.0),(5565.0,4584.0),(
7929.0,7142.0),(5092.0,6249.0),(1395.0,6935.0)

128,(9138.0,1330.0),(5906.0,1076.0),(3782.0,2492.0),(8199.0,4662.0),(7436.0,2472.0),(2162.0,4952.0),(5543.0,4356.0)

,(8084.0,7343.0),(5061.0,6258.0),(1336.0,6607.0)

129,(9143.0,1344.0),(6142.0,1278.0),(3462.0,2501.0),(8532.0,4662.0),(7198.0,2354.0),(2177.0,5057.0),(5223.0,4288.0)

,(8117.0,7660.0),(4731.0,6221.0),(1349.0,6680.0)

130,(9030.0,1030.0),(6340.0,1278.0),(3796.0,2492.0),(8866.0,4662.0),(6866.0,2324.0),(2132.0,4727.0),(5116.0,4294.0),

(8082.0,7329.0),(4405.0,6158.0),(1415.0,7006.0)

131,(8916.0,717.0),(6124.0,1278.0),(4105.0,2458.0),(9162.0,4699.0),(6856.0,2628.0),(2099.0,4395.0),(5403.0,4172.0),(

7914.0,7142.0),(4083.0,6079.0),(1456.0,7088.0)

132,(8803.0,404.0),(6426.0,1231.0),(4097.0,2128.0),(9162.0,5032.0),(6842.0,2366.0),(2106.0,4468.0),(5540.0,4062.0),(

7581.0,7133.0),(4025.0,6075.0),(1418.0,7013.0)

133,(8576.0,264.0),(6550.0,921.0),(4115.0,2339.0),(9162.0,5145.0),(6855.0,2610.0),(1988.0,4692.0),(5588.0,4054.0),(7

248.0,7131.0),(4109.0,5889.0),(1430.0,7036.0)

134,(8262.0,305.0),(6592.0,606.0),(4118.0,2295.0),(9162.0,4812.0),(6728.0,2842.0),(1663.0,4766.0),(5598.0,4354.0),(6

915.0,7141.0),(4223.0,5633.0),(1509.0,7358.0)

135,(8262.0,638.0),(6521.0,287.0),(4104.0,2536.0),(9162.0,4844.0),(6551.0,2936.0),(1555.0,5021.0),(5586.0,4435.0),(6

750.0,7314.0),(4473.0,5520.0),(1570.0,7686.0)

136,(8262.0,972.0),(6431.0,288.0),(4104.0,2870.0),(9162.0,5178.0),(6451.0,2668.0),(1377.0,5190.0),(5663.0,4486.0),(6

750.0,7647.0),(4169.0,5658.0),(1592.0,7858.0)

137,(8262.0,1305.0),(6097.0,288.0),(3818.0,2949.0),(8906.0,5247.0),(6376.0,2611.0),(1072.0,5166.0),(5857.0,4218.0),(

6750.0,7931.0),(4075.0,5428.0),(1581.0,7771.0)

138,(8281.0,1638.0),(5832.0,355.0),(4104.0,2911.0),(8573.0,5235.0),(6670.0,2748.0),(1007.0,4840.0),(6024.0,4094.0),(

6750.0,7597.0),(4152.0,5227.0),(1581.0,7773.0)

139,(8294.0,1844.0),(5832.0,688.0),(4104.0,2821.0),(8240.0,5224.0),(6752.0,2602.0),(938.0,4514.0),(5960.0,4417.0),(6

750.0,7264.0),(4065.0,5343.0),(1709.0,7497.0)

140,(8274.0,1512.0),(5832.0,453.0),(4160.0,3133.0),(8328.0,5227.0),(6464.0,2433.0),(932.0,4487.0),(5971.0,4397.0),(6

535.0,7131.0),(4102.0,5674.0),(1937.0,7254.0)

141,(8270.0,1448.0),(5999.0,288.0),(4493.0,3149.0),(8662.0,5238.0),(6291.0,2179.0),(1052.0,4755.0),(6027.0,4072.0),(

6500.0,7188.0),(4001.0,5962.0),(2019.0,7076.0)

142,(8290.0,1781.0),(6332.0,288.0),(4173.0,3133.0),(8995.0,5250.0),(6398.0,2394.0),(1385.0,4776.0),(5849.0,4239.0),(

6748.0,7145.0),(3732.0,6006.0),(1686.0,7085.0)

143,(8281.0,2113.0),(6664.0,270.0),(4104.0,2835.0),(9164.0,5422.0),(6685.0,2562.0),(1714.0,4754.0),(5956.0,4050.0),(

6499.0,7186.0),(3986.0,5789.0),(1477.0,7194.0)

144,(8350.0,2356.0),(6998.0,270.0),(4104.0,2502.0),(9170.0,5755.0),(6987.0,2696.0),(2039.0,4681.0),(5885.0,3902.0),(

6533.0,7131.0),(4104.0,5865.0),(1541.0,7522.0)

145,(8683.0,2349.0),(7331.0,270.0),(4418.0,2495.0),(9176.0,6088.0),(7289.0,2833.0),(2154.0,4907.0),(6115.0,3746.0),(

6634.0,7138.0),(4131.0,5978.0),(1527.0,7450.0)

146,(9016.0,2343.0),(7424.0,500.0),(4590.0,2339.0),(9284.0,6315.0),(7029.0,2706.0),(2124.0,4662.0),(5917.0,3816.0),(

6658.0,7139.0),(4139.0,5671.0),(1464.0,7123.0)

147,(9189.0,2491.0),(7420.0,606.0),(4346.0,2250.0),(9617.0,6305.0),(6720.0,2740.0),(2093.0,4330.0),(5864.0,4050.0),(

6509.0,7129.0),(4443.0,5533.0),(1431.0,7038.0)

148,(9166.0,2756.0),(7433.0,273.0),(4113.0,2358.0),(9950.0,6305.0),(6516.0,2839.0),(2112.0,4533.0),(5701.0,4050.0),(

6175.0,7121.0),(4253.0,5620.0),(1356.0,6715.0)

149,(9177.0,2678.0),(7420.0,599.0),(4104.0,2691.0),(10200.0,6316.0),(6498.0,2704.0),(2105.0,4458.0),(6033.0,4047.0)

,(5842.0,7114.0),(4104.0,5857.0),(1330.0,6572.0)

150,(9170.0,2727.0),(7203.0,731.0),(4104.0,2808.0),(9867.0,6302.0),(6263.0,2480.0),(2100.0,4405.0),(6329.0,3893.0),(

5722.0,7112.0),(3859.0,6056.0),(1389.0,6900.0)

151,(9037.0,2673.0),(6870.0,750.0),(4094.0,2917.0),(9840.0,6025.0),(6162.0,2162.0),(2049.0,4678.0),(6390.0,4076.0),(6055.0,7119.0),(3871.0,5886.0),(1360.0,6739.0)

152,(8705.0,2696.0),(6998.0,743.0),(3802.0,2986.0),(10126.0,5937.0),(6146.0,2146.0),(1724.0,4752.0),(6512.0,3821.0),(6389.0,7126.0),(4100.0,5660.0),(1326.0,6548.0)

153,(8372.0,2718.0),(7331.0,724.0),(3850.0,3316.0),(10027.0,5951.0),(5855.0,2254.0),(1732.0,4750.0),(6832.0,3728.0),(6511.0,7221.0),(4064.0,5329.0),(1384.0,6876.0)

154,(8093.0,2661.0),(7168.0,733.0),(3531.0,3330.0),(9829.0,6125.0),(5532.0,2338.0),(2057.0,4677.0),(6527.0,3817.0),(6470.0,7287.0),(4173.0,5184.0),(1626.0,6951.0)

155,(8118.0,2919.0),(6835.0,752.0),(3201.0,3363.0),(9652.0,6304.0),(5536.0,2666.0),(1865.0,4720.0),(6392.0,4059.0),(6137.0,7274.0),(4067.0,5138.0),(1955.0,6903.0)

156,(8285.0,2964.0),(6701.0,498.0),(3149.0,3049.0),(9318.0,6314.0),(5461.0,2932.0),(1564.0,4814.0),(6390.0,4393.0),(5804.0,7261.0),(4151.0,5126.0),(2257.0,6872.0)

157,(7955.0,2923.0),(6666.0,364.0),(3132.0,2826.0),(8985.0,6313.0),(5495.0,2925.0),(1549.0,5147.0),(6395.0,4726.0),(5613.0,7128.0),(4050.0,5198.0),(2186.0,7081.0)

158,(7763.0,2716.0),(6750.0,686.0),(3230.0,2782.0),(8652.0,6304.0),(5165.0,2928.0),(1251.0,5195.0),(6390.0,4662.0),(5614.0,7085.0),(4128.0,5068.0),(2484.0,7144.0)

159,(7802.0,2861.0),(6867.0,998.0),(3349.0,2796.0),(8636.0,6304.0),(5281.0,2942.0),(1048.0,5043.0),(6273.0,4860.0),(5601.0,6752.0),(4107.0,5216.0),(2492.0,7478.0)

160,(7525.0,2969.0),(6793.0,817.0),(3124.0,2658.0),(8969.0,6312.0),(5478.0,2929.0),(997.0,4787.0),(5940.0,4860.0),(5587.0,6419.0),(4106.0,5010.0),(2499.0,7811.0)

161,(7315.0,3175.0),(6701.0,497.0),(3290.0,2502.0),(9177.0,6441.0),(5511.0,2922.0),(1062.0,5114.0),(5661.0,4860.0),(5586.0,6405.0),(4081.0,5085.0),(2496.0,7659.0)

162,(7359.0,2946.0),(6545.0,283.0),(3456.0,2670.0),(9170.0,6774.0),(5182.0,2930.0),(1135.0,5439.0),(5994.0,4860.0),(5600.0,6738.0),(4092.0,5212.0),(2488.0,7326.0)

163,(7201.0,2876.0),(6564.0,281.0),(3261.0,2786.0),(9162.0,7107.0),(5158.0,3032.0),(1212.0,5763.0),(6327.0,4860.0),(5614.0,7072.0),(4128.0,5221.0),(2486.0,7227.0)

164,(7410.0,2946.0),(6706.0,517.0),(3132.0,2857.0),(9168.0,6851.0),(5163.0,2971.0),(1159.0,5539.0),(6382.0,5111.0),(5610.0,7110.0),(4109.0,5017.0),(2494.0,7560.0)

165,(7441.0,2977.0),(6746.0,672.0),(3145.0,3019.0),(9175.0,6517.0),(5021.0,2733.0),(1083.0,5215.0),(6147.0,5220.0),(5277.0,7110.0),(4099.0,5214.0),(2501.0,7893.0)

166,(7389.0,2925.0),(6662.0,349.0),(3184.0,3350.0),(9177.0,6184.0),(5019.0,2710.0),(1017.0,4888.0),(5813.0,5220.0),(4944.0,7110.0),(4083.0,5078.0),(2177.0,7914.0)

167,(7308.0,3185.0),(6638.0,270.0),(3453.0,3268.0),(9172.0,5851.0),(5151.0,2931.0),(948.0,4562.0),(5622.0,5084.0),(646.0,7142.0),(3845.0,5214.0),(1844.0,7928.0)

168,(7389.0,2925.0),(6567.0,280.0),(3225.0,3186.0),(9166.0,5517.0),(5294.0,2943.0),(1002.0,4318.0),(5526.0,4875.0),(4675.0,7475.0),(3512.0,5235.0),(1590.0,7847.0)

169,(7443.0,2999.0),(6796.0,270.0),(3134.0,2916.0),(9162.0,5184.0),(5623.0,2922.0),(1274.0,4213.0),(5624.0,5046.0),(4703.0,7807.0),(3565.0,5518.0),(1657.0,7552.0)

170,(7368.0,3102.0),(7129.0,270.0),(3456.0,2992.0),(9162.0,4851.0),(5955.0,2950.0),(1251.0,3881.0),(5456.0,5211.0),(4899.0,7948.0),(3668.0,5835.0),(1885.0,7309.0)

171,(7339.0,3142.0),(7432.0,298.0),(3440.0,3320.0),(9162.0,4806.0),(5664.0,2926.0),(1046.0,3752.0),(5296.0,5054.0),(5232.0,7935.0),(3680.0,5871.0),(2094.0,7074.0)

172,(7420.0,2973.0),(7419.0,632.0),(3173.0,3434.0),(9162.0,5139.0),(5537.0,2703.0),(824.0,3857.0),(5565.0,4869.0),(556.0,7921.0),(3577.0,5554.0),(1761.0,7083.0)

173,(7140.0,2849.0),(7171.0,733.0),(3114.0,3761.0),(9162.0,5039.0),(5526.0,2370.0),(876.0,4186.0),(5562.0,4609.0),(5

899.0,7928.0),(3473.0,5237.0),(1487.0,7091.0)

174,(7445.0,2984.0),(6838.0,752.0),(3173.0,3433.0),(9162.0,4705.0),(5535.0,2643.0),(1077.0,4304.0),(5561.0,4870.0),(6232.0,7939.0),(3140.0,5224.0),(1820.0,7081.0)

175,(7768.0,2908.0),(6701.0,501.0),(3176.0,3268.0),(9162.0,4951.0),(5484.0,2927.0),(1269.0,4137.0),(5293.0,5056.0),(6565.0,7950.0),(3042.0,5454.0),(2134.0,7035.0)

176,(7532.0,2967.0),(6665.0,361.0),(3222.0,3186.0),(9190.0,5257.0),(5472.0,3132.0),(1246.0,3804.0),(5152.0,5343.0),(6898.0,7956.0),(2763.0,5538.0),(2201.0,6947.0)

177,(7340.0,2876.0),(6749.0,683.0),(3466.0,3104.0),(9523.0,5275.0),(5580.0,3435.0),(969.0,3755.0),(5093.0,5630.0),(7232.0,7956.0),(2432.0,5574.0),(2279.0,7090.0)

178,(7327.0,2863.0),(6866.0,995.0),(3244.0,2940.0),(9835.0,5291.0),(5580.0,3768.0),(983.0,3754.0),(5158.0,5337.0),(7422.0,7790.0),(2101.0,5611.0),(2356.0,7097.0)

179,(7379.0,2915.0),(6794.0,820.0),(3127.0,2697.0),(9502.0,5273.0),(5257.0,3780.0),(1315.0,3732.0),(5172.0,5319.0),(7431.0,7460.0),(1769.0,5647.0),(2187.0,7081.0)

180,(7426.0,2976.0),(6704.0,1011.0),(3123.0,2638.0),(9169.0,5256.0),(5407.0,3780.0),(1645.0,3683.0),(5159.0,5335.0),(7380.0,7131.0),(1713.0,5726.0),(2519.0,7110.0)

181,(7134.0,2846.0),(6624.0,1328.0),(3132.0,2860.0),(9162.0,4930.0),(5419.0,3780.0),(1964.0,3648.0),(5171.0,5322.0),(7050.0,7137.0),(1618.0,5829.0),(2852.0,7110.0)

182,(7439.0,2982.0),(6705.0,1005.0),(3115.0,2527.0),(9162.0,4726.0),(5085.0,3777.0),(2029.0,3975.0),(5089.0,5610.0),(6782.0,7145.0),(1285.0,5816.0),(3186.0,7110.0)

183,(7351.0,3125.0),(6797.0,826.0),(3421.0,2502.0),(9162.0,5060.0),(4752.0,3765.0),(2080.0,4229.0),(5228.0,5869.0),(7116.0,7135.0),(1495.0,5824.0),(3511.0,7100.0)

184,(7356.0,3119.0),(6889.0,1140.0),(3520.0,2250.0),(9299.0,5263.0),(4868.0,3769.0),(2015.0,3902.0),(5081.0,5573.0),(7390.0,7196.0),(1718.0,5697.0),(3576.0,6773.0)

185,(7452.0,2994.0),(6752.0,1341.0),(3853.0,2250.0),(9632.0,5280.0),(5201.0,3780.0),(1950.0,3575.0),(5088.0,5606.0),(7441.0,7525.0),(1588.0,5828.0),(3640.0,6446.0)

186,(7481.0,3326.0),(6673.0,1134.0),(4109.0,2186.0),(9960.0,5259.0),(5534.0,3780.0),(1886.0,3248.0),(5173.0,5318.0),(7712.0,7624.0),(1255.0,5815.0),(3704.0,6119.0)

187,(7502.0,3577.0),(6754.0,810.0),(4115.0,2219.0),(9914.0,5272.0),(5613.0,4034.0),(1822.0,2921.0),(5318.0,5035.0),(8044.0,7660.0),(1264.0,6112.0),(3682.0,6231.0)

188,(7474.0,3245.0),(6873.0,1012.0),(4121.0,2247.0),(9846.0,5553.0),(5580.0,3703.0),(1877.0,3201.0),(5441.0,4933.0),(8091.0,7410.0),(1309.0,6443.0),(3618.0,6558.0)

189,(7398.0,2934.0),(6876.0,1340.0),(4007.0,2486.0),(9846.0,5887.0),(5580.0,3369.0),(1941.0,3528.0),(5274.0,5195.0),(7996.0,7144.0),(1274.0,6186.0),(3554.0,6885.0)

190,(7451.0,2989.0),(6880.0,1029.0),(3674.0,2495.0),(9846.0,5731.0),(5438.0,3075.0),(2005.0,3855.0),(5278.0,5165.0),(7662.0,7135.0),(1229.0,5856.0),(3614.0,7110.0)

191,(7267.0,3266.0),(6805.0,753.0),(3456.0,2616.0),(9846.0,5398.0),(5538.0,2917.0),(2071.0,4182.0),(5283.0,5122.0),(7380.0,7077.0),(1262.0,6102.0),(3948.0,7110.0)

192,(7118.0,3564.0),(7138.0,735.0),(3314.0,2792.0),(9879.0,5282.0),(5209.0,2933.0),(2110.0,4513.0),(5151.0,5195.0),(7380.0,6744.0),(1307.0,6432.0),(4281.0,7110.0)

193,(7250.0,3301.0),(7360.0,723.0),(3132.0,2921.0),(9548.0,5276.0),(5022.0,2739.0),(2107.0,4478.0),(4818.0,5184.0),(7380.0,6411.0),(1596.0,6441.0),(4614.0,7110.0)

194,(7428.0,3020.0),(7027.0,741.0),(3159.0,3120.0),(9215.0,5258.0),(5018.0,2704.0),(2098.0,4385.0),(4675.0,4956.0),(7566.0,6268.0),(1927.0,6396.0),(4948.0,7110.0)

195,(7286.0,3229.0),(6750.0,827.0),(3132.0,2788.0),(9162.0,4976.0),(5165.0,2942.0),(2130.0,4717.0),(4627.0,4626.0),(7899.0,6276.0),(2257.0,6351.0),(4762.0,7110.0)

196,(7137.0,3527.0),(6669.0,1150.0),(3112.0,2456.0),(9162.0,4681.0),(5041.0,3177.0),(2229.0,5019.0),(4649.0,4348.0),
(8003.0,6279.0),(2587.0,6307.0),(4614.0,6897.0)

197,(6840.0,3546.0),(6484.0,1375.0),(3098.0,2123.0),(9162.0,5014.0),(5152.0,3210.0),(2293.0,5174.0),(4980.0,4305.0),
,(7670.0,6271.0),(2876.0,6177.0),(4569.0,6566.0)

198,(7075.0,3310.0),(6614.0,1334.0),(3211.0,2070.0),(9162.0,5164.0),(5153.0,3218.0),(2162.0,4966.0),(4988.0,4533.0),
(7380.0,6307.0),(3073.0,5955.0),(4524.0,6236.0)

199,(7121.0,2991.0),(6805.0,1344.0),(3294.0,2089.0),(9162.0,4830.0),(4945.0,3346.0),(2326.0,5177.0),(4997.0,4403.0),
,(7380.0,6640.0),(2741.0,5926.0),(4557.0,6477.0)

200,(7297.0,2919.0),(6660.0,1185.0),(3466.0,2250.0),(9162.0,4826.0),(4932.0,3341.0),(2206.0,4996.0),(5150.0,4572.0),
(7380.0,6973.0),(2409.0,5896.0),(4602.0,6807.0)

201,(7611.0,2948.0),(6741.0,861.0),(3294.0,2103.0),(9162.0,5159.0),(5074.0,3337.0),(2126.0,4684.0),(5208.0,4572.0),(
7559.0,7132.0),(2077.0,5866.0),(4640.0,7082.0)

202,(7689.0,2928.0),(6853.0,963.0),(3197.0,2070.0),(9162.0,5018.0),(4977.0,3183.0),(2095.0,4352.0),(5096.0,4572.0),(
7892.0,7141.0),(1745.0,5836.0),(4594.0,6751.0)

203,(7389.0,2925.0),(6807.0,852.0),(3097.0,2108.0),(9162.0,4685.0),(5165.0,3329.0),(2039.0,4024.0),(5261.0,4572.0),(
8074.0,6984.0),(1412.0,5821.0),(4549.0,6421.0)

204,(7441.0,2977.0),(6710.0,534.0),(3111.0,2441.0),(9162.0,4971.0),(5161.0,2998.0),(1974.0,3697.0),(5043.0,4572.0),(
8096.0,6652.0),(1243.0,5957.0),(4545.0,6094.0)

205,(7410.0,2946.0),(6582.0,278.0),(3134.0,2772.0),(9211.0,5258.0),(5114.0,3171.0),(1694.0,3676.0),(5376.0,4577.0),(
8080.0,6892.0),(1287.0,6287.0),(4610.0,5775.0)

206,(7443.0,2714.0),(6527.0,286.0),(3456.0,2798.0),(9544.0,5276.0),(4927.0,3347.0),(1364.0,3725.0),(5418.0,4584.0),(
7984.0,7143.0),(1338.0,6617.0),(4621.0,5802.0)

207,(7434.0,2380.0),(6537.0,376.0),(3458.0,2464.0),(9875.0,5283.0),(4641.0,3280.0),(1256.0,3952.0),(5463.0,4352.0),(
7651.0,7135.0),(1397.0,6945.0),(4537.0,6122.0)

208,(7443.0,2704.0),(6614.0,700.0),(3591.0,2250.0),(9944.0,5263.0),(4626.0,2947.0),(1297.0,4266.0),(5430.0,4339.0),(
7441.0,7129.0),(1427.0,7030.0),(4268.0,6117.0)

209,(7416.0,2952.0),(6555.0,451.0),(3925.0,2250.0),(9931.0,5267.0),(4945.0,2920.0),(1630.0,4266.0),(5365.0,4538.0),(
7774.0,7138.0),(1508.0,7352.0),(3940.0,6069.0)

210,(7434.0,2970.0),(6604.0,275.0),(3985.0,2250.0),(9601.0,5279.0),(5277.0,2941.0),(1964.0,4266.0),(5593.0,4356.0),(
8066.0,7102.0),(1560.0,7620.0),(3830.0,6046.0)

211,(7442.0,2679.0),(6526.0,286.0),(3652.0,2250.0),(9268.0,5261.0),(5542.0,2853.0),(1878.0,4266.0),(5488.0,4595.0),(
8089.0,6769.0),(1496.0,7293.0),(4157.0,6081.0)

212,(7434.0,2405.0),(6837.0,270.0),(3462.0,2404.0),(9162.0,5029.0),(5531.0,2520.0),(1545.0,4266.0),(5532.0,4603.0),(
8088.0,6775.0),(1407.0,6990.0),(4131.0,5977.0)

213,(7444.0,2739.0),(7170.0,270.0),(3220.0,2502.0),(9162.0,4696.0),(5674.0,2301.0),(1344.0,4266.0),(5660.0,4828.0),(
8066.0,7107.0),(1448.0,7072.0),(3934.0,6069.0)

214,(7533.0,2967.0),(7431.0,340.0),(3341.0,2502.0),(9162.0,4961.0),(5996.0,2217.0),(1677.0,4266.0),(5871.0,4571.0),(
7769.0,7138.0),(1769.0,7083.0),(3835.0,6048.0)

215,(7857.0,2906.0),(7417.0,673.0),(3674.0,2495.0),(9162.0,5217.0),(5704.0,2293.0),(2010.0,4266.0),(5688.0,4794.0),(
7435.0,7129.0),(2103.0,7074.0),(4093.0,5877.0)

216,(8049.0,2939.0),(7130.0,735.0),(4007.0,2486.0),(9162.0,4884.0),(5519.0,2191.0),(2111.0,4521.0),(5565.0,4621.0),(
7102.0,7135.0),(1775.0,7083.0),(4249.0,6111.0)

217,(7788.0,2809.0),(6797.0,754.0),(4104.0,2720.0),(9162.0,4773.0),(5458.0,1908.0),(2147.0,4852.0),(5766.0,4362.0),(
6769.0,7145.0),(1451.0,7078.0),(4466.0,6176.0)

218,(7885.0,2628.0),(6883.0,1037.0),(3966.0,2932.0),(9162.0,5106.0),(5139.0,1869.0),(2130.0,4716.0),(5684.0,4461.0)

,(7063.0,7137.0),(1410.0,6996.0),(4147.0,6083.0)

219,(8118.0,2759.0),(6858.0,1348.0),(3635.0,2969.0),(9162.0,5072.0),(5137.0,1868.0),(2098.0,4385.0),(5653.0,4498.0),
(7382.0,7144.0),(1439.0,7054.0),(3961.0,6070.0)

220,(8258.0,2962.0),(6722.0,1339.0),(3608.0,2971.0),(9162.0,4738.0),(5246.0,1908.0),(2107.0,4478.0),(5851.0,4233.0),
,(7433.0,7474.0),(1435.0,7047.0),(4124.0,5951.0)

221,(8118.0,2947.0),(6887.0,1171.0),(3818.0,3092.0),(8905.0,4662.0),(5510.0,1979.0),(2142.0,4810.0),(6058.0,4052.0),
(7420.0,7388.0),(1413.0,7002.0),(4323.0,5753.0)

222,(8084.0,2635.0),(6808.0,854.0),(3758.0,3330.0),(8572.0,4662.0),(5524.0,2312.0),(2136.0,4758.0),(6390.0,4085.0),(
7449.0,7129.0),(1515.0,6967.0),(4622.0,5609.0)

223,(8129.0,2952.0),(6994.0,743.0),(3424.0,3331.0),(8238.0,4662.0),(5821.0,2263.0),(2153.0,4895.0),(6670.0,4007.0),(
7783.0,7138.0),(1845.0,6919.0),(4936.0,5497.0)

224,(8459.0,2946.0),(7326.0,724.0),(3277.0,3352.0),(8258.0,4662.0),(5879.0,2248.0),(2255.0,5171.0),(6466.0,3852.0),(
8116.0,7146.0),(2174.0,6871.0),(5089.0,5613.0)

225,(8784.0,2873.0),(7425.0,476.0),(3468.0,3207.0),(8591.0,4662.0),(5556.0,2332.0),(2200.0,5166.0),(6682.0,3852.0),(
8449.0,7146.0),(2124.0,7049.0),(5091.0,5621.0)

226,(9110.0,2801.0),(7428.0,396.0),(3468.0,3127.0),(8925.0,4662.0),(5817.0,2264.0),(2264.0,5172.0),(7016.0,3852.0),(
8782.0,7146.0),(2248.0,6884.0),(5147.0,5349.0)

227,(9165.0,2764.0),(7416.0,709.0),(3473.0,3180.0),(9162.0,4758.0),(6130.0,2154.0),(1931.0,5173.0),(6690.0,3852.0),(
9116.0,7146.0),(2201.0,7083.0),(5135.0,5194.0)

228,(9176.0,2686.0),(7429.0,376.0),(3584.0,2974.0),(9162.0,5091.0),(6140.0,2149.0),(1598.0,5182.0),(6761.0,4050.0),(
8874.0,7146.0),(2434.0,7105.0),(4917.0,5312.0)

229,(9172.0,2719.0),(7424.0,496.0),(3814.0,3069.0),(9330.0,5264.0),(6175.0,2203.0),(1560.0,4901.0),(7003.0,3940.0),(
8541.0,7146.0),(2103.0,7076.0),(4634.0,5455.0)

230,(9163.0,2782.0),(7306.0,726.0),(3921.0,3334.0),(9663.0,5282.0),(6276.0,2521.0),(1346.0,4774.0),(7025.0,4055.0),(
8207.0,7146.0),(1875.0,7319.0),(4675.0,5439.0)

231,(9174.0,2704.0),(6974.0,744.0),(4128.0,3199.0),(9991.0,5255.0),(6535.0,2721.0),(1013.0,4753.0),(7356.0,4096.0),(
8083.0,7334.0),(1647.0,7562.0),(4922.0,5269.0)

232,(9196.0,2371.0),(6736.0,633.0),(4150.0,3348.0),(10324.0,5242.0),(6754.0,2913.0),(1298.0,4771.0),(7443.0,3941.0),
,(8117.0,7666.0),(1712.0,7494.0),(5179.0,5197.0)

233,(9181.0,2640.0),(6652.0,310.0),(4228.0,3348.0),(10257.0,5245.0),(6516.0,2893.0),(1563.0,4853.0),(7584.0,4104.0),
,(8092.0,7973.0),(1940.0,7250.0),(5280.0,5150.0)

234,(9183.0,2608.0),(6715.0,552.0),(3895.0,3333.0),(9926.0,5268.0),(6670.0,2748.0),(1550.0,5183.0),(7917.0,4104.0),(
7759.0,7965.0),(2160.0,7001.0),(5396.0,4970.0)

235,(9194.0,2404.0),(6890.0,749.0),(4110.0,3311.0),(9858.0,5288.0),(6671.0,2945.0),(1883.0,5174.0),(7840.0,4104.0),(
7426.0,7956.0),(2369.0,6742.0),(5630.0,4921.0)

236,(9169.0,2736.0),(7223.0,730.0),(4115.0,3281.0),(9902.0,5275.0),(6516.0,2756.0),(2216.0,5167.0),(7507.0,4104.0),(
7443.0,7654.0),(2588.0,6491.0),(5581.0,5218.0)

237,(9165.0,2765.0),(7275.0,727.0),(4132.0,3086.0),(9846.0,5421.0),(6282.0,2538.0),(2284.0,5074.0),(7448.0,3910.0),(
7410.0,7324.0),(2735.0,6323.0),(5277.0,5177.0)

238,(9176.0,2687.0),(6942.0,746.0),(4264.0,2922.0),(9846.0,5755.0),(6181.0,2220.0),(2185.0,5105.0),(7438.0,3968.0),(
7245.0,7131.0),(2516.0,6574.0),(5282.0,5134.0)

239,(9197.0,2354.0),(6925.0,747.0),(4597.0,2934.0),(9833.0,6087.0),(6043.0,1917.0),(2197.0,4987.0),(7421.0,4071.0),(
6912.0,7141.0),(2296.0,6825.0),(5287.0,5091.0)

240,(9180.0,2657.0),(7258.0,728.0),(4599.0,2629.0),(9690.0,6303.0),(5806.0,1687.0),(2354.0,5144.0),(7305.0,3875.0),(
6746.0,6975.0),(1981.0,6899.0),(5290.0,5067.0)

End

3. 第六问中白天连续 4 个小时内的各警车的正常巡逻位置数据

10, 0.191, 0.436, 0.265, 0.564, 0.593, 1.616, 12.525175346877306
0,(7434.0,2376.0),(1818.0,2898.0),(5364.0,2952.0),(9882.0,4032.0),(5274.0,5202.0),(1224.0,5814.0),(7380.0,6264.0),(14040.0,6840.0),(10296.0,7182.0),(3528.0,7902.0)
1,(7101.0,2345.0),(1881.0,3225.0),(5032.0,2917.0),(9549.0,4057.0),(4940.0,5184.0),(1147.0,5489.0),(7713.0,6272.0),(13712.0,6903.0),(9962.0,7171.0),(3861.0,7917.0)
2,(6910.0,2328.0),(1945.0,3553.0),(4699.0,2930.0),(9217.0,4083.0),(5157.0,4983.0),(1088.0,5239.0),(8046.0,6280.0),(13384.0,6967.0),(9629.0,7160.0),(4194.0,7932.0)
3,(7242.0,2358.0),(1914.0,3391.0),(4603.0,2675.0),(9197.0,4381.0),(5134.0,4970.0),(1165.0,5564.0),(7856.0,6275.0),(13546.0,6936.0),(9296.0,7150.0),(4527.0,7947.0)
4,(7438.0,2516.0),(1850.0,3064.0),(4590.0,2342.0),(9168.0,4612.0),(5133.0,4963.0),(1234.0,5889.0),(7522.0,6267.0),(13873.0,6872.0),(9361.0,7152.0),(4703.0,7812.0)
5,(7447.0,2850.0),(1791.0,2736.0),(4349.0,2250.0),(9183.0,4502.0),(5206.0,4872.0),(1278.0,6220.0),(7380.0,6454.0),(14203.0,6840.0),(9694.0,7162.0),(4675.0,7480.0)
6,(7446.0,2792.0),(1737.0,2407.0),(4113.0,2355.0),(9162.0,4834.0),(4987.0,5123.0),(1304.0,6409.0),(7380.0,6787.0),(14298.0,6840.0),(10027.0,7173.0),(4647.0,7147.0)
7,(7436.0,2459.0),(1684.0,2078.0),(4104.0,2688.0),(9162.0,5167.0),(4751.0,5184.0),(1259.0,6079.0),(7380.0,7121.0),(14114.0,6840.0),(10300.0,7246.0),(4669.0,7404.0)
8,(7441.0,2625.0),(1717.0,2282.0),(4104.0,2809.0),(9162.0,5011.0),(5084.0,5192.0),(1208.0,5749.0),(7429.0,7450.0),(14388.0,6840.0),(10326.0,7578.0),(4697.0,7736.0)
9,(7451.0,2958.0),(1771.0,2611.0),(4104.0,2908.0),(9162.0,4677.0),(5291.0,5059.0),(1132.0,5425.0),(7423.0,7411.0),(14054.0,6840.0),(10311.0,7388.0),(4603.0,7950.0)
10,(7443.0,2684.0),(1811.0,2855.0),(4104.0,2575.0),(9192.0,4436.0),(5152.0,5343.0),(1103.0,5303.0),(7426.0,7129.0),(14358.0,6840.0),(10305.0,7307.0),(4270.0,7935.0)
11,(7434.0,2401.0),(1757.0,2526.0),(4121.0,2257.0),(9176.0,4556.0),(5180.0,5310.0),(1180.0,5627.0),(7759.0,7137.0),(14144.0,6840.0),(10331.0,7639.0),(3937.0,7920.0)
12,(7444.0,2734.0),(1703.0,2197.0),(4112.0,2378.0),(9175.0,4557.0),(5252.0,5036.0),(1191.0,5676.0),(8035.0,7145.0),(14269.0,6840.0),(10307.0,7327.0),(3604.0,7905.0)
13,(7449.0,2908.0),(1659.0,1867.0),(4106.0,2454.0),(9162.0,4890.0),(5014.0,4910.0),(1115.0,5351.0),(7702.0,7136.0),(14233.0,6840.0),(10483.0,7179.0),(3522.0,7645.0)
14,(7439.0,2575.0),(1650.0,1776.0),(4108.0,2182.0),(9162.0,5223.0),(5257.0,4808.0),(1044.0,5025.0),(7369.0,7128.0),(14179.0,6840.0),(10816.0,7174.0),(3514.0,7311.0)
15,(7300.0,2363.0),(1664.0,1924.0),(4119.0,2284.0),(9162.0,4955.0),(5196.0,5005.0),(1000.0,4805.0),(7035.0,7137.0),(14322.0,6840.0),(11149.0,7169.0),(3378.0,7110.0)
16,(6968.0,2333.0),(1644.0,1719.0),(4114.0,2350.0),(9167.0,4622.0),(5464.0,4914.0),(1065.0,5131.0),(6797.0,7144.0),(14090.0,6840.0),(11483.0,7165.0),(3045.0,7110.0)
17,(7043.0,2340.0),(1670.0,1980.0),(4095.0,2117.0),(9184.0,4492.0),(5491.0,4892.0),(1341.0,5191.0),(7130.0,7135.0),(14412.0,6840.0),(11331.0,7167.0),(2711.0,7110.0)
18,(7375.0,2370.0),(1638.0,1663.0),(4106.0,2447.0),(9183.0,4501.0),(5282.0,5131.0),(1421.0,5188.0),(7296.0,7130.0),(14078.0,6840.0),(10998.0,7172.0),(2379.0,7100.0)
19,(7160.0,2351.0),(1671.0,1994.0),(4117.0,2312.0),(9168.0,4612.0),(5535.0,5215.0),(1088.0,5201.0),(6963.0,7139.0),(14334.0,6840.0),(10664.0,7176.0),(2164.0,7079.0)
20,(6851.0,2323.0),(1642.0,1705.0),(4116.0,2323.0),(9196.0,4403.0),(5363.0,5206.0),(1404.0,5189.0),(6870.0,7142.0),(1416

8.0,6840.0),(10331.0,7181.0),(2471.0,7108.0)
 21,(7183.0,2353.0),(1666.0,1938.0),(4105.0,2459.0),(9171.0,4590.0),(5205.0,5009.0),(1357.0,5191.0),(7203.0,7133.0),(1424
 5.0,6840.0),(10318.0,7478.0),(2139.0,7077.0)
 22,(7352.0,2368.0),(1648.0,1762.0),(4112.0,2373.0),(9180.0,4524.0),(5062.0,4934.0),(1069.0,5147.0),(7223.0,7132.0),(1425
 7.0,6840.0),(10319.0,7488.0),(2404.0,7102.0)
 23,(7020.0,2338.0),(1660.0,1881.0),(4121.0,2262.0),(9187.0,4469.0),(4955.0,5158.0),(1003.0,4821.0),(6890.0,7141.0),(1415
 5.0,6840.0),(10297.0,7207.0),(2489.0,7363.0)
 24,(6991.0,2335.0),(1654.0,1818.0),(4110.0,2398.0),(9164.0,4645.0),(5134.0,4970.0),(1041.0,5009.0),(6942.0,7140.0),(1434
 6.0,6840.0),(10323.0,7539.0),(2497.0,7696.0)
 25,(7323.0,2365.0),(1654.0,1824.0),(4107.0,2434.0),(9192.0,4435.0),(5157.0,4983.0),(1053.0,5067.0),(7276.0,7130.0),(1406
 6.0,6840.0),(10314.0,7427.0),(2499.0,7774.0)
 26,(7212.0,2355.0),(1659.0,1875.0),(4118.0,2298.0),(9162.0,4766.0),(5111.0,4958.0),(992.0,4763.0),(7150.0,7134.0),(14399.
 0,6840.0),(10302.0,7268.0),(2491.0,7441.0)
 27,(6881.0,2325.0),(1649.0,1768.0),(4115.0,2337.0),(9162.0,5100.0),(5121.0,4963.0),(1057.0,5090.0),(6817.0,7144.0),(1410
 2.0,6840.0),(10328.0,7600.0),(2481.0,7109.0)
 28,(7130.0,2348.0),(1665.0,1932.0),(4104.0,2472.0),(9162.0,5078.0),(5170.0,4990.0),(1299.0,5193.0),(7015.0,7138.0),(1431
 0.0,6840.0),(10310.0,7366.0),(2149.0,7078.0)
 29,(7405.0,2373.0),(1643.0,1711.0),(4113.0,2359.0),(9162.0,4745.0),(5098.0,4952.0),(1463.0,5187.0),(7348.0,7128.0),(1419
 2.0,6840.0),(10307.0,7329.0),(2393.0,7101.0)
 30,(7073.0,2343.0),(1671.0,1988.0),(4119.0,2276.0),(9162.0,4912.0),(4935.0,5180.0),(1130.0,5200.0),(7077.0,7136.0),(1422
 1.0,6840.0),(10331.0,7638.0),(2726.0,7110.0)
 31,(6938.0,2330.0),(1638.0,1657.0),(4109.0,2411.0),(9162.0,5245.0),(5158.0,4932.0),(1024.0,4924.0),(6750.0,7151.0),(1428
 1.0,6840.0),(10305.0,7305.0),(3060.0,7110.0)
 32,(7270.0,2361.0),(1671.0,1986.0),(4088.0,2082.0),(9162.0,4933.0),(5181.0,4903.0),(1020.0,4906.0),(6750.0,7484.0),(1413
 1.0,6840.0),(10312.0,7390.0),(3393.0,7110.0)
 33,(7265.0,2360.0),(1643.0,1713.0),(4111.0,2386.0),(9162.0,4724.0),(5261.0,4804.0),(1111.0,5200.0),(6750.0,7817.0),(1437
 0.0,6840.0),(10326.0,7577.0),(3293.0,7110.0)
 34,(6933.0,2330.0),(1665.0,1929.0),(4121.0,2251.0),(9162.0,5057.0),(5237.0,5027.0),(1444.0,5187.0),(6750.0,7760.0),(1404
 2.0,6840.0),(10300.0,7244.0),(2959.0,7110.0)
 35,(7078.0,2343.0),(1427.0,2033.0),(4111.0,2384.0),(9162.0,5121.0),(5500.0,4884.0),(1318.0,5192.0),(6750.0,7427.0),(1437
 5.0,6840.0),(10316.0,7451.0),(2626.0,7110.0)
 36,(7410.0,2373.0),(1095.0,2057.0),(4106.0,2448.0),(9162.0,4787.0),(5286.0,5100.0),(1061.0,5108.0),(6801.0,7144.0),(1412
 6.0,6840.0),(10321.0,7515.0),(2294.0,7091.0)
 37,(7125.0,2347.0),(762.0,2082.0),(4117.0,2312.0),(9189.0,4456.0),(5281.0,5142.0),(996.0,4782.0),(7135.0,7134.0),(14286.
 0,6840.0),(10296.0,7183.0),(2249.0,7087.0)
 38,(6886.0,2326.0),(433.0,2105.0),(4116.0,2323.0),(9162.0,4657.0),(5179.0,4995.0),(1292.0,4770.0),(7291.0,7130.0),(14216
 .0,6840.0),(10321.0,7512.0),(2386.0,7100.0)
 39,(7218.0,2356.0),(766.0,2081.0),(4105.0,2458.0),(9190.0,4448.0),(5212.0,5013.0),(1563.0,4846.0),(6958.0,7140.0),(14197
 .0,6840.0),(10316.0,7454.0),(2157.0,7078.0)
 40,(7317.0,2365.0),(1098.0,2057.0),(4112.0,2373.0),(9162.0,4779.0),(4968.0,5144.0),(1548.0,5179.0),(6874.0,7142.0),(1430
 5.0,6840.0),(10300.0,7241.0),(2484.0,7115.0)
 41,(6985.0,2335.0),(1430.0,2033.0),(4104.0,2706.0),(9162.0,5113.0),(5211.0,5198.0),(1219.0,5196.0),(7207.0,7132.0),(1410
 7.0,6840.0),(10326.0,7573.0),(2491.0,7448.0)
 42,(7026.0,2338.0),(1584.0,2022.0),(4104.0,2823.0),(9162.0,5065.0),(5182.0,4997.0),(1274.0,5194.0),(7404.0,7287.0),(1439
 4.0,6840.0),(10312.0,7393.0),(2499.0,7781.0)

43,(7358.0,2369.0),(1252.0,2046.0),(4104.0,2890.0),(9162.0,4732.0),(5086.0,4946.0),(1488.0,5186.0),(7448.0,7575.0),(1406
1.0,6840.0),(10305.0,7302.0),(2497.0,7688.0)

44,(7177.0,2352.0),(919.0,2070.0),(4104.0,2557.0),(9196.0,4401.0),(4937.0,5177.0),(1155.0,5199.0),(7398.0,7245.0),(14351
.0,6840.0),(10330.0,7634.0),(2489.0,7355.0)

45,(6845.0,2322.0),(587.0,2094.0),(4120.0,2275.0),(9168.0,4610.0),(5158.0,4983.0),(1029.0,4949.0),(7165.0,7134.0),(14150
.0,6840.0),(10307.0,7332.0),(2485.0,7197.0)

46,(7166.0,2351.0),(609.0,2093.0),(4113.0,2360.0),(9183.0,4503.0),(5134.0,4970.0),(1121.0,4760.0),(6832.0,7143.0),(14262
.0,6840.0),(10309.0,7363.0),(2493.0,7530.0)

47,(7369.0,2370.0),(941.0,2069.0),(4104.0,2471.0),(9184.0,4489.0),(5124.0,4965.0),(1454.0,4781.0),(7000.0,7138.0),(14240
.0,6840.0),(10328.0,7604.0),(2501.0,7864.0)

48,(7037.0,2339.0),(1274.0,2044.0),(4115.0,2336.0),(9166.0,4624.0),(5114.0,4960.0),(1555.0,5009.0),(7333.0,7129.0),(1417
3.0,6840.0),(10302.0,7271.0),(2495.0,7606.0)

49,(6974.0,2334.0),(1606.0,2020.0),(4118.0,2299.0),(9194.0,4414.0),(5155.0,4982.0),(1389.0,5190.0),(7093.0,7136.0),(1432
9.0,6840.0),(10314.0,7424.0),(2487.0,7273.0)

50,(7306.0,2364.0),(1647.0,1751.0),(4107.0,2435.0),(9173.0,4578.0),(5137.0,4972.0),(1075.0,5179.0),(6759.0,7145.0),(1408
3.0,6840.0),(10323.0,7543.0),(2314.0,7093.0)

51,(7229.0,2357.0),(1661.0,1892.0),(4110.0,2397.0),(9178.0,4535.0),(4900.0,5184.0),(1010.0,4852.0),(7073.0,7136.0),(1441
7.0,6840.0),(10298.0,7210.0),(2228.0,7085.0)

52,(6897.0,2327.0),(1653.0,1808.0),(4121.0,2261.0),(9189.0,4457.0),(4865.0,5184.0),(1035.0,4978.0),(7353.0,7128.0),(1408
5.0,6840.0),(10319.0,7485.0),(2561.0,7110.0)

53,(7114.0,2346.0),(1655.0,1835.0),(4112.0,2374.0),(9162.0,4656.0),(5112.0,4987.0),(1059.0,5098.0),(7020.0,7138.0),(1432
7.0,6840.0),(10319.0,7481.0),(2894.0,7110.0)

54,(7421.0,2374.0),(1658.0,1864.0),(4105.0,2458.0),(9190.0,4446.0),(5116.0,4961.0),(993.0,4771.0),(6812.0,7144.0),(14174
.0,6840.0),(10298.0,7214.0),(3227.0,7110.0)

55,(7090.0,2344.0),(1650.0,1778.0),(4107.0,2178.0),(9177.0,4546.0),(5191.0,4758.0),(1051.0,5059.0),(7146.0,7134.0),(1423
8.0,6840.0),(10324.0,7546.0),(3458.0,7110.0)

56,(6921.0,2329.0),(1664.0,1921.0),(4119.0,2288.0),(9174.0,4567.0),(5058.0,4712.0),(1267.0,5194.0),(7380.0,7028.0),(1426
4.0,6840.0),(10314.0,7420.0),(3125.0,7110.0)

57,(7253.0,2359.0),(1644.0,1722.0),(4114.0,2346.0),(9193.0,4425.0),(5208.0,4870.0),(1495.0,5186.0),(7380.0,6695.0),(1414
9.0,6840.0),(10303.0,7275.0),(2792.0,7110.0)

58,(7282.0,2362.0),(1670.0,1978.0),(4104.0,2482.0),(9162.0,4688.0),(4989.0,5121.0),(1162.0,5198.0),(7380.0,6362.0),(1435
3.0,6840.0),(10328.0,7607.0),(2509.0,7110.0)

59,(6950.0,2332.0),(1638.0,1665.0),(4114.0,2349.0),(9162.0,5022.0),(5099.0,5000.0),(1030.0,4955.0),(7380.0,6499.0),(1405
9.0,6840.0),(10309.0,7359.0),(2842.0,7110.0)

60,(7061.0,2342.0),(1672.0,1997.0),(4119.0,2285.0),(9162.0,5156.0),(5371.0,4991.0),(1014.0,4875.0),(7380.0,6832.0),(1439
3.0,6840.0),(10307.0,7336.0),(3175.0,7110.0)

61,(7393.0,2372.0),(1642.0,1703.0),(4108.0,2181.0),(9162.0,4823.0),(5626.0,4832.0),(1079.0,5201.0),(7417.0,7128.0),(1410
9.0,6840.0),(10330.0,7631.0),(3509.0,7110.0)

62,(7142.0,2349.0),(1666.0,1940.0),(4101.0,2148.0),(9162.0,4834.0),(5477.0,4660.0),(1412.0,5189.0),(7751.0,7137.0),(1430
3.0,6840.0),(10304.0,7298.0),(3177.0,7110.0)

63,(6869.0,2324.0),(1417.0,2034.0),(4095.0,2115.0),(9162.0,5167.0),(5215.0,4860.0),(1349.0,5191.0),(8043.0,7145.0),(1419
8.0,6840.0),(10512.0,7178.0),(2844.0,7110.0)

64,(7201.0,2354.0),(1084.0,2058.0),(4106.0,2445.0),(9162.0,5011.0),(5282.0,5052.0),(1067.0,5140.0),(7710.0,7136.0),(1421
4.0,6840.0),(10845.0,7174.0),(2510.0,7110.0)

65,(7334.0,2366.0),(752.0,2082.0),(4117.0,2309.0),(9162.0,4677.0),(4987.0,4896.0),(1002.0,4813.0),(7377.0,7128.0),(14288

.0,6840.0),(11178.0,7169.0),(2179.0,7080.0)

66,(7002.0,2336.0),(444.0,2105.0),(4116.0,2325.0),(9162.0,4979.0),(4994.0,4900.0),(1260.0,4768.0),(7043.0,7137.0),(14125.0,6840.0),(11512.0,7164.0),(2364.0,7098.0)

67,(7009.0,2337.0),(776.0,2081.0),(4105.0,2461.0),(9162.0,5199.0),(5288.0,5056.0),(1564.0,4815.0),(6789.0,7144.0),(14377.0,6840.0),(11302.0,7167.0),(2488.0,7323.0)

68,(7341.0,2367.0),(1109.0,2056.0),(4112.0,2370.0),(9162.0,4865.0),(5001.0,4903.0),(1549.0,5148.0),(7122.0,7135.0),(14044.0,6840.0),(10969.0,7172.0),(2496.0,7656.0)

69,(7194.0,2354.0),(1441.0,2032.0),(4120.0,2264.0),(9162.0,4791.0),(5024.0,5082.0),(1250.0,5195.0),(7304.0,7130.0),(14369.0,6840.0),(10636.0,7177.0),(2499.0,7813.0)

70,(6862.0,2324.0),(1664.0,1916.0),(4110.0,2400.0),(9162.0,5124.0),(5064.0,5039.0),(1048.0,5042.0),(6971.0,7139.0),(14133.0,6840.0),(10302.0,7181.0),(2492.0,7480.0)

71,(7149.0,2350.0),(1645.0,1727.0),(4108.0,2431.0),(9162.0,5053.0),(5247.0,5033.0),(997.0,4788.0),(6862.0,7142.0),(14279.0,6840.0),(10321.0,7507.0),(2484.0,7147.0)

72,(7386.0,2371.0),(1669.0,1972.0),(4118.0,2296.0),(9162.0,4720.0),(5019.0,4912.0),(1062.0,5114.0),(7195.0,7133.0),(14222.0,6840.0),(10317.0,7460.0),(2189.0,7081.0)

73,(7054.0,2341.0),(1639.0,1670.0),(4115.0,2339.0),(9197.0,4394.0),(4987.0,5122.0),(1324.0,5192.0),(7231.0,7132.0),(14190.0,6840.0),(10300.0,7236.0),(2354.0,7097.0)

74,(6957.0,2332.0),(1672.0,2002.0),(4104.0,2474.0),(9170.0,4599.0),(5101.0,4999.0),(1438.0,5188.0),(6898.0,7141.0),(14312.0,6840.0),(10325.0,7568.0),(2281.0,7090.0)

75,(7289.0,2362.0),(1354.0,2039.0),(4113.0,2357.0),(9181.0,4515.0),(5199.0,5006.0),(1105.0,5201.0),(6934.0,7140.0),(14101.0,6840.0),(10312.0,7398.0),(2262.0,7088.0)

76,(7246.0,2358.0),(1022.0,2063.0),(4119.0,2278.0),(9162.0,4847.0),(5068.0,4937.0),(1019.0,4900.0),(7268.0,7131.0),(14401.0,6840.0),(10304.0,7297.0),(2374.0,7099.0)

77,(6914.0,2328.0),(690.0,2087.0),(4109.0,2413.0),(9162.0,5180.0),(5222.0,5019.0),(1025.0,4930.0),(7158.0,7134.0),(14068.0,6840.0),(10330.0,7629.0),(2169.0,7080.0)

78,(7097.0,2345.0),(506.0,2100.0),(4109.0,2418.0),(9162.0,4998.0),(5255.0,5037.0),(1136.0,5199.0),(6825.0,7143.0),(14345.0,6840.0),(10307.0,7337.0),(2466.0,7108.0)

79,(7429.0,2375.0),(838.0,2076.0),(4119.0,2282.0),(9162.0,4664.0),(5001.0,5108.0),(1469.0,5187.0),(7007.0,7138.0),(14157.0,6840.0),(10309.0,7358.0),(2134.0,7076.0)

80,(7443.0,2704.0),(1171.0,2052.0),(4104.0,2615.0),(9190.0,4449.0),(5162.0,5196.0),(1293.0,5193.0),(7340.0,7129.0),(14255.0,6840.0),(10328.0,7609.0),(2409.0,7102.0)

81,(7450.0,2938.0),(1503.0,2028.0),(4104.0,2883.0),(9177.0,4543.0),(5129.0,5370.0),(1056.0,5084.0),(7674.0,7135.0),(14246.0,6840.0),(10303.0,7276.0),(2489.0,7368.0)

82,(7440.0,2605.0),(1511.0,2027.0),(4104.0,2850.0),(9174.0,4570.0),(5086.0,5595.0),(991.0,4757.0),(8007.0,7144.0),(14166.0,6840.0),(10314.0,7419.0),(2497.0,7701.0)

83,(7437.0,2479.0),(1179.0,2051.0),(4104.0,2648.0),(9162.0,4903.0),(5181.0,5310.0),(1054.0,5073.0),(7787.0,7138.0),(14336.0,6840.0),(10324.0,7548.0),(2498.0,7769.0)

84,(7446.0,2813.0),(846.0,2075.0),(4104.0,2652.0),(9162.0,5236.0),(5150.0,5346.0),(1282.0,5194.0),(7454.0,7129.0),(14077.0,6840.0),(10298.0,7215.0),(2491.0,7436.0)

85,(7447.0,2829.0),(514.0,2100.0),(4104.0,2845.0),(9162.0,4942.0),(5092.0,5625.0),(1615.0,5182.0),(7419.0,7383.0),(14410.0,6840.0),(10318.0,7480.0),(2484.0,7117.0)

86,(7437.0,2496.0),(682.0,2087.0),(4104.0,2887.0),(9169.0,4609.0),(5089.0,5609.0),(1948.0,5172.0),(7433.0,7478.0),(14092.0,6840.0),(10319.0,7487.0),(2491.0,7450.0)

87,(7440.0,2588.0),(1014.0,2063.0),(4104.0,2611.0),(9182.0,4504.0),(5138.0,5359.0),(2110.0,5168.0),(7383.0,7149.0),(14321.0,6840.0),(10298.0,7209.0),(2499.0,7783.0)

88,(7450.0,2921.0),(1347.0,2039.0),(4104.0,2690.0),(9185.0,4488.0),(5192.0,5297.0),(1777.0,5177.0),(7691.0,7136.0),(1418.0,6840.0),(10323.0,7541.0),(2497.0,7687.0)

89,(7444.0,2720.0),(1668.0,2016.0),(4104.0,2808.0),(9166.0,4625.0),(5082.0,5578.0),(1444.0,5187.0),(8025.0,7144.0),(1423.0,6840.0),(10314.0,7426.0),(2489.0,7353.0)

90,(7434.0,2387.0),(1336.0,2040.0),(4104.0,2907.0),(9162.0,4958.0),(5076.0,5545.0),(1110.0,5200.0),(7769.0,7138.0),(1427.0,6840.0),(10384.0,7180.0),(2395.0,7101.0)

91,(7443.0,2697.0),(1003.0,2064.0),(4104.0,2789.0),(9162.0,5219.0),(5096.0,5408.0),(1382.0,5190.0),(7436.0,7129.0),(1414.2,6840.0),(10717.0,7176.0),(2148.0,7078.0)

92,(7450.0,2945.0),(671.0,2088.0),(4104.0,2511.0),(9162.0,4886.0),(5234.0,5248.0),(1380.0,5190.0),(7422.0,7401.0),(14360.0,6840.0),(11051.0,7171.0),(2480.0,7109.0)

93,(7440.0,2612.0),(525.0,2099.0),(4104.0,2844.0),(9176.0,4554.0),(5070.0,5515.0),(1112.0,5200.0),(7587.0,7610.0),(14053.0,6840.0),(11384.0,7166.0),(2155.0,7078.0)

94,(7436.0,2472.0),(857.0,2075.0),(4104.0,2869.0),(9175.0,4560.0),(5063.0,5482.0),(1445.0,5187.0),(7919.0,7646.0),(14386.0,6840.0),(11541.0,7024.0),(2388.0,7100.0)

95,(7446.0,2806.0),(1190.0,2051.0),(4104.0,2536.0),(9192.0,4433.0),(5059.0,5459.0),(1317.0,5192.0),(8104.0,7535.0),(14116.0,6840.0),(11465.0,6699.0),(2247.0,7087.0)

96,(7447.0,2836.0),(1522.0,2026.0),(4104.0,2764.0),(9180.0,4100.0),(5165.0,5759.0),(1175.0,5198.0),(8069.0,7203.0),(14297.0,6840.0),(11389.0,6374.0),(2295.0,7092.0)

97,(7437.0,2503.0),(1655.0,1835.0),(4104.0,2882.0),(9197.0,4380.0),(5144.0,5723.0),(1508.0,5185.0),(8092.0,7419.0),(14205.0,6840.0),(11101.0,6318.0),(2340.0,7096.0)

98,(7440.0,2581.0),(1653.0,1808.0),(4104.0,2832.0),(9182.0,4124.0),(5067.0,5500.0),(1254.0,5195.0),(8109.0,7584.0),(14207.0,6840.0),(10767.0,6318.0),(2203.0,7083.0)

99,(7449.0,2914.0),(1661.0,1891.0),(4104.0,2499.0),(9194.0,4326.0),(5060.0,5467.0),(1238.0,5195.0),(8075.0,7253.0),(14294.0,6840.0),(10434.0,6318.0),(2485.0,7161.0)

100,(7444.0,2727.0),(1647.0,1751.0),(4104.0,2802.0),(9178.0,3993.0),(5061.0,5473.0),(1524.0,5184.0),(7838.0,7140.0),(14118.0,6840.0),(10382.0,6318.0),(2492.0,7494.0)

101,(7434.0,2394.0),(1667.0,1948.0),(4104.0,2912.0),(9174.0,3660.0),(5261.0,5216.0),(1191.0,5197.0),(7505.0,7131.0),(14384.0,6840.0),(10716.0,6318.0),(2500.0,7828.0)

102,(7120.0,2347.0),(1641.0,1695.0),(4104.0,2795.0),(9169.0,3327.0),(5276.0,5178.0),(1036.0,4984.0),(7588.0,7133.0),(14050.0,6840.0),(11049.0,6318.0),(2496.0,7642.0)

103,(6891.0,2326.0),(1672.0,2005.0),(4105.0,2461.0),(9164.0,2993.0),(5153.0,4981.0),(1008.0,4846.0),(7921.0,7142.0),(14362.0,6840.0),(11369.0,6318.0),(2488.0,7309.0)

104,(7223.0,2356.0),(1639.0,1673.0),(4112.0,2370.0),(9163.0,2919.0),(5138.0,4972.0),(1074.0,5173.0),(7873.0,7140.0),(14140.0,6840.0),(11035.0,6318.0),(2617.0,7110.0)

105,(7437.0,2497.0),(1669.0,1970.0),(4120.0,2265.0),(9168.0,3252.0),(5130.0,4967.0),(1020.0,4903.0),(7540.0,7132.0),(14273.0,6840.0),(10702.0,6318.0),(2951.0,7110.0)

106,(7447.0,2830.0),(1645.0,1730.0),(4110.0,2400.0),(9173.0,3586.0),(5209.0,4869.0),(1025.0,4927.0),(7207.0,7132.0),(14229.0,6840.0),(10369.0,6318.0),(3284.0,7110.0)

107,(7446.0,2811.0),(1663.0,1913.0),(4108.0,2431.0),(9177.0,3919.0),(5259.0,4806.0),(1069.0,5149.0),(6873.0,7142.0),(14183.0,6840.0),(10254.0,6523.0),(3402.0,7110.0)

108,(7437.0,2478.0),(1651.0,1786.0),(4112.0,2204.0),(9189.0,4252.0),(5536.0,4623.0),(1004.0,4822.0),(6959.0,7140.0),(14318.0,6840.0),(10275.0,6856.0),(3068.0,7110.0)

109,(7204.0,2355.0),(1658.0,1856.0),(4119.0,2238.0),(9172.0,4583.0),(5631.0,4901.0),(1041.0,5007.0),(7292.0,7130.0),(14094.0,6840.0),(10296.0,7189.0),(2735.0,7110.0)

110,(6872.0,2324.0),(1656.0,1843.0),(4117.0,2228.0),(9195.0,4409.0),(5616.0,5205.0),(1215.0,5196.0),(7625.0,7134.0),(144

08.0,6840.0),(10322.0,7521.0),(2565.0,7110.0)
 111,(7139.0,2349.0),(1652.0,1800.0),(4109.0,2407.0),(9180.0,4094.0),(5633.0,4873.0),(1547.0,5184.0),(7958.0,7143.0),(140
 74.0,6840.0),(10316.0,7446.0),(2899.0,7110.0)
 112,(7396.0,2372.0),(1662.0,1900.0),(4087.0,2077.0),(9193.0,4427.0),(5568.0,4869.0),(1214.0,5196.0),(7835.0,7139.0),(143
 38.0,6840.0),(10301.0,7250.0),(3232.0,7110.0)
 113,(7064.0,2342.0),(1646.0,1743.0),(4094.0,2110.0),(9174.0,4566.0),(5594.0,4721.0),(1278.0,5194.0),(7502.0,7131.0),(141
 64.0,6840.0),(10326.0,7582.0),(3454.0,7110.0)
 114,(6947.0,2331.0),(1614.0,2020.0),(4100.0,2144.0),(9177.0,4548.0),(5379.0,4722.0),(1484.0,5186.0),(7590.0,7133.0),(142
 49.0,6840.0),(10311.0,7385.0),(3120.0,7110.0)
 115,(7279.0,2361.0),(1282.0,2044.0),(4116.0,2324.0),(9190.0,4445.0),(5143.0,4951.0),(1151.0,5199.0),(7923.0,7142.0),(142
 53.0,6840.0),(10425.0,7180.0),(2787.0,7110.0)
 116,(7439.0,2553.0),(949.0,2068.0),(4117.0,2311.0),(9162.0,4669.0),(4951.0,5185.0),(1028.0,4945.0),(7870.0,7140.0),(1415
 9.0,6840.0),(10758.0,7175.0),(2513.0,7110.0)
 117,(7449.0,2887.0),(617.0,2092.0),(4104.0,2520.0),(9162.0,5002.0),(5275.0,5191.0),(1126.0,4760.0),(7537.0,7132.0),(1434
 2.0,6840.0),(11092.0,7170.0),(2847.0,7110.0)
 118,(7445.0,2755.0),(579.0,2095.0),(4104.0,2854.0),(9162.0,5176.0),(5444.0,4931.0),(1458.0,4781.0),(7204.0,7133.0),(1407
 0.0,6840.0),(11425.0,7166.0),(3180.0,7110.0)
 119,(7435.0,2422.0),(911.0,2071.0),(4104.0,2860.0),(9162.0,4842.0),(5600.0,4741.0),(1555.0,5013.0),(6871.0,7142.0),(1440
 3.0,6840.0),(11389.0,7166.0),(3506.0,7110.0)
 120,(7148.0,2350.0),(1244.0,2047.0),(4104.0,2743.0),(9182.0,4510.0),(5615.0,4794.0),(1385.0,5190.0),(6961.0,7139.0),(140
 98.0,6840.0),(11055.0,7171.0),(3172.0,7110.0)
 121,(6863.0,2324.0),(1576.0,2023.0),(4109.0,2409.0),(9169.0,4603.0),(5620.0,5125.0),(1107.0,5200.0),(7295.0,7130.0),(143
 14.0,6840.0),(10722.0,7175.0),(2839.0,7110.0)
 122,(7195.0,2354.0),(1650.0,1781.0),(4108.0,2422.0),(9197.0,4393.0),(5627.0,4981.0),(1441.0,5188.0),(7628.0,7134.0),(141
 88.0,6840.0),(10389.0,7180.0),(2506.0,7110.0)
 123,(7340.0,2367.0),(1658.0,1862.0),(4114.0,2213.0),(9162.0,4724.0),(5443.0,4931.0),(1321.0,5192.0),(7961.0,7143.0),(142
 25.0,6840.0),(10536.0,7178.0),(2795.0,7110.0)
 124,(7008.0,2337.0),(1656.0,1838.0),(4121.0,2253.0),(9162.0,5058.0),(5396.0,4970.0),(1171.0,5198.0),(7833.0,7139.0),(142
 77.0,6840.0),(10869.0,7173.0),(3128.0,7110.0)
 125,(7003.0,2336.0),(1652.0,1805.0),(4111.0,2382.0),(9162.0,5120.0),(5573.0,4868.0),(1504.0,5185.0),(7500.0,7131.0),(141
 35.0,6840.0),(11202.0,7169.0),(3461.0,7110.0)
 126,(7335.0,2367.0),(1552.0,2024.0),(4089.0,2086.0),(9162.0,4787.0),(5520.0,4876.0),(1258.0,5195.0),(7412.0,7338.0),(143
 66.0,6840.0),(11535.0,7164.0),(3224.0,7110.0)
 127,(7200.0,2354.0),(1219.0,2048.0),(4089.0,2086.0),(9162.0,4870.0),(5623.0,5078.0),(1234.0,5196.0),(7524.0,7603.0),(140
 46.0,6840.0),(11278.0,7168.0),(2891.0,7110.0)
 128,(6868.0,2324.0),(887.0,2072.0),(4111.0,2382.0),(9162.0,5203.0),(5625.0,5028.0),(1528.0,5184.0),(7856.0,7639.0),(1437
 9.0,6840.0),(10945.0,7172.0),(2558.0,7110.0)
 129,(7143.0,2349.0),(555.0,2097.0),(4121.0,2253.0),(9162.0,4975.0),(5625.0,5024.0),(1195.0,5197.0),(8048.0,7660.0),(1412
 2.0,6840.0),(10612.0,7177.0),(2743.0,7110.0)
 130,(7392.0,2372.0),(641.0,2090.0),(4114.0,2213.0),(9164.0,4641.0),(5478.0,5212.0),(1297.0,5193.0),(7717.0,7624.0),(1429
 0.0,6840.0),(10313.0,7181.0),(3076.0,7110.0)
 131,(7060.0,2342.0),(973.0,2066.0),(4104.0,2545.0),(9187.0,4472.0),(5402.0,5208.0),(1465.0,5187.0),(7441.0,7530.0),(1421
 2.0,6840.0),(10646.0,7177.0),(3409.0,7110.0)
 132,(6951.0,2332.0),(1306.0,2042.0),(4104.0,2878.0),(9180.0,4521.0),(5621.0,5100.0),(1132.0,5199.0),(7391.0,7200.0),(142
 01.0,6840.0),(10979.0,7172.0),(3276.0,7110.0)

133,(7283.0,2362.0),(1638.0,2018.0),(4104.0,2835.0),(9171.0,4593.0),(5542.0,4873.0),(1360.0,5191.0),(7639.0,7134.0),(14301.0,6840.0),(11313.0,7167.0),(2943.0,7110.0)

134,(7252.0,2359.0),(1376.0,2037.0),(4104.0,2718.0),(9162.0,4926.0),(5289.0,5074.0),(1401.0,5189.0),(7972.0,7143.0),(14111.0,6840.0),(11501.0,7165.0),(2610.0,7110.0)

135,(6920.0,2329.0),(1044.0,2061.0),(4104.0,2582.0),(9162.0,5252.0),(5478.0,5212.0),(1077.0,5191.0),(8088.0,7386.0),(14390.0,6840.0),(11168.0,7169.0),(2277.0,7090.0)

136,(7091.0,2344.0),(711.0,2085.0),(4104.0,2915.0),(9162.0,4919.0),(5420.0,5209.0),(1012.0,4864.0),(8112.0,7617.0),(14057.0,6840.0),(10835.0,7174.0),(2266.0,7089.0)

137,(7423.0,2375.0),(484.0,2102.0),(4104.0,2817.0),(9172.0,4586.0),(5460.0,5211.0),(1208.0,4765.0),(8078.0,7285.0),(14355.0,6840.0),(10501.0,7179.0),(2369.0,7099.0)

138,(7112.0,2346.0),(817.0,2078.0),(4104.0,2681.0),(9179.0,4527.0),(5438.0,5210.0),(1541.0,4786.0),(7871.0,7140.0),(14146.0,6840.0),(10288.0,7054.0),(2173.0,7080.0)

139,(6898.0,2327.0),(1149.0,2054.0),(4114.0,2348.0),(9188.0,4465.0),(5271.0,5046.0),(1551.0,5096.0),(7538.0,7132.0),(14266.0,6840.0),(10267.0,6721.0),(2462.0,7107.0)

140,(7230.0,2357.0),(1482.0,2029.0),(4104.0,2483.0),(9182.0,4133.0),(4995.0,4900.0),(1302.0,5193.0),(7204.0,7133.0),(14236.0,6840.0),(10246.0,6389.0),(2130.0,7076.0)

141,(7305.0,2364.0),(1659.0,1875.0),(4114.0,2348.0),(9196.0,4371.0),(5006.0,5102.0),(1058.0,5093.0),(6871.0,7142.0),(14177.0,6840.0),(10258.0,6579.0),(2413.0,7103.0)

142,(6973.0,2334.0),(1649.0,1767.0),(4104.0,2681.0),(9167.0,4622.0),(5082.0,5019.0),(992.0,4766.0),(6961.0,7139.0),(14325.0,6840.0),(10279.0,6912.0),(2489.0,7372.0)

143,(7038.0,2340.0),(1665.0,1932.0),(4104.0,2817.0),(9184.0,4492.0),(5223.0,5020.0),(1052.0,5063.0),(7294.0,7130.0),(14087.0,6840.0),(10300.0,7244.0),(2497.0,7705.0)

144,(7370.0,2370.0),(1643.0,1711.0),(4104.0,2915.0),(9183.0,4501.0),(5044.0,4925.0),(1042.0,5013.0),(7132.0,7135.0),(14414.0,6840.0),(10326.0,7577.0),(2498.0,7765.0)

145,(7165.0,2351.0),(1671.0,1989.0),(4104.0,2582.0),(9168.0,4612.0),(5054.0,4930.0),(1003.0,4817.0),(6798.0,7144.0),(14081.0,6840.0),(10312.0,7390.0),(2491.0,7431.0)

146,(6846.0,2322.0),(1638.0,1657.0),(4121.0,2250.0),(9162.0,4945.0),(5064.0,4935.0),(1068.0,5144.0),(7034.0,7137.0),(14331.0,6840.0),(10305.0,7305.0),(2472.0,7108.0)

147,(7178.0,2352.0),(1671.0,1986.0),(4111.0,2385.0),(9162.0,5232.0),(5226.0,5021.0),(1026.0,4932.0),(7367.0,7128.0),(14170.0,6840.0),(10331.0,7638.0),(2140.0,7077.0)

148,(7436.0,2452.0),(1643.0,1714.0),(4106.0,2446.0),(9162.0,4899.0),(4979.0,5132.0),(1019.0,4898.0),(7059.0,7137.0),(14242.0,6840.0),(10307.0,7329.0),(2402.0,7102.0)

149,(7446.0,2786.0),(1665.0,1929.0),(4104.0,2779.0),(9174.0,4567.0),(5194.0,5197.0),(1075.0,5178.0),(6773.0,7145.0),(14260.0,6840.0),(10310.0,7366.0),(2489.0,7361.0)

150,(7448.0,2856.0),(1649.0,1770.0),(4104.0,2897.0),(9177.0,4547.0),(5020.0,5188.0),(1010.0,4852.0),(7107.0,7135.0),(14153.0,6840.0),(10328.0,7600.0),(2497.0,7695.0)

151,(7438.0,2523.0),(1659.0,1872.0),(4104.0,2817.0),(9162.0,4879.0),(4744.0,5184.0),(1035.0,4978.0),(7319.0,7129.0),(14349.0,6840.0),(10302.0,7268.0),(2499.0,7775.0)

152,(7439.0,2561.0),(1655.0,1827.0),(4104.0,2484.0),(9162.0,5213.0),(5077.0,5191.0),(1059.0,5098.0),(6986.0,7139.0),(14063.0,6840.0),(10314.0,7427.0),(2491.0,7442.0)

153,(7449.0,2894.0),(1654.0,1816.0),(4104.0,2817.0),(9162.0,4965.0),(5291.0,5065.0),(993.0,4771.0),(6846.0,7143.0),(14397.0,6840.0),(10323.0,7539.0),(2483.0,7109.0)

154,(7444.0,2747.0),(1541.0,2025.0),(4104.0,2897.0),(9165.0,4632.0),(5287.0,5092.0),(1051.0,5059.0),(7179.0,7133.0),(14105.0,6840.0),(10297.0,7207.0),(2151.0,7078.0)

155,(7435.0,2414.0),(1209.0,2049.0),(4104.0,2780.0),(9186.0,4481.0),(5282.0,5135.0),(1043.0,5017.0),(7513.0,7131.0),(143

07.0,6840.0),(10319.0,7488.0),(2392.0,7101.0)
 156,(7140.0,2349.0),(876.0,2073.0),(4104.0,2521.0),(9182.0,4511.0),(5008.0,5188.0),(1002.0,4813.0),(7846.0,7140.0),(1419
 4.0,6840.0),(10318.0,7478.0),(2489.0,7351.0)
 157,(6871.0,2324.0),(544.0,2097.0),(4104.0,2854.0),(9169.0,4602.0),(5105.0,4994.0),(1067.0,5139.0),(7948.0,7142.0),(1421
 8.0,6840.0),(10331.0,7181.0),(2497.0,7684.0)
 158,(7203.0,2355.0),(652.0,2090.0),(4104.0,2860.0),(9197.0,4393.0),(5193.0,5003.0),(1349.0,5191.0),(7615.0,7134.0),(1428
 4.0,6840.0),(10665.0,7176.0),(2499.0,7786.0)
 159,(7436.0,2477.0),(984.0,2065.0),(4104.0,2742.0),(9162.0,4724.0),(5074.0,4940.0),(1413.0,5189.0),(7282.0,7130.0),(1412
 9.0,6840.0),(10998.0,7172.0),(2491.0,7452.0)
 160,(7446.0,2810.0),(1317.0,2041.0),(4109.0,2409.0),(9162.0,5057.0),(4946.0,5168.0),(1079.0,5201.0),(6948.0,7140.0),(143
 73.0,6840.0),(11331.0,7167.0),(2484.0,7119.0)
 161,(7447.0,2831.0),(1649.0,2017.0),(4108.0,2422.0),(9162.0,5121.0),(4930.0,5198.0),(1014.0,4875.0),(6615.0,7137.0),(140
 40.0,6840.0),(11483.0,7165.0),(2161.0,7079.0)
 162,(7437.0,2498.0),(1643.0,1708.0),(4104.0,2755.0),(9162.0,4787.0),(4819.0,5184.0),(1030.0,4955.0),(6676.0,7141.0),(143
 73.0,6840.0),(11149.0,7169.0),(2381.0,7100.0)
 163,(7440.0,2586.0),(1665.0,1934.0),(4104.0,2873.0),(9189.0,4456.0),(5152.0,5195.0),(1063.0,5121.0),(7009.0,7138.0),(141
 29.0,6840.0),(10816.0,7174.0),(2254.0,7088.0)
 164,(7449.0,2919.0),(1648.0,1765.0),(4104.0,2841.0),(9162.0,4657.0),(5283.0,5123.0),(998.0,4794.0),(7342.0,7129.0),(1428
 3.0,6840.0),(10483.0,7179.0),(2289.0,7091.0)
 165,(7444.0,2723.0),(1660.0,1878.0),(4104.0,2508.0),(9162.0,4991.0),(5278.0,5166.0),(1046.0,5036.0),(7676.0,7135.0),(142
 18.0,6840.0),(10307.0,7327.0),(2346.0,7096.0)
 166,(7434.0,2390.0),(1654.0,1821.0),(4116.0,2324.0),(9162.0,5187.0),(5080.0,5427.0),(1244.0,5195.0),(8009.0,7144.0),(141
 94.0,6840.0),(10331.0,7639.0),(2197.0,7082.0)
 167,(7443.0,2695.0),(1654.0,1821.0),(4104.0,2657.0),(9162.0,4854.0),(5071.0,5520.0),(1518.0,5185.0),(7785.0,7138.0),(143
 08.0,6840.0),(10305.0,7307.0),(2485.0,7155.0)
 168,(7450.0,2947.0),(1536.0,2025.0),(4104.0,2841.0),(9180.0,4521.0),(5077.0,5553.0),(1185.0,5197.0),(7452.0,7129.0),(141
 05.0,6840.0),(10503.0,7179.0),(2492.0,7488.0)
 169,(7441.0,2614.0),(1203.0,2050.0),(4104.0,2891.0),(9171.0,4592.0),(5208.0,5278.0),(1307.0,5193.0),(7640.0,7134.0),(143
 97.0,6840.0),(10836.0,7174.0),(2500.0,7821.0)
 170,(7436.0,2470.0),(871.0,2074.0),(4104.0,2606.0),(9196.0,4401.0),(5122.0,5378.0),(1455.0,5187.0),(7974.0,7143.0),(1406
 4.0,6840.0),(11169.0,7169.0),(2496.0,7648.0)
 171,(7446.0,2803.0),(538.0,2098.0),(4104.0,2694.0),(9162.0,4713.0),(5084.0,5584.0),(1122.0,5200.0),(8089.0,7388.0),(1434
 9.0,6840.0),(11503.0,7164.0),(2488.0,7315.0)
 172,(7447.0,2838.0),(657.0,2089.0),(4104.0,2811.0),(9162.0,5047.0),(5102.0,5648.0),(1370.0,5190.0),(8112.0,7616.0),(1415
 3.0,6840.0),(11514.0,6908.0),(2356.0,7097.0)
 173,(7437.0,2505.0),(989.0,2065.0),(4104.0,2902.0),(9162.0,5131.0),(5208.0,5833.0),(1703.0,5179.0),(8078.0,7284.0),(1425
 9.0,6840.0),(11438.0,6583.0),(2186.0,7081.0)
 174,(7439.0,2579.0),(1322.0,2041.0),(4104.0,2785.0),(9162.0,4798.0),(5073.0,5532.0),(2037.0,5170.0),(8083.0,7338.0),(142
 42.0,6840.0),(11389.0,6376.0),(2484.0,7144.0)
 175,(7449.0,2912.0),(1654.0,2017.0),(4104.0,2515.0),(9188.0,4466.0),(5128.0,5694.0),(2021.0,5170.0),(8117.0,7665.0),(141
 70.0,6840.0),(11465.0,6701.0),(2492.0,7478.0)
 176,(7444.0,2730.0),(1360.0,2038.0),(4104.0,2849.0),(9182.0,4134.0),(5182.0,5788.0),(1688.0,5180.0),(8083.0,7333.0),(143
 32.0,6840.0),(11541.0,7025.0),(2499.0,7811.0)
 177,(7434.0,2396.0),(1028.0,2062.0),(4104.0,2865.0),(9196.0,4370.0),(5063.0,5481.0),(1355.0,5191.0),(7919.0,7142.0),(140
 81.0,6840.0),(11530.0,6977.0),(2496.0,7659.0)

178,(7443.0,2688.0),(695.0,2086.0),(4104.0,2532.0),(9167.0,4623.0),(5256.0,5222.0),(1137.0,5199.0),(7586.0,7133.0),(1441
 4.0,6840.0),(11454.0,6652.0),(2488.0,7326.0)
 179,(7451.0,2954.0),(500.0,2101.0),(4104.0,2769.0),(9184.0,4491.0),(5074.0,5434.0),(1470.0,5186.0),(7506.0,7131.0),(1408
 8.0,6840.0),(11378.0,6328.0),(2367.0,7098.0)
 180,(7441.0,2621.0),(833.0,2076.0),(4104.0,2886.0),(9183.0,4502.0),(5069.0,5511.0),(1559.0,4928.0),(7839.0,7140.0),(1432
 5.0,6840.0),(11449.0,6632.0),(2176.0,7080.0)
 181,(7436.0,2463.0),(1165.0,2052.0),(4104.0,2828.0),(9184.0,4170.0),(5236.0,5245.0),(1557.0,4980.0),(8075.0,7254.0),(141
 77.0,6840.0),(11525.0,6956.0),(2459.0,7107.0)
 182,(7446.0,2797.0),(1498.0,2028.0),(4104.0,2494.0),(9194.0,4334.0),(5549.0,5216.0),(1418.0,5188.0),(8109.0,7586.0),(142
 35.0,6840.0),(11546.0,7046.0),(2127.0,7076.0)
 183,(7447.0,2845.0),(1658.0,1859.0),(4115.0,2337.0),(9162.0,4658.0),(5349.0,5205.0),(1085.0,5201.0),(8092.0,7418.0),(142
 66.0,6840.0),(11470.0,6722.0),(2415.0,7103.0)
 184,(7438.0,2512.0),(1650.0,1783.0),(4104.0,2670.0),(9194.0,4327.0),(5531.0,5215.0),(1015.0,4880.0),(8070.0,7205.0),(141
 46.0,6840.0),(11394.0,6397.0),(2490.0,7374.0)
 185,(7238.0,2358.0),(1664.0,1916.0),(4104.0,2828.0),(9185.0,4176.0),(5368.0,5206.0),(1029.0,4950.0),(8104.0,7536.0),(143
 56.0,6840.0),(11124.0,6318.0),(2497.0,7708.0)
 186,(6906.0,2328.0),(1645.0,1727.0),(4104.0,2905.0),(9191.0,4274.0),(5209.0,5012.0),(1064.0,5126.0),(8097.0,7467.0),(140
 57.0,6840.0),(10791.0,6318.0),(2498.0,7762.0)
 187,(7105.0,2346.0),(1669.0,1973.0),(4104.0,2593.0),(9188.0,4230.0),(5242.0,5030.0),(999.0,4799.0),(8053.0,7145.0),(1439
 0.0,6840.0),(10457.0,6318.0),(2491.0,7429.0)
 188,(7430.0,2375.0),(1639.0,1670.0),(4121.0,2260.0),(9187.0,4220.0),(5025.0,4915.0),(1045.0,5031.0),(7720.0,7136.0),(141
 12.0,6840.0),(10249.0,6435.0),(2497.0,7110.0)
 189,(7098.0,2345.0),(1672.0,2002.0),(4110.0,2396.0),(9191.0,4283.0),(4983.0,5127.0),(1048.0,5045.0),(7387.0,7128.0),(143
 01.0,6840.0),(10270.0,6768.0),(2831.0,7110.0)
 190,(6913.0,2328.0),(1642.0,1698.0),(4107.0,2435.0),(9168.0,4615.0),(5105.0,4994.0),(996.0,4785.0),(7054.0,7137.0),(1420
 1.0,6840.0),(10290.0,7100.0),(3164.0,7110.0)
 191,(7245.0,2358.0),(1666.0,1945.0),(4112.0,2200.0),(9196.0,4405.0),(5211.0,4765.0),(1061.0,5111.0),(6778.0,7145.0),(142
 11.0,6840.0),(10315.0,7433.0),(3497.0,7110.0)
 192,(7290.0,2362.0),(1647.0,1754.0),(4120.0,2266.0),(9162.0,4736.0),(4974.0,4754.0),(1321.0,5192.0),(7112.0,7135.0),(142
 90.0,6840.0),(10323.0,7534.0),(3188.0,7110.0)
 193,(6958.0,2332.0),(1661.0,1889.0),(4112.0,2369.0),(9162.0,5069.0),(5156.0,4982.0),(1441.0,5188.0),(7314.0,7129.0),(141
 22.0,6840.0),(10297.0,7202.0),(2855.0,7110.0)
 194,(7053.0,2341.0),(1653.0,1811.0),(4105.0,2463.0),(9162.0,5108.0),(5429.0,4943.0),(1108.0,5200.0),(6981.0,7139.0),(143
 80.0,6840.0),(10320.0,7494.0),(2522.0,7110.0)
 195,(7385.0,2371.0),(1655.0,1832.0),(4116.0,2327.0),(9162.0,4775.0),(5330.0,5025.0),(1384.0,5190.0),(6851.0,7143.0),(140
 46.0,6840.0),(10318.0,7473.0),(2190.0,7082.0)
 196,(7150.0,2350.0),(1525.0,2026.0),(4117.0,2308.0),(9162.0,4881.0),(5136.0,5194.0),(1378.0,5190.0),(7184.0,7133.0),(143
 66.0,6840.0),(10299.0,7222.0),(2353.0,7097.0)
 197,(6861.0,2323.0),(1193.0,2050.0),(4107.0,2443.0),(9162.0,5215.0),(5018.0,5089.0),(1073.0,5167.0),(7241.0,7131.0),(141
 36.0,6840.0),(10324.0,7555.0),(2686.0,7110.0)
 198,(7193.0,2354.0),(860.0,2074.0),(4111.0,2388.0),(9162.0,4963.0),(4993.0,4899.0),(1007.0,4841.0),(6908.0,7141.0),(1427
 7.0,6840.0),(10313.0,7412.0),(3019.0,7110.0)
 199,(7342.0,2367.0),(528.0,2099.0),(4121.0,2247.0),(9166.0,4630.0),(5242.0,4827.0),(1232.0,4767.0),(6924.0,7141.0),(1422
 5.0,6840.0),(10303.0,7283.0),(3353.0,7110.0)
 200,(7010.0,2337.0),(668.0,2088.0),(4115.0,2219.0),(9185.0,4483.0),(5006.0,4693.0),(1564.0,4787.0),(7257.0,7131.0),(1418

7.0,6840.0),(10329.0,7616.0),(3333.0,7110.0)
 201,(7001.0,2336.0),(1000.0,2064.0),(4109.0,2416.0),(9182.0,4509.0),(5243.0,4777.0),(1550.0,5119.0),(7169.0,7134.0),(143
 14.0,6840.0),(10309.0,7351.0),(3000.0,7110.0)
 202,(7333.0,2366.0),(1333.0,2040.0),(4109.0,2415.0),(9169.0,4604.0),(5067.0,4936.0),(1279.0,5194.0),(6835.0,7143.0),(140
 98.0,6840.0),(10308.0,7345.0),(2666.0,7110.0)
 203,(7440.0,2608.0),(1665.0,2016.0),(4119.0,2280.0),(9162.0,4937.0),(5074.0,5028.0),(1053.0,5070.0),(6997.0,7138.0),(144
 04.0,6840.0),(10329.0,7622.0),(2334.0,7095.0)
 204,(7450.0,2941.0),(1641.0,1692.0),(4104.0,2613.0),(9162.0,5241.0),(4809.0,5184.0),(991.0,4760.0),(7330.0,7129.0),(1407
 0.0,6840.0),(10304.0,7290.0),(2209.0,7083.0)
 205,(7443.0,2701.0),(1667.0,1951.0),(4104.0,2885.0),(9162.0,4907.0),(4955.0,5185.0),(1057.0,5087.0),(7096.0,7136.0),(143
 42.0,6840.0),(10313.0,7406.0),(2485.0,7167.0)
 206,(7434.0,2383.0),(1647.0,1749.0),(4104.0,2847.0),(9173.0,4575.0),(5275.0,5187.0),(1037.0,4990.0),(6763.0,7145.0),(141
 60.0,6840.0),(10325.0,7561.0),(2492.0,7500.0)
 207,(7444.0,2716.0),(1661.0,1894.0),(4104.0,2650.0),(9178.0,4539.0),(5216.0,5198.0),(1007.0,4840.0),(7070.0,7136.0),(142
 53.0,6840.0),(10299.0,7229.0),(2500.0,7834.0)
 208,(7450.0,2925.0),(1652.0,1805.0),(4104.0,2650.0),(9189.0,4454.0),(4926.0,5231.0),(1073.0,5167.0),(7356.0,7128.0),(142
 49.0,6840.0),(10317.0,7467.0),(2495.0,7636.0)
 209,(7440.0,2592.0),(1656.0,1837.0),(4104.0,2848.0),(9162.0,4660.0),(4987.0,5186.0),(1021.0,4909.0),(7023.0,7138.0),(141
 63.0,6840.0),(10320.0,7500.0),(2488.0,7303.0)
 210,(7437.0,2492.0),(1520.0,2027.0),(4104.0,2885.0),(9162.0,4993.0),(5279.0,5156.0),(1023.0,4921.0),(6809.0,7144.0),(143
 38.0,6840.0),(10297.0,7195.0),(2344.0,7096.0)
 211,(7447.0,2825.0),(1187.0,2051.0),(4104.0,2613.0),(9162.0,5185.0),(5363.0,5206.0),(1070.0,5155.0),(7142.0,7134.0),(140
 74.0,6840.0),(10322.0,7528.0),(2199.0,7082.0)
 212,(7446.0,2817.0),(855.0,2075.0),(4119.0,2280.0),(9162.0,4851.0),(5535.0,5215.0),(1005.0,4828.0),(7394.0,7222.0),(1440
 7.0,6840.0),(10315.0,7439.0),(2437.0,7105.0)
 213,(7437.0,2484.0),(522.0,2099.0),(4104.0,2551.0),(9180.0,4519.0),(5282.0,5131.0),(1244.0,4767.0),(7445.0,7552.0),(1409
 4.0,6840.0),(10371.0,7180.0),(2106.0,7074.0)
 214,(7209.0,2355.0),(673.0,2088.0),(4104.0,2885.0),(9170.0,4594.0),(5288.0,5088.0),(1565.0,4799.0),(7408.0,7310.0),(1431
 8.0,6840.0),(10704.0,7176.0),(2438.0,7105.0)
 215,(6877.0,2325.0),(1006.0,2064.0),(4104.0,2829.0),(9197.0,4399.0),(5232.0,5024.0),(1550.0,5132.0),(7230.0,7132.0),(141
 84.0,6840.0),(11037.0,7171.0),(2197.0,7082.0)
 216,(7134.0,2348.0),(1338.0,2040.0),(4104.0,2712.0),(9169.0,4608.0),(5058.0,4932.0),(1266.0,5194.0),(6897.0,7141.0),(142
 29.0,6840.0),(11371.0,7166.0),(2346.0,7096.0)
 217,(7401.0,2373.0),(1671.0,2016.0),(4112.0,2379.0),(9182.0,4505.0),(5068.0,5035.0),(1051.0,5058.0),(6935.0,7140.0),(142
 73.0,6840.0),(11443.0,7165.0),(2289.0,7091.0)
 218,(7069.0,2342.0),(1641.0,1687.0),(4106.0,2453.0),(9185.0,4487.0),(5020.0,5087.0),(994.0,4772.0),(7268.0,7131.0),(1413
 9.0,6840.0),(11110.0,7170.0),(2253.0,7088.0)
 219,(6942.0,2331.0),(1668.0,1956.0),(4116.0,2317.0),(9166.0,4626.0),(5280.0,5051.0),(1059.0,5099.0),(7158.0,7134.0),(143
 62.0,6840.0),(10777.0,7175.0),(2586.0,7110.0)
 220,(7274.0,2361.0),(1646.0,1743.0),(4116.0,2318.0),(9194.0,4417.0),(5478.0,4902.0),(1308.0,5193.0),(6825.0,7143.0),(140
 50.0,6840.0),(10443.0,7179.0),(2919.0,7110.0)
 221,(7261.0,2360.0),(1662.0,1899.0),(4106.0,2453.0),(9162.0,4748.0),(5361.0,4999.0),(1454.0,5187.0),(7008.0,7138.0),(143
 83.0,6840.0),(10310.0,7367.0),(3252.0,7110.0)
 222,(6929.0,2330.0),(1652.0,1800.0),(4096.0,2123.0),(9162.0,5081.0),(5629.0,4844.0),(1121.0,5200.0),(7341.0,7129.0),(141
 18.0,6840.0),(10328.0,7600.0),(3433.0,7110.0)

223,(7081.0,2343.0),(1656.0,1843.0),(4103.0,2156.0),(9162.0,5097.0),(5488.0,4654.0),(1022.0,4915.0),(7424.0,7419.0),(142
 94.0,6840.0),(10302.0,7268.0),(3100.0,7110.0)
 224,(7413.0,2374.0),(1658.0,1857.0),(4109.0,2189.0),(9162.0,4763.0),(5223.0,4850.0),(1022.0,4915.0),(7428.0,7443.0),(142
 08.0,6840.0),(10542.0,7178.0),(2767.0,7110.0)
 225,(7122.0,2347.0),(1651.0,1786.0),(4119.0,2277.0),(9162.0,4893.0),(5006.0,4906.0),(1072.0,5162.0),(7365.0,7128.0),(142
 05.0,6840.0),(10876.0,7173.0),(2534.0,7110.0)
 226,(6889.0,2326.0),(1663.0,1913.0),(4113.0,2358.0),(9162.0,5226.0),(5283.0,5053.0),(1006.0,4835.0),(7032.0,7137.0),(142
 97.0,6840.0),(11209.0,7169.0),(2867.0,7110.0)
 227,(7221.0,2356.0),(1645.0,1729.0),(4104.0,2474.0),(9162.0,4951.0),(5228.0,4845.0),(1038.0,4995.0),(6800.0,7144.0),(141
 15.0,6840.0),(11542.0,7164.0),(3200.0,7110.0)
 228,(7314.0,2365.0),(1669.0,1970.0),(4115.0,2338.0),(9167.0,4618.0),(5148.0,4944.0),(1202.0,5197.0),(7133.0,7135.0),(143
 86.0,6840.0),(11272.0,7168.0),(3485.0,7110.0)
 229,(6982.0,2334.0),(1639.0,1673.0),(4118.0,2297.0),(9184.0,4495.0),(5191.0,4891.0),(1535.0,5184.0),(7380.0,7040.0),(140
 53.0,6840.0),(10938.0,7172.0),(3152.0,7110.0)
 230,(7029.0,2339.0),(1672.0,2004.0),(4107.0,2432.0),(9183.0,4498.0),(5270.0,4791.0),(1227.0,5196.0),(7380.0,6707.0),(143
 59.0,6840.0),(10605.0,7177.0),(2819.0,7110.0)
 231,(7361.0,2369.0),(1641.0,1695.0),(4110.0,2399.0),(9168.0,4616.0),(5033.0,4919.0),(1043.0,5019.0),(7380.0,6374.0),(141
 42.0,6840.0),(10297.0,7205.0),(2485.0,7110.0)
 232,(7441.0,2636.0),(1667.0,1948.0),(4120.0,2263.0),(9162.0,4949.0),(5049.0,5055.0),(1001.0,4811.0),(7380.0,6487.0),(142
 70.0,6840.0),(10323.0,7538.0),(2815.0,7110.0)
 233,(7451.0,2969.0),(1647.0,1752.0),(4104.0,2596.0),(9162.0,5229.0),(4914.0,5342.0),(1067.0,5138.0),(7380.0,6820.0),(142
 32.0,6840.0),(10315.0,7429.0),(3148.0,7110.0)
 234,(7442.0,2672.0),(1661.0,1891.0),(4104.0,2902.0),(9162.0,4896.0),(4914.0,5342.0),(1027.0,4938.0),(7354.0,7128.0),(141
 81.0,6840.0),(10302.0,7266.0),(3482.0,7110.0)
 235,(7435.0,2412.0),(1466.0,2031.0),(4104.0,2831.0),(9175.0,4563.0),(4757.0,5184.0),(1018.0,4892.0),(7020.0,7138.0),(143
 21.0,6840.0),(10328.0,7599.0),(3204.0,7110.0)
 236,(7444.0,2745.0),(1133.0,2055.0),(4104.0,2667.0),(9176.0,4550.0),(5007.0,5187.0),(1076.0,5185.0),(6812.0,7144.0),(140
 91.0,6840.0),(10310.0,7368.0),(2871.0,7110.0)
 237,(7449.0,2897.0),(801.0,2079.0),(4104.0,2633.0),(9191.0,4442.0),(5282.0,5136.0),(1011.0,4858.0),(7145.0,7134.0),(1441
 0.0,6840.0),(10307.0,7327.0),(2537.0,7110.0)
 238,(7439.0,2564.0),(468.0,2103.0),(4104.0,2864.0),(9162.0,4671.0),(5164.0,5196.0),(1214.0,4766.0),(7478.0,7130.0),(1407
 7.0,6840.0),(10331.0,7639.0),(2205.0,7083.0)
 239,(7438.0,2520.0),(727.0,2084.0),(4104.0,2868.0),(9162.0,5005.0),(5032.0,5189.0),(1547.0,4786.0),(7811.0,7139.0),(1433
 5.0,6840.0),(10305.0,7307.0),(2338.0,7096.0)
 240,(7448.0,2854.0),(1060.0,2060.0),(4104.0,2630.0),(9162.0,5173.0),(5214.0,5271.0),(1551.0,5102.0),(7982.0,7143.0),(141
 66.0,6840.0),(10503.0,7179.0),(2488.0,7296.0)

End

